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# Linear Algebra I-II

## Theorems & Lemmas

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# 1 Vector Spaces

## 1.1 Vector Spaces and Linear Subspaces

### Definition 1.1: Vector Space

A **vector space** over a field  $K$  is a set  $V$ , endowed with two operations

$$\begin{aligned} + : V \times V &\rightarrow V, & (v_1, v_2) &\mapsto v_1 + v_2, \\ \cdot : K \times V &\rightarrow V, & (\alpha, v) &\mapsto \alpha \cdot v \end{aligned}$$

s.t. the following conditions (axioms) hold:

- (V1)  $\forall v_1, v_2, v_3 \in V : v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$ ,
- (V2)  $\exists$  an element  $0 \in V : 0 + v = v \quad \forall v \in V$ ,
- (V3)  $\forall v \in V \exists v' \in V : v + v' = 0$ ,
- (V4)  $\forall v_1, v_2 \in V : v_1 + v_2 = v_2 + v_1$ ,
- (V5)  $\forall a, b \in K \forall v \in V : a \cdot (b \cdot v) = (a \cdot b) \cdot v$ ,
- (V6)  $\forall v \in V : 1 \cdot v = v$ ,
- (V7)  $\forall a \in K \forall v_1, v_2 \in V : a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2$ ,
- (V8)  $\forall a, b \in K \forall v \in V : (a + b) \cdot v = a \cdot v + b \cdot v$ .

### Lemma 1.2: Properties of Vector Spaces

Let  $V$  be a vector space over a field  $K$ . Then:

- (a) The 0 vector from (V2) is unique. To avoid confusion (e.g. with  $0 \in K$ ) we will sometimes denote it by  $0_V$ .
- (b) The element  $v'$  (corresponding to  $v$ ) from (V3) is unique. We'll denote it by  $-v$ . We'll also write  $w - v := w + (-v)$  for all  $w \in V$ ,
- (c)  $\forall v \in V : 0 \cdot v = 0$ ,
- (d)  $\forall a \in K : a \cdot 0 = 0$ ,
- (e)  $\forall v \in V : -1 \cdot v = -v$ ,
- (f)  $\forall v \in V : -(-v) = v$ ,
- (g) If  $a \cdot v = 0$  for some  $a \in K, v \in V$ , then either  $a = 0$  or  $v = 0$ ,
- (h) Associativity for  $+$  holds also for some of  $n$  elements.  $v_1 + \dots + v_n$  makes sense no matter how we put the brackets  $v_1 + (v_2 + (\dots + (v_{n-1} + v_n)))$  or  $(v_1 + v_2) + (\dots)$  etc.

**Example: The Coordinate Space  $K^n$** 

Let  $K$  be a field and  $n \in \mathbb{N}$ .

$$K^n = \{(a_1, \dots, a_n) \mid a_i \in K, 1 \leq i \leq n\}.$$

$K^n$  is a vector space over  $K$  if we endow it with the following operations:

- Let  $v = (a_1, \dots, a_n), w = (b_1, \dots, b_n) \in K^n$ .

$$v + w = (a_1, \dots, a_n) + (b_1, \dots, b_n) := (a_1 + b_1, \dots, a_n + b_n),$$

- For  $a \in K$

$$a \cdot v = a \cdot (a_1, \dots, a_n) := (a \cdot a_1, \dots, a \cdot a_n)$$

- $0 := (0, \dots, 0) \in K^n$

With these operations  $K^n$  becomes a vector space over  $K$ .

**Definition 1.3: Linear Subspace**

Let  $V$  be a vector space over  $K$ . A subset  $W \subseteq V$  is called a **linear subspace** of  $V$  (or just subspace of  $V$ ), if the following holds

(LSS1)  $W \neq \emptyset$ ,

(LSS2)  $\forall w_1, w_2 \in W : w_1 + w_2 \in W$ ,

(LSS3)  $\forall a \in K, w \in W : a \cdot w \in W$ .

**Lemma 1.4: Linear Subspace is a Vector Space**

*Let  $V$  be a vector space over  $K$  and  $W \subseteq V$  a subspace. Then  $W$  is a vector space on its own, when endowed with the operations from  $V$ .*

**Lemma 1.5: Criterion for a Linear Subspace**

*Let  $V$  be a vector space over  $K$ , and  $W \subseteq V$  a subset. Then  $W$  is a linear subspace of  $V$  if and only if the following holds:*

(1)  $0_V \in W$ ,

(2)  $\forall a_1, a_2 \in K, w_1, w_2 \in W : a_1 w_1 + a_2 w_2 \in W$ .

**Example: Spaces of functions**

Fix a field  $K$  and a non-empty set  $S$ . Define

$$K^S = \{f : S \rightarrow K \mid f \text{ is a function}\}.$$

Define  $+$  and  $\cdot$  on  $K^S$  as follows. Let  $f, g \in K^S$ ,  $a \in K$ . Define

$$\begin{aligned}(f + g) : S &\rightarrow K, & S \ni x &\mapsto f(x) + g(x), \\ (a \cdot f) : S &\rightarrow K, & S \ni x &\mapsto a \cdot f(x).\end{aligned}$$

Now Let  $S = [0, 1]$  and take  $K = \mathbb{R}$ . Define  $C(S) := \{f \in \mathbb{R}^S \mid f \text{ is continuous on } [0, 1]\} \subseteq \mathbb{R}^S$ . Then  $C(S) \subseteq \mathbb{R}^S$  is a linear subspace.

## 1.2 Span of Sets

### Definition 1.6: Span

Let  $S \subseteq V$  be a subset. We define

$$\text{Sp}(S) := \bigcap_{W \in \mathcal{N}} W,$$

where

$$\mathcal{N} := \{W \mid W \text{ is a subspace of } V \text{ and } S \subseteq W\},$$

i.e., the collection of all subspaces  $W \subseteq V$  that contain  $S$ .  $\text{Sp}(S)$  is called the span of  $S$  (in  $V$ ).

### Lemma 1.7: Properties of the Span

- (1)  $\text{Sp}(S) \subseteq V$  is a linear subspace.
- (2) If  $W \subseteq V$  is a linear subspace and  $S \subseteq W$ , then  $\text{Sp}(S) \subseteq W$ .

In other words,  $\text{Sp}(S)$  is a linear subspace of  $V$ , and moreover among those subspaces that contain  $S$ ,  $\text{Sp}(S)$  is the smallest one.

### Definition 1.8: Linear Combination

Let  $n \in \mathbb{N}$ ,  $a_1, \dots, a_n \in K$ ,  $v_1, \dots, v_n \in V$ . A **linear combination** of the vectors  $v_1, \dots, v_n$  with coefficients  $a_1, \dots, a_n$  is the vector

$$a_1 v_1 + \dots + a_n v_n = \sum_{i=1}^n a_i v_i \in V.$$

It is important to note that in this course consider only finitely many elements in the sums of linear combinations.

### Lemma 1.9: Alternate Definition of Span

Let  $\emptyset \neq S \subseteq V$ . Then

$$\text{Sp}(S) = \{a_1 v_1 + \dots + a_n v_n \mid n \in \mathbb{N}, a_i \in K, v_i \in S \quad \forall 1 \leq i \leq n\}.$$

**Remark 1.10.** The set  $S$  can be finite or infinite. Regardless of that, each linear combination uses only finite numbers of elements from  $S$ .

Let  $n \in \mathbb{N}$ ,  $v_1, \dots, v_n \in V$ . We'll write

$$\text{Sp}(v_1, \dots, v_n) := \text{Sp}(\{v_1, \dots, v_n\}).$$

**Remark 1.11.** *It obviously holds that*

$$\text{Sp}(v_1, \dots, v_n) = \left\{ \sum_{i=1}^n \alpha_i v_i \mid \alpha_i \in K \quad \forall 1 \leq i \leq n \right\}.$$

**Remark 1.12.** *It follows from the definition of span that*

$$\text{Sp}(\emptyset) = \{0_V\}.$$

#### Definition 1.13: Generating Set

Let  $V$  be a vect. sp. over  $K$  and  $S \subseteq V$ . We say that  $S$  **generates** (or **spans**)  $V$  if  $\text{Sp}(S) = V$ .  $S$  is called a generating set of  $V$  if  $\text{Sp}(S) = W$ , where  $W \subseteq V$  is a subspace of  $V$ . We say that  $S$  **generates** (or **spans**)  $W$ .

#### Definition 1.14: Finite Dimensional

A vector space  $V$  is called **finite-dimensional** if  $\exists$  a **finite** set  $S \subseteq V$  such that  $\text{Sp}(S) = V$ . Otherwise  $V$  is called **infinite-dimensional**.

## 1.3 Bases of Vector Spaces

### 1.3.1 Linear Independence

Let  $V$  be a vector space over  $K$ .

#### Definition 1.15: Linear Independence

Let  $v_1, \dots, v_n$  be a list of  $n$  vectors in  $V$ . We say that  $v_1, \dots, v_n$  is **linearly independent** if the only linear combination of  $v_1, \dots, v_n$  that gives  $0 \in V$  is the trivial one. In other words,

$$\forall a_1, \dots, a_n \in K : a_1 v_1 + \dots + a_n v_n = 0 \implies a_1 = 0, \dots, a_n = 0.$$

By definition, we say that the empty list of vectors  $\emptyset$  is linearly independent. We say that a list of vectors is linearly dependent if, the list is not linearly independent.

**Remark 1.16.** (1)  $0$  can always be written as a linear combination of any finite list of vectors  $0v_1 + \dots + 0v_n = 0$ , i.e., the trivial linear combination.

(2) If the list  $v_1, \dots, v_n$  has two vectors that are the same, say  $v_k = v_l$  for  $1 \leq k < l \leq n$ , then  $v_1, \dots, v_n$  is **not** linearly independent. Indeed,  $v_k + (-1)v_l = 0$ .

(3) The order of the vectors in the list  $v_1, \dots, v_n$  has no effect on linear independence. So, if there are no repetitions in  $v_1, \dots, v_n$  we can talk about linear independence of the set  $\{v_1, \dots, v_n\}$ .

**Definition 1.17: Linear Independence in the Context of Families**

Let  $\mathcal{F} = \{v_i\}_{i \in I}$  be a family of vectors in  $V$ , indexed by a set  $I$ . We say that  $\mathcal{F}$  is linearly independent if  $\forall n \in \mathbb{N}$  and any sequence of distinct indices  $i_1, \dots, i_n \in I$ , the list of vectors  $v_{i_1}, \dots, v_{i_n}$  is linearly independent. Of course, if in  $\mathcal{F}$  there are any repetitions of vectors, i.e., if  $\exists i, j, i \neq j$ , s.t.  $v_i = v_j$ , then  $\mathcal{F}$  cannot be linearly independent.

**Lemma 1.18: Subsets of Linearly Independent Sets are Linearly Independent**

*If  $\emptyset \neq S \subseteq V$  is linearly independent. Then every non-empty subset  $S' \subseteq S$  is also linearly independent.*

**Definition 1.19: Linear Dependence**

A family  $\mathcal{F} = \{v_i\}_{i \in I}$  of vectors in  $V$  is called **linearly dependent** if it is not linearly independent.

**1.3.2 Basis of a Vector Space****Definition 1.20: Basis of a Vector Space**

A subset  $S \subseteq V$  of a vector space is called a basis of  $V$  if the following holds:

- (1)  $S$  is linearly independent.
- (2)  $S$  generates (or spans)  $V$ , i.e.,  $\text{Sp}(S) = V$ .

**Proposition 1.21: Equivalent Formulation of basis**

*A subset  $S \subseteq V$  is a basis of  $V$  if and only if every  $v \in V$  can be written in a unique way as a linear combination of vectors in  $S$ .*

*Proof.* For simplicity we will assume that  $|S| < \infty$ . The case  $|S| = \infty$  is similar.

$\Leftarrow$ : We assume that every  $v \in V$  can be written in a unique way as a linear combination of elements from  $S$ . Now, "can be written" means that  $\text{Sp}(S) = V$ . Next write  $S = \{v_1, \dots, v_n\}$ . Consider the vector  $0 \in V$ . Clearly

$$0 = 0 \cdot v_1 + \dots + 0 \cdot v_n.$$

Since 0 can be written in a unique way as a linear combination of  $v_1, \dots, v_n$  it follows that

$$0 = a_1 v_1 + \dots + a_n v_n \quad \implies \quad a_1 = \dots = a_n = 0.$$

Therefore,  $S$  is linearly independent, which proves that  $S$  is a basis of  $V$ .

$\Rightarrow$ : We assume that  $S$  is a basis for  $V$ . By assumption  $\text{Sp}(S) = V$ , so this means every  $v \in V$  can be expressed as a linear combination of vectors from  $S$ . It remains to show that this unique.

Assume by contradiction that  $\exists v \in V$  which has two different ways to be written as a linear combination of vectors from  $S$ , i.e.,

$$v = a_1 v_1 + \dots + a_n v_n = b_1 v_1 + \dots + b_n v_n$$



and  $a_k \neq b_k$  for some  $1 \leq k \leq n$ . Then

$$0 = (a_1 - b_1)v_1 + \dots + \underbrace{(a_k - b_k)}_{\neq 0} + \dots + (a_n - b_n)v_n$$

and we deduce that 0 can be written as a non-trivial linear combination of elements from  $S$ . Therefore,  $S$  is linearly dependent, which is a contradiction to the fact that  $S$  is a basis of  $V$ .  $\square$

### 1.3.3 Example: Bases

$V = M_{m \times n}(K)$ ,  $\{E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n\}$  is a basis, where

$$E_{ij} = \begin{pmatrix} 0 & \dots & 0 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 1 & \dots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \dots & 0 & \dots & 0 \end{pmatrix},$$

where the 1 is at the position  $(i, j)$  in the matrix.

#### Lemma 1.22: A

Let  $v_1, \dots, v_m \in V$  be a list of vectors which is linearly dependent. Then  $\exists 1 \leq j \leq m$  s.t.:

- (1)  $v_j \in \text{Sp}(v_1, \dots, v_{j-1})$ .
- (2)  $\text{Sp}(v_1, \dots, v_{j-1}, v_{j+1}, \dots, v_m) = \text{Sp}(v_1, \dots, v_m)$ . In other words, we can drop one of the vectors from the list without changing the span.

#### Lemma 1.23: B

Same assumptions as in Lemma 1.22, but we assume in addition that  $v_1, \dots, v_k$  are linearly independent, where  $1 \leq k \leq m$  is some index. Then  $k < j$ .

#### Lemma 1.24: C

Let  $w_1, \dots, w_n \in V$  s.t.  $\text{Sp}(w_1, \dots, w_n) = V$ , and let  $v \in V$ . Then  $v, w_1, \dots, w_n$  is linearly dependent.

#### Lemma 1.25: D

Let  $v_1, \dots, v_n \in V$  be s.t.  $\text{Sp}(v_1, \dots, v_n) = V$ , and let  $u_1, \dots, u_m \in V$  be linearly independent. Then  $m \leq n$ . Moreover, one can delete  $m$  vectors from the list  $v_1, \dots, v_n$  s.t. the remaining list, say  $v'_1, \dots, v'_{n-m}$  together with  $u_1, \dots, u_m$  still span  $V$ , i.e.,  $\text{Sp}(u_1, \dots, u_m, v'_1, \dots, v'_{n-m}) = V$ .

#### Lemma 1.26: E

Let  $w_1, \dots, w_l \in V$  and assume that  $w_j \notin \text{Sp}(w_1, \dots, w_{j-1}) \quad \forall 1 \leq j \leq l$ . Then the list  $w_1, \dots, w_l$  is linearly independent.

**Lemma 1.27: F**

*Suppose that  $\text{Sp}(v_1, \dots, v_n) = V$ . Then  $\exists$  a subset of  $\{v_1, \dots, v_n\}$  which is a basis for  $V$ .*

**Theorem 1.28: Bases Have the Same Number of Elements**

*Let  $V$  be a finite dimensional vector space over  $K$ . Then  $V$  has a basis with finitely many elements. Moreover, every basis of  $V$  is finite and has the same number of elements.*

**Definition 1.29: Dimension of a Vector Space**

Let  $V$  be a finite dimensional vector space over  $K$ . We define the dimension of  $V$  to be the number  $n \in \mathbb{N}$  of elements in a basis (any basis !) of  $V$ . We write  $\dim V := n$ . In case  $V$  is not finite dimensional we write  $\dim V = \infty$ .

**Remark 1.30.** •  $\dim V$  is well defined as it does not depend on a specific choice of basis for  $V$ .

- $\dim\{0\} = 0$ , b.c.  $\emptyset$  is a basis for the space  $\{0\}$ , and  $|\emptyset| = 0$ .

**Summary**

Let  $V$  be a finite dimensional vector space over  $K$ . Then:

- (1) Every finite list of vectors that spans  $V$  contains a sublist which is a basis for  $V$ .
- (2) Every linearly independent list can be extended to a basis of  $V$ .

**Theorem 1.31: Equivalent Statements for Bases**

*Let  $V$  be a vector space of  $\dim V = n \in \mathbb{N}$ . Let  $v_1, \dots, v_n \in V$ . Then the following statements are equivalent:*

- (1)  $v_1, \dots, v_n$  are linearly independent.
- (2)  $v_1, \dots, v_n$  generate  $V$ , i.e.,  $\text{Sp}(v_1, \dots, v_n) = V$ .
- (3)  $v_1, \dots, v_n$  is a basis for  $V$ .

**Theorem 1.32: Consequences of Bases**

*Suppose that  $n = \dim V < \infty$ . Let  $v_1, \dots, v_k \in V$ . Then:*

- (1) If  $k < n$ , the list  $v_1, \dots, v_k$  **cannot** span  $V$ .
- (2) If  $k > n$ , the list  $v_1, \dots, v_k$  **cannot** be linearly independent.

**Proposition 1.33: Dimension of Subspaces**

*Let  $V$  be a finite dimensional vector space over  $K$ . Then every subspace  $U \subseteq V$  is also finite dimensional. Moreover,  $\dim U \leq \dim V$  with equality if and only if  $U = V$ .*

**1.4 Row and Column Spaces**

Row and Column Spaces

### Definition 1.34: Row and Column Space

Let  $A \in M_{m \times n}(K)$  and  $u_1, \dots, u_m \in K^n$  the rows of  $A$  and  $v_1, \dots, v_n \in K^m$  the columns of  $A$ , i.e.,

$$A = \begin{pmatrix} - & u_1 & - \\ & \vdots & \\ - & u_m & - \end{pmatrix} = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}.$$

We define two spaces

$$\text{RowS}(A) := \text{Sp}(u_1, \dots, u_m) \subseteq K^n$$

$$\text{ColS}(A) := \text{Sp}(v_1, \dots, v_n) \subseteq K^m.$$

### Lemma 1.35: Row Equivalent Matrices

If  $A, B \in M_{m \times n}(K)$  are row-equivalent matrices then  $\text{RowS}(A) = \text{RowS}(B)$ .

**Remark 1.36.** The statement of the Lemma does **not** hold for the column space.

#### 1.4.1 Transpose of a Matrix

### Definition 1.37: Transpose of a Matrix

Let  $A = (a_{i,j}) \in M_{m \times n}(K)$ . We define the transposed matrix of  $A$  as  $A^T := (b_{i,j}) \in M_{n \times m}(K)$ , where  $b_{i,j} = a_{j,i}$ .

### Lemma 1.38: Properties of the Transpose

or every  $A, B \in M_{m \times n}(K)$ ,  $\lambda \in K$  we have

1.  $(A + B)^T = A^T + B^T$ .
2.  $(\lambda A)^T = \lambda A^T$ .
3.  $(A^T)^T = A$ .
4.  $\text{RowS}(A^T) = \text{ColS}(A)$ ,  $\text{ColS}(A^T) = \text{RowS}(A)$ .

## 1.5 Sums of Vector Spaces

### Definition 1.39: Sum of Vector Spaces

Let  $V$  be a vector space over  $K$  and  $U, W \subseteq V$  linear subspaces. We define the sum  $U + W$  of  $U, W$  as

$$U + W := \{u + w \mid u \in U, w \in W\}.$$

In a similar way we can define sums of more than two subspaces. If  $U_1, \dots, U_r \subseteq V$  are subspaces, define

$$U_1 + \dots + U_r := \{u_1 + \dots + u_r \mid u_1 \in U_1, \dots, u_r \in U_r\}.$$

### Proposition 1.40: Properties of the Sum of Vector Spaces

Let  $U, W \subseteq V$  be linear subspaces. Then:

1.  $U + W = \text{Sp}(U \cup W)$ . In particular,  $U + W \subseteq V$  is a subspace and  $U, W \subseteq U + W$  are subspaces of  $U + W$ .
2. If  $U, W$  are finite dimensional then so is  $U + W$ . Moreover, one can find a basis for  $U + W$  as follows:
  - (a) Choose a basis  $p_1, \dots, p_k$  for  $U \cap W$ , where  $k = \dim(U \cap W)$ .
  - (b) Extend  $p_1, \dots, p_k$  to a basis  $p_1, \dots, p_k, u_1, \dots, u_{l-k}$  of  $U$ , where  $l = \dim U$ .
  - (c) Extend  $p_1, \dots, p_k$  to a basis  $p_1, \dots, p_k, w_1, \dots, w_{m-k}$  of  $W$ , where  $m = \dim W$ .

Then  $p_1, \dots, p_k, u_1, \dots, u_{l-k}, w_1, \dots, w_{m-k}$  is a basis for  $U + W$ .

3. In particular, we have  $\dim(U + W) = \dim U + \dim W - \dim(U \cap W)$ .

### Corollary 1.41: Consequences of the Properties

Let  $U, W \subseteq V$  be two finite dimensional linear subspaces of a vector space  $V$ . The following are equivalent:

1.  $\dim(U + W) = \dim U + \dim W$ .
2.  $\dim(U \cap W) = 0$ .
3.  $U \cap W = \{0\}$ .
4. Every  $v \in U + W$  can be written in a **unique** way as  $v = u + w$  with  $u \in U, w \in W$ .
5. If  $u + w = 0$ , where  $u \in U$  and  $w \in W$ , then  $u = w = 0$ .

### Definition 1.42: Linear Complement

Let  $V$  be a vector space over  $K$  and  $U \subseteq V$  a linear subspace. A linear subspace  $W \subseteq V$  is called a **linear complement** of  $U$  if  $U + W = V$  and  $U \cap W = \{0\}$ . Note that  $U, W$  satisfy all the five equivalent statements from the corollary 1.41.

### Proposition 1.43: Existence of a Complement

Let  $V$  be a finite dimensional vector space and  $U \subseteq V$  a subspace. Then there exists a subspace  $W \subseteq V$  which is a complement to  $U$ .

## 2 Linear Maps

### Definition 2.1: Linear Map

Let  $V, W$  be vector spaces over  $K$ . A map  $T : V \rightarrow W$  is called linear map if:

1.  $\forall v_1, v_2 \in V : T(v_1 + v_2) = T(v_1) + T(v_2)$
2.  $\forall \alpha \in K, \forall v \in V : T(\alpha \cdot v) = \alpha \cdot T(v)$

This means that  $T : V \rightarrow W$  is a map that respects the structures of vector spaces on  $(V, +, \cdot), (W, +, \cdot)$ . We denote the set of all linear maps  $T : V \rightarrow W$  by  $\text{Hom}(V, W)$ . Sometimes  $\text{hom}_K(V, W)$ .

### Proposition 2.2: Properties of Linear Maps

Let  $V, W$  be vector spaces over a field  $K$ , and let  $T : V \rightarrow W$  be a linear map. Then:

1.  $\forall n \in \mathbb{N}, a_1, \dots, a_n \in K, v_1, \dots, v_n \in V$  we have

$$T\left(\sum_{i=1}^n a_i v_i\right) = \sum_{i=1}^n a_i T(v_i).$$

2.  $T(0_V) = 0_W$ .

### Theorem 2.3: A unique Linear Map

Let  $V$  be a finite dimensional vector space and  $v_1, \dots, v_n \in V$  a basis for  $V$ . Let  $w_1, \dots, w_n \in W$  be arbitrary vectors. Then there exists a unique linear map  $T : V \rightarrow W$  s.t.

$$T(v_i) = w_i \quad \forall 1 \leq i \leq n.$$

### Lemma 2.4: Matrix Representation of the Unique Linear Map

For every linear transformation  $T : K^n \rightarrow K^m$  there exists a unique matrix  $A \in M_{m \times n}(K)$  s.t.  $T = T_A$ .

## 2.1 Kernel & Image

### Definition 2.5: Kernel and Image

Let  $V, W$  be vector spaces over  $K$  and  $T : V \rightarrow W$  a linear map.

1. The set

$$\ker(T) := \{v \in V \mid Tv = 0\} \subseteq V$$

is a linear subspace of  $V$ . It is called the **kernel of  $T$** .

2. The set

$$\text{Im}(T) := \{Tv \mid v \in V\} \subseteq W$$

is a linear subspace of  $W$ . It is called the **image of  $T$** .

With the notation from set theory, we have

$$\ker(T) = T^{-1}(\{0\}), \quad \text{Im}(T) = T(V).$$

### Proposition 2.6: Equivalences for Injective and Surjective

Let  $T : V \rightarrow W$  be a linear map. Then:

1.  $T$  is injective  $\iff \ker(T) = \{0\}$ .
2.  $T$  is surjective  $\iff \text{Im}(T) = W$ .

### 2.1.1 About Linear Maps

#### Definition 2.7: Isomorphism

A linear map  $T : V \rightarrow W$  is called an **isomorphism** if there exists a linear map  $S : W \rightarrow V$  s.t.  $S \circ T = \text{id}_V$ ,  $T \circ S = \text{id}_W$ . We say that  $V$  is isomorphic to  $W$  if there exists an isomorphism  $T : V \rightarrow W$ . We write  $V \cong W$ .

#### Lemma 2.8: Linearity of the Inverse Map

Let  $T : V \rightarrow W$  be a linear map which is bijective. Then the inverse map  $T^{-1} : W \rightarrow V$  is linear. In other words, a linear bijective map is an isomorphism.

#### Lemma 2.9: Composition of Linear Maps

Let  $T : V \rightarrow W$  and  $S : W \rightarrow U$  be linear maps. Then the composition  $S \circ T : V \rightarrow U$  is also linear.

#### Definition 2.10: Endomorphism / Automorphism

A linear map  $T : V \rightarrow V$  is sometimes called an **endomorphism** of  $V$ . We denote the set of all endomorphisms of  $V$  by  $\text{End}(V)$ . (So,  $\text{End}(V) = \text{Hom}(V, V)$ .)

A linear map  $T : V \rightarrow V$  which is an isomorphism is also called an **automorphism** of  $V$ .

### Lemma 2.11: Span of a Basis

Let  $T : V \rightarrow W$  be a linear map. Let  $v_1, \dots, v_n$  be a basis of  $V$ . Then  $\text{Im}(T) = \text{Sp}(Tv_1, \dots, Tv_n)$

### Theorem 2.12: Rank Theorem

Let  $V, W$  be vector spaces, where  $V$  is finite dimensional and let  $T : V \rightarrow W$  be a linear map. Then

$$\dim \ker(T) + \dim \text{Im}(T) = \dim V.$$

*Proof.* Define  $n := \dim V$ . Let  $u_1, \dots, u_k$  be a basis for  $\ker(T)$ . Note that  $\ker(T)$  is finite dimensional, b.c. it is a subspace of  $V$ , which is finite dimensional. Extend  $u_1, \dots, u_k$  to a basis  $u_1, \dots, u_k, v_1, \dots, v_{n-k}$  of  $V$ , which we can do by a previous result. By Lemma 2.11, we have that

$$\text{Im}(T) = \text{Sp}(\underbrace{T u_1, \dots, T u_k}_{=0}, T v_1, \dots, T v_{n-k}) = \text{Sp}(T v_1, \dots, T v_{n-k}).$$

We'll show now that  $T v_1, \dots, T v_{n-k}$  is linearly independent. To see this, assume that

$$\sum_{i=1}^{n-k} a_i T v_i = 0,$$

for some  $a_1, \dots, a_{n-k} \in K$ . Since  $T$  is linear we can write this linear combination as

$$T\left(\sum_{i=1}^{n-k} a_i v_i\right) = 0 \quad \implies \quad \sum_{i=1}^{n-k} a_i v_i \in \ker(T).$$

Now,  $u_1, \dots, u_k$  form a basis for  $\ker(T)$ . So,  $\exists b_1, \dots, b_k \in K$  s.t.

$$\begin{aligned} \sum_{i=1}^{n-k} a_i v_i &= b_1 u_1 + \dots + b_k u_k \\ \implies b_1 u_1 + \dots + b_k u_k + (-a_1) v_1 + \dots + (-a_{n-k}) v_{n-k} &= 0. \end{aligned}$$

Since,  $u_1, \dots, u_k, v_1, \dots, v_{n-k}$  form a basis for  $V$ , we have that  $b_1 = \dots = b_k = a_1 = \dots = a_{n-k} = 0$ . This shows that  $T v_1, \dots, T v_{n-k}$  are linearly independent. Therefore,  $T v_1, \dots, T v_{n-k}$  form a basis for  $\text{Im}(T)$ . Thus,

$$\begin{aligned} \dim \text{Im}(T) &= n - k = \dim V - \dim \ker(T) \\ \implies \dim \ker(T) + \dim \text{Im}(T) &= \dim V. \end{aligned}$$

□

**Corollary 2.13: Consequences of the Rank Theorem**

Let  $T : V \rightarrow W$  be a linear map, where  $V, W$  are finite dimensional vector spaces. Then the following holds:

1. If  $\dim W < \dim V$  then  $T$  is **not** injective.
2. If  $\dim W > \dim V$  then  $T$  is **not** surjective.
3. If  $\dim W = \dim V$  then  $T$  is bijective  $\iff T$  is surjective  $\iff T$  is injective.

**Corollary 2.14: Equivalence for an Isomorphism**

Two finite dimensional vector spaces  $V, W$  are isomorphic if and only if  $\dim V = \dim W$ .

**Theorem 2.15: Image of a Set**

Let  $T : V \rightarrow W$  be an isomorphism between two vector spaces  $V, W$ . Let  $S$  be a family of vector in  $V$ . Write  $T(S) := \{Tv\}_{v \in S}$ . Then:

1.  $S$  is linearly independent if and only if  $T(S)$  is linearly independent.
2.  $S$  spans  $V$  if and only if  $T(S)$  spans  $W$ .
3.  $S$  is a basis of  $V$  if and only if  $T(S)$  is a basis of  $W$ .

What is the significance of isomorphisms? If two vector spaces  $V, W$  are isomorphic,  $V \cong W$  (i.e.,  $\exists T : V \rightarrow W$  bijection), then  $V$  and  $W$  have the same linear algebraic properties (like dimension). We can think of  $V$  and  $W$  as the same space represented in two different ways.

However, identifying  $V$  with  $W$  usually requires a choice of an isomorphism  $T : V \rightarrow W$ . Usually there is no preferred (canonical) choice. It is therefore better, in general, not to identify different vector spaces that are isomorphic.

**Definition 2.16: Rank**

Let  $T : V \rightarrow W$  be a linear map. We define the **rank** of  $T$  by

$$\text{rank}(T) := \dim \text{Im}(T)$$

**Lemma 2.17: Rank of the Composition**

Let  $T : V \rightarrow W$ ,  $S : W \rightarrow U$  be linear maps, where  $U, V, W$  are finite dimensional vector spaces. Then

1.  $\text{rank}(S \circ T) \leq \min\{\text{rank}(S), \text{rank}(T)\}$ .
2. If  $S$  is injective, then  $\text{rank}(S \circ T) = \text{rank}(T)$ .
3. If  $T$  is surjective, then  $\text{rank}(S \circ T) = \text{rank}(S)$ .

**2.2 Writing Linear Maps using Bases/Coordinates**

In this Section we'll assume all vector spaces to be finite dimensional unless otherwise stated. We'll write bases of a vector space  $V$  as an ordered list  $\mathcal{B} = (v_1, \dots, v_n)$ .



**Definition 2.18: Coordinate Map**

Let  $V$  be a vector space over  $K$  and  $\mathcal{B} = (v_1, \dots, v_n)$  a basis of  $V$ . Let  $v \in V$ . We define the coordinate-vector of  $v$  w.r.t. the basis  $\mathcal{B}$  as

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in K^n,$$

where  $a_1, \dots, a_n \in K$  are the unique scalars in  $K$  s.t.  $v = a_1 v_1 + \dots + a_n v_n$ . Note that  $[v]_{\mathcal{B}}$  is uniquely defined since every  $v \in V$  has a unique representation as a linear combination of  $v_1, \dots, v_n$ .

Define a map  $\Phi_{\mathcal{B}} : V \rightarrow K^n$  as

$$\Phi_{\mathcal{B}}(v) := [v]_{\mathcal{B}} \in K^n.$$

**Proposition 2.19:  $\Phi_{\mathcal{B}}$  is an Isomorphism**

$\Phi_{\mathcal{B}}$  is an isomorphism

**Remark 2.20.** The inverse map to  $\Phi_{\mathcal{B}}$  is given by

$$\Phi_{\mathcal{B}}^{-1} : K^n \rightarrow V, \quad \Phi_{\mathcal{B}}^{-1} \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} := a_1 v_1 + \dots + a_n v_n.$$

**Corollary 2.21: Isomorphism Coordinate Space**

Let  $V$  be a vector space over  $K$  with  $\dim V = n \in \mathbb{Z}_{\geq 0}$ . Then  $V \cong K^n$ .

*Proof.* Choose a basis  $\mathcal{B}$  for  $V$ . Then  $\Phi_{\mathcal{B}} : V \rightarrow K^n$  is an isomorphism. □

**Remark 2.22.** 1. The corollary does not hold for  $\dim V = \infty$ , at least not as stated.

2. We've seen that  $V \cong K^n$ , if  $V$  is finite dimensional, but in general there does **not** exist a canonical isomorphism  $V \xrightarrow{\cong} K^n$ . For example  $\Phi_{\mathcal{B}}$  depends on the choice of basis  $\mathcal{B}$ .

Now we arrive at the main question. Let  $V, W$  be vector spaces over  $K$ ,  $n := \dim V, m := \dim W$ , and let  $T : V \rightarrow W$  be a linear map. Fix a basis  $\mathcal{B}$  for  $V$  and a basis  $\mathcal{C}$  for  $W$ . Consider the diagram

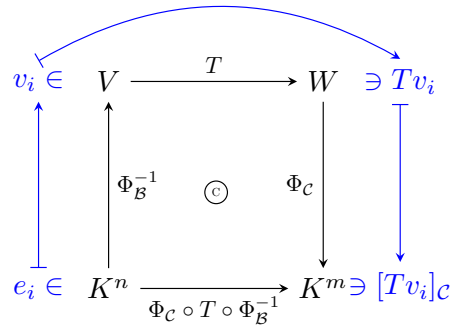
$$\begin{array}{ccc} V & \xrightarrow{T} & W \\ \Phi_{\mathcal{B}}^{-1} \uparrow & \text{\textcircled{C}} & \downarrow \Phi_{\mathcal{C}} \\ K^n & \xrightarrow{\Phi_{\mathcal{C}} \circ T \circ \Phi_{\mathcal{B}}^{-1}} & K^m \end{array}$$

The 'C' means that the diagram is commutative.

Since  $\Phi_B^{-1}, T, \Phi_C$  are all linear maps, the map  $\Phi_C \circ T \circ \Phi_B^{-1} : K^n \rightarrow K^m$  is also linear. By Lemma 2.4, any linear map  $K^n \rightarrow K^m$  can be written using a matrix  $A \in M_{m \times n}(K)$ . In other words, there exists a unique matrix  $A \in M_{m \times n}(K)$  s.t.  $\Phi_C \circ T \circ \Phi_B^{-1} = T_A$ . The matrix  $A$  is called the **representation matrix of  $T$  w.r.t. the bases  $\mathcal{B}$  and  $\mathcal{C}$** . We denote this matrix by  $[T]_{\mathcal{C}}^{\mathcal{B}} \in M_{m \times n}(K)$ .

### 2.2.1 Entries of the Representation Matrix

Let  $T : V \rightarrow W$  be a linear map and  $\mathcal{B} = (v_1, \dots, v_n)$  a basis for  $V$ , and  $\mathcal{C} = (w_1, \dots, w_m)$  a basis for  $W$ .



$T_A(e_i) = \Phi_C \circ T \circ \Phi_B^{-1}(e_i) = \Phi_C \circ T v_i = [T v_i]_{\mathcal{C}}$ . But these are precisely the columns of  $A$ . In other words

$$A = [T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [T v_1]_{\mathcal{C}} & \dots & [T v_n]_{\mathcal{C}} \\ | & & | \end{pmatrix} \in M_{m \times n}.$$

Another way to describe this matrix is: Let  $A = (a_{i,j})$ , where  $a_{i,j}$  are defined uniquely by

$$T v_j = \sum_{i=1}^m a_{i,j} w_i.$$

#### Proposition 2.23: Representation matrix

Let  $T : V \rightarrow W$  be a linear map between two finite dimensional vector spaces  $V, W$ . Let  $\mathcal{B}, \mathcal{C}$  be bases for  $V, W$  respectively. Then

$$\forall v \in V : [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [v]_{\mathcal{B}} = [T v]_{\mathcal{C}}.$$

### 2.2.2 Representing Composition of Linear Maps by Matrices

Let  $T : V \rightarrow W, S : W \rightarrow U$  be linear maps. Let  $\mathcal{A} = (v_1, \dots, v_n)$  be a basis for  $V$ ,  $\mathcal{B} = (w_1, \dots, w_m)$  a basis for  $W$  and  $\mathcal{C} = (u_1, \dots, u_p)$  a basis for  $U$ .

Write  $[T]_{\mathcal{B}}^{\mathcal{A}} = (a_{ij})$ ,  $[S]_{\mathcal{C}}^{\mathcal{B}} = (b_{ij})$ ,  $[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = (c_{ij})$ . By definition we have that

$$T(v_j) = \sum_{k=1}^m a_{kj} w_k \quad \forall 1 \leq j \leq n \quad (2.1)$$

$$S(w_j) = \sum_{i=1}^p b_{ij} u_i \quad \forall 1 \leq j \leq m \quad (2.2)$$

$$S \circ T(v_j) = \sum_{i=1}^p c_{ij} u_i \quad \forall 1 \leq j \leq n \quad (2.3)$$

But we can calculate  $S \circ T(v_j)$  by using Equations (2.1), (2.2). Then we have

$$\begin{aligned} S \circ T(v_j) &= S\left(\sum_{k=1}^m a_{kj} w_k\right) = \sum_{k=1}^m a_{kj} S(w_k) = \sum_{k=1}^m a_{kj} \left(\sum_{i=1}^p b_{ik} u_i\right) \\ &= \sum_{k=1}^m \sum_{i=1}^p a_{kj} b_{ik} u_i = \sum_{i=1}^p \sum_{k=1}^m a_{kj} b_{ik} u_i = \sum_{i=1}^p \left(\sum_{k=1}^m b_{ik} a_{kj}\right) u_i \\ \implies c_{ij} &= \sum_{k=1}^m b_{ik} a_{kj}. \end{aligned}$$

### 3 Matrices

#### Definition 3.1: Matrix Multiplication

Let  $A \in M_{m \times n}(K)$ ,  $B \in M_{n \times p}(K)$ . We define a new matrix  $A \cdot B \in M_{m \times p}(K)$  which is called the product of  $A$  and  $B$ . Write  $A = (a_{ij})$ ,  $B = (b_{ij})$ . The matrix  $A \cdot B$  has entries  $(c_{ij})$ , where

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}.$$

**Important:**  $A \cdot B$  is defined only when the number of columns of  $A$  equals the number of rows of  $B$ .

**Conclusion:** Let  $T : V \rightarrow W$ ,  $S : W \rightarrow U$  be linear maps between finite dimensional vector spaces. Let  $\mathcal{A}, \mathcal{B}, \mathcal{C}$  be bases for  $V, W, U$  respectively. Then

$$[S \circ T]_{\mathcal{C}}^{\mathcal{A}} = [S]_{\mathcal{C}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{A}}.$$

### Definition 3.2: Kronecker-Delta/Identity Matrix

Let  $\mathcal{I}$  be an index set. Define for all  $i, j \in \mathcal{I}$

$$\delta_{ij} := \begin{cases} 1 & \text{if } i = j, \\ 0 & \text{if } i \neq j. \end{cases}$$

This is called the **kronecker-delta**.

We define the **identity matrix**  $I_n \in M_{n \times n}(K)$  as

$$I_n := \delta_{ij}, \quad I_n = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

**Remark 3.3.** Let  $V$  be a finite dimensional vector space and  $n := \dim V$ . Let  $\text{id} : V \rightarrow V$  be the identity map. Let  $\mathcal{B}$  be any basis for  $V$ . Then  $[\text{id}]_{\mathcal{B}}^{\mathcal{B}} = I_n$ . In other words, the matrix representation of the identity map w.r.t  $\mathcal{B}$  is the identity matrix  $I_n$ .

### Proposition 3.4: Associativity and Distributivity of Matrix Multiplication

$$1. \forall A \in M_{m \times n}(K), B \in M_{n \times p}(K), C \in M_{p \times q}(K) : (AB)C = A(BC) \in M_{m \times q}.$$

$$2. \forall A, B \in M_{m \times n}(K), C \in M_{n \times p}(K), C' \in M_{q \times m}(K) \text{ we have}$$

$$(A + B)C = AC + BC \in M_{m \times p}(K),$$

$$C'(A + B) = C'A + C'B \in M_{q \times n}(K).$$

$$3. \forall A \in M_{m \times n}(K) : I_m A = A I_n = A.$$

$$4. \forall \alpha \in K, A \in M_{m \times n}(K), B \in M_{n \times p}(K) : \alpha(AB) = (\alpha A)B = A(\alpha B).$$

**Remark 3.5.** Matrix multiplication is in general **not** commutative, i.e.,  $\exists A, B$  s.t.  $AB \neq BA$ .

In  $M_{n \times n}(K)$  we can add/subtract matrices and also multiply matrices. There is also a unit  $I_n$ . These operations satisfy properties like associativity, distributivity, etc. From this we get that  $M_{n \times n}(K)$  is a (non-commutative) **ring** with a unity.

### Definition 3.6: Invertibility

An  $n \times n$  matrix  $A \in M_{n \times n}(K)$  is **invertible** if  $\exists B \in M_{n \times n}(K)$  s.t.  $AB = BA = I_n$ .

**Remark 3.7.** If  $A$  is invertible, then there exists only one matrix  $B$  s.t.  $AB = BA = I_n$ .

*Proof.* Assume  $AB = BA = I_n$  and  $AC = CA = I_n$ . Then

$$B = BI_n = B(AC) = (BA)C = I_n C = C.$$

□

**Definition 3.8: The Inverse Matrix**

If  $A$  is invertible, the unique matrix  $B$  s.t.  $AB = BA = I_n$  is called **the inverse of  $A$**  and is denoted by  $A^{-1}$ . So, we have  $AA^{-1} = A^{-1}A = I_n$ .

We denote the set of all invertible  $n \times n$  matrices by

$$GL_n(K) = GL(n, K) := \{A \in M_{n \times n}(K) \mid A \text{ is invertible}\}.$$

This set is called the general linear group.

**Proposition 3.9: Structure of the General Linear Group**

$GL_n(K)$ , endowed with matrix multiplication  $(A, B) \mapsto A \cdot B$  is a group. The unity is  $I_n$ . The inverse of  $A \in GL_n(K)$  is  $A^{-1}$ . Moreover,

$$\forall A, B \in GL_n(K) : (A^{-1})^{-1} = A, \quad (AB)^{-1} = B^{-1}A^{-1}.$$

**Corollary 3.10: Unique Solution for Linear System of Equations**

Let  $A \in GL_n(K)$ . Then  $\forall b \in K^n$  the linear system of equations  $Ax = b$  has a unique solution, namely  $x = A^{-1}b$ .

**Proposition 3.11: Invertibility and Isomorphisms**

Let  $A \in M_{n \times n}(K)$ . Then  $A \in GL_n(K)$  if and only if  $T_A : K^n \rightarrow K^n$  is an isomorphism. Moreover,  $(T_A)^{-1} = T_{A^{-1}}$ .

### Definition 3.12: Upper/Lower Triangular and Diagonal Matrices

A matrix  $A \in M_{n \times n}(K)$ ,  $A = (a_{ij})$ , is called

- **Upper triangular** if  $a_{ij} = 0 \quad \forall i > j$ , i.e.,

$$\begin{pmatrix} * & \dots & \dots & * \\ 0 & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

- **Lower Triangular** if  $a_{ij} = 0 \quad \forall i < j$ , i.e.,

$$\begin{pmatrix} * & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & 0 \\ * & \dots & \dots & * \end{pmatrix}.$$

- **Diagonal** if  $a_{ij} = 0 \quad \forall i \neq j$ , i.e.,

$$\begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix}.$$

### Lemma 3.13: Multiplication of Triangular/Diagonal Matrices

Let  $A, B$  be two diagonal (respectively both upper triangular, respectively both lower triangular). Then  $A \cdot B$  is of the same type.

### Corollary 3.14: Change of Bases for a Linear Map

Let  $V, W$  be finite dimensional vector spaces over  $K$ . Let  $n := \dim V, m := \dim W$ . Let  $\mathcal{B}, \mathcal{B}'$  be two bases for  $V$ , and  $\mathcal{C}, \mathcal{C}'$  be two bases for  $W$ . Then

$$[id_V]_{\mathcal{B}}^{\mathcal{B}'} \in GL_n(K), [id_W]_{\mathcal{C}'}^{\mathcal{C}} \in GL_m(K), \text{ and} \\ [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}, [id_V]_{\mathcal{C}}^{\mathcal{C}'} = ([id_V]_{\mathcal{C}'}^{\mathcal{C}})^{-1}.$$

Let  $T : V \rightarrow W$  be a linear map. Then

$$[T]_{\mathcal{C}'}^{\mathcal{B}'} = [id_W]_{\mathcal{C}'}^{\mathcal{C}} \cdot [T]_{\mathcal{C}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

### Definition 3.15: Transition Matrix/Change of Bases Matrix

Let  $V$  be a finite dimensional vector space over  $K$ ,  $n := \dim V$ . Let  $\mathcal{B}, \mathcal{B}'$  be two bases of  $V$ . The matrix  $[id_V]_{\mathcal{B}'}^{\mathcal{B}} \in GL_n(K)$  is called the **change of bases matrix** between  $\mathcal{B}$  and  $\mathcal{B}'$ , or the **transition matrix** from basis  $\mathcal{B}$  to basis  $\mathcal{B}'$ .

If  $\mathcal{B} = (v_1, \dots, v_n), \mathcal{B}' = (v'_1, \dots, v'_n)$  then

$$[id_V]_{\mathcal{B}'}^{\mathcal{B}} = \begin{pmatrix} | & & | \\ [v_1]_{\mathcal{B}'} & \dots & [v_n]_{\mathcal{B}'} \\ | & & | \end{pmatrix}.$$

### Proposition 3.16: Inverse of a Matrix with Bases

Let  $T : V \rightarrow W$  be an isomorphism between two finite dimensional vector spaces. Let  $\mathcal{B}, \mathcal{C}$  be bases for  $V, W$  respectively. Then  $[T]_{\mathcal{C}}^{\mathcal{B}}$  is an invertible matrix and

$$[T^{-1}]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{C}}^{\mathcal{B}})^{-1}.$$

### Corollary 3.17: Change of Basis for an Endomorphism

Let  $V$  be a finite dimensional vector space and  $T : V \rightarrow V$  a linear map (an endomorphism). Let  $\mathcal{B}, \mathcal{B}'$  be two bases for  $V$ . Then

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'}.$$

### Definition 3.18: Equivalence and Similarity of Matrices

1. Let  $A, B \in M_{m \times n}(K)$ . We say that  $A$  and  $B$  are **equivalent** if  $\exists P \in GL_m(K), Q \in GL_n(K)$  s.t.  $B = PAQ$ .
2. Let  $A, B \in M_{n \times n}(K)$ . We say that  $A$  and  $B$  are **similar** if  $\exists P \in GL_n(K)$  s.t.  $B = P^{-1}AP$ .

**Remark 3.19.** 1. Equivalence between  $m \times n$  matrices gives an equivalence relation on  $M_{m \times n}(K)$ .

2.  $A, B \in M_{m \times n}(K)$  are equivalent if and only if there exists two vector spaces  $V, W$  with  $n := \dim V$ ,  $m := \dim W$  and two bases  $\mathcal{B}, \mathcal{B}'$  of  $V$  and two bases  $\mathcal{C}, \mathcal{C}'$  for  $W$  s.t.  $A = [T]_{\mathcal{C}}^{\mathcal{B}}$ ,  $B = [T]_{\mathcal{C}'}^{\mathcal{B}'}$ . In other words,  $A$  is equivalent to  $B$  if and only if  $A$  and  $B$  are matrix representations of the same linear map w.r.t. different choices of bases for  $V$  and  $W$ .

3. Similarity between  $n \times n$  matrices gives an equivalence relation on  $M_{n \times n}(K)$ .

4.  $M, N \in M_{n \times n}(K)$  are similar if and only if there exists an  $n$ -dimensional vector space  $V$  and  $T : V \rightarrow V$  and two bases  $\mathcal{B}, \mathcal{B}'$  of  $V$  s.t.  $M = [T]_{\mathcal{B}}^{\mathcal{B}}$ ,  $N = [T]_{\mathcal{B}'}^{\mathcal{B}'}$ .

**Proposition 3.20: Existence of 'Nice' Bases**

Let  $V, W$  be finite dimensional vector spaces over  $K$ , of dimension  $n := \dim V$ ,  $m := \dim W$ . Let  $T : V \rightarrow W$  be a linear map with  $\text{rank}(T) = r$ . (Recall that  $\text{rank}(T) := \dim \text{Im}(T)$ .) Then there exists a basis  $\mathcal{B} = (v_1, \dots, v_n)$  of  $V$  and a basis  $\mathcal{C} = (w_1, \dots, w_m)$  of  $W$  s.t.

$$[T]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

**Corollary 3.21**

Let  $A \in M_{m \times n}(K)$ . Then  $\exists P \in GL_m(K), Q \in GL_n(K)$  s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix},$$

where  $r = \text{col-rank}(A)$ . In particular,  $\text{col-rank}(A) \leq \min\{m, n\}$ .

**Corollary 3.22: Condition for Equivalence of Matrices**

Let  $A, B \in M_{m \times n}(K)$ . Then  $A \sim B$  if and only if  $\text{col-rank}(A) = \text{col-rank}(B)$ .

Moreover,  $\forall 0 < r \leq \min\{m, n\}$  there is precisely one equivalence class of matrices  $A$  with  $\text{col-rank} = r$ .

**3.1 col-rank = row-rank****Theorem 3.23: col-rank = row-rank**

Let  $A \in M_{m \times n}(K)$ . Then  $\text{col-rank}(A) = \text{row-rank}(A)$ .

**Remark 3.24.** 1. Because of this Theorem we'll denote from now on  $\text{rank}(A) := \text{col-rank}(A) = \text{row-rank}(A)$ .

2. Suppose  $A'$  is a row-reduced echelon matrix which is row-equivalent to  $A$ . Then we have that  $\text{rank}(A) = \# \text{ pivots in } A'$ . Indeed,  $\text{rank}(A) = \text{row-rank}(A) = \text{row-rank}(A') = \# \text{ pivots in } A'$ .
3. Recall that for  $T : V \rightarrow W$  we defined  $\text{rank}(T) := \dim \text{Im}(T)$ . Also, if  $A \in M_{m \times n}(K)$ , then  $\text{rank}(T_A) = \dim \text{Im}(T) = \text{col-rank}(A)$ .

We now present some Lemmas for the proof of the Theorem.

**Lemma 3.25: Isomorphisms preserve the Rank**

Let  $T : V \rightarrow W$  be a linear map, and  $S : W \xrightarrow{\cong} U$ ,  $L : P \xrightarrow{\cong} V$  be isomorphisms. Then

1.  $\text{rank}(S \circ T) = \text{rank}(T)$ .
2.  $\text{rank}(T \circ S) = \text{rank}(T)$ .

**Lemma 3.26**

Let  $A \in M_{m \times n}(K)$ . Then  $A$  is invertible if and only if  $T_A K^n \rightarrow K^n$  is an isomorphism.



### Lemma 3.27: Equivalent Matrices Have the Same Rank

Let  $A, B \in M_{m \times n}(K)$  s.t.  $B = PAQ$ , where  $P \in GL_m(K)$ ,  $Q \in GL_n(K)$ . Then

1.  $\text{col-rank}(A) = \text{col-rank}(B)$
2.  $\text{row-rank}(A) = \text{row-rank}(B)$ .

*Proof. Proof of Theorem 3.23:* By Corollary 3.21,  $\exists P \in GL_m(K), Q \in GL_n(K)$  s.t.

$$PAQ = \begin{pmatrix} I_r & 0_{r, n-r} \\ 0_{m-r, r} & 0_{m-r, n-r} \end{pmatrix}.$$

Denote this matrix by  $B$ . For  $B$  we have  $\text{col-rank}(B) = \text{row-rank}(B) = r$ . By Lemma 3.27, we have that  $\text{col-rank}(A) = \text{col-rank}(B) = r$ , and  $\text{row-rank}(A) = \text{row-rank}(B) = r$ . Therefore,  $\text{col-rank}(A) = \text{row-rank}(A)$ .  $\square$

**Remark 3.28.** Although  $\text{row-rank}(A) = \text{col-rank}(A)$ , the spaces  $\text{ColS}(A)$  and  $\text{RowS}(A)$  are **different**. First of all if  $A \in M_{m \times n}$ , then  $\text{ColS}(A) \subseteq K^m$  and  $\text{RowS}(A) \subseteq K^n$ . But even when  $n = m$ , these two spaces are **different**. They have the same dimension, but are different.

## 3.2 More on Matrices and Linear Equations

Let  $A \in M_{m \times n}(K)$ , and consider the system of linear equations  $Ax = 0$ . Denote by

$$\text{Sol}(A, 0) := \{x \in K^n \mid Ax = 0\} = \ker(T_A),$$

the **space** of solutions  $Ax = 0$ . Note that  $\dim \text{Sol}(A, 0) = n - \text{rank}(A)$  (b.c.  $\dim \ker(T_A) = n - \dim \text{Im}(T_A)$ ). Let  $A'$  be a row-reduced echelon matrix which is row-equivalent to  $A$  (e.g.  $A'$  is obtained from  $A$  by Gauss-elimination). # pivots of  $A' = \text{rank}(A)$ ,  $n - \text{rank}(A) = \#$  free variables. Also, we have that  $\text{Sol}(A, 0) = \text{Sol}(A', 0)$ .

$$\begin{pmatrix} 0 & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & 0 & 1 & * & \dots & * \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Denote by  $x_{i_1}, \dots, x_{i_l}$  the free variables, where  $1 \leq i_1 < \dots < i_l \leq n$ ,  $l = n - \text{rank}(A)$  and by  $x_{j_1}, \dots, x_{j_k}$ , where  $1 \leq j_1 < \dots < j_k \leq n$ ,  $r = \text{rank}(A)$  the pivots. For each choice of values for the free variables  $x_{i_1}, \dots, x_{i_l} \in K$ , the pivot variables  $x_{j_1}, \dots, x_{j_k}$  are uniquely determined. So, we obtain a map

$$\Phi : K^l \rightarrow \text{Sol}(A, 0)$$

$$\Phi(\lambda_1, \dots, \lambda_l) = \text{the unique solution of } Ax = 0,$$

with  $x_{i_1} = \lambda_1, \dots, x_{i_l} = \lambda_l$ .

### Proposition 3.29: $\Phi$ is an Isomorphism

The map  $\Phi$  is linear. Moreover, it is an isomorphism.

## Corollary 3.30

Let  $b \in K^m$ ,  $A \in M_{m \times n}(K)$ . Then

1.  $\text{Sol}(A, b) := \{x \in K^n \mid Ax = b\} \neq \emptyset$  iff  $b \in \text{ColS}(A)$ .
2. If  $\text{Sol}(A, b) \neq \emptyset$  and  $y \in \text{Sol}(A, b)$  then

$$\text{Sol}(A, b) = y + \text{Sol}(A, 0) := \{y + x \mid x \in \text{Sol}(A, 0)\}.$$

## Proposition 3.31

Let  $A \in M_{m \times n}(K)$ ,  $b \in K^m$ . The following statements are equivalent:

1.  $\text{rank}(A|b) = \text{rank}(A)$ .
2. The system of equations  $Ax = b$  has a solution.

## Proposition 3.32

1.  $\text{rank}(A|b) = \text{rank}(A) = n$ .
2. The system of equations  $Ax = b$  has a unique solution.

## 3.3 Elementary Row Operations Revisited

Let  $A \in M_{n \times p}(K)$ . We learned about elementary row operations which are used as part of the Gauss elimination. All of these operations can be expressed by matrix multiplications.

Denote by  $E_{ij}$  the matrix in Example 1.3.3.

**Type I:** Let  $1 \leq i \leq n$ ,  $1 \leq j \leq n$ ,  $i \neq j$ . Let  $\alpha \in K$ . Define  $Q_{ij}(\alpha)$  to be the matrix obtained from  $I_n$  after applying  $R_i + \alpha R_j \rightarrow R_i$ . We have  $Q_{ij}(\alpha) = I_n + \alpha E_{ij}$ .

**Type II:** Let  $1 \leq i \leq n$ ,  $1 \leq j \leq n$ . Define  $P_{ij}$  to be the matrix obtained from  $I_n$  by permuting (exchanging) row  $\# i$  and row  $\# j$ , i.e. applying  $R_i \leftrightarrow R_j$ . We have  $P_{ij} = I_n - E_{ii} - E_{jj} + E_{ij} + E_{ji}$ .

**Type III:** Let  $1 \leq i \leq n$  and  $0 \neq \alpha \in K$ . Denote by  $S_i(\alpha)$  the matrix obtained from  $I_n$  by applying  $\alpha R_i \rightarrow R_i$ .

## Lemma 3.33: Elementary Row Operations as Matrix Multiplication

Let  $A \in M_{n \times p}(K)$ .

1. Multiplying  $A$  from the left with  $Q_{ij}(\alpha)$  results in the row operation  $R_i + \alpha R_j \rightarrow R_i$ :  $A \xrightarrow{R_i + \alpha R_j \rightarrow R_i} Q_{ij}(\alpha)A$ .
2. Multiplying  $A$  from the left with  $P_{ij}$  results in the row operation  $R_i \leftrightarrow R_j$ :  $A \xrightarrow{R_i \leftrightarrow R_j} P_{ij}A$ .
3. Multiplying  $A$  from the left with  $S_i(\alpha)$  results in the row operation  $\alpha R_i \rightarrow R_i$ :  $A \xrightarrow{\alpha R_i \rightarrow R_i} S_i(\alpha)A$ .

The same holds for elementary column operations, which is done by right hand side multiplications of the matrices.

**Lemma 3.34: Elementary Matrices are Invertible**

Every elementary matrix is invertible, and the inverse of each of them is an elementary matrix of the same type, i.e.

$$Q_{ij}(\alpha)^{-1} = Q_{ij}(-\alpha), \quad P_{ij}^{-1} = P_{ji}, \quad S_{ij}(\alpha)^{-1} = S_{ij}(\alpha^{-1}).$$

**Theorem 3.35**

$\forall A \in GL_n(K)$ ,  $\exists k \geq 1$  and elementary matrices  $T_1, \dots, T_k$  s.t.  $T_k \cdot \dots \cdot T_1 \cdot A = I_n$ . Furthermore, we have that

$$A = T_1^{-1} \cdot \dots \cdot T_k^{-1}, \quad A^{-1} = T_k \cdot \dots \cdot T_1.$$

In particular,  $A$  can be written as a finite product of elementary matrices.

**Theorem 3.36: Equivalent condition for Invertibility**

Let  $A \in M_{n \times n}(K)$ . Place  $A$  together with  $I_n$  in the following way  $(A|I_n)$ . Then  $A$  is invertible if and only if  $(A|I_n)$  can be brought via a sequence of elementary row-operations to the form  $(I_n|B)$  for some matrix  $B \in M_{n \times n}(K)$ . In this case we have  $A^{-1} = B$ .

## 4 Constructing New Vector Spaces Out of Old Ones

**Proposition 4.1: The Vector Space of Linear Functions**

Let  $V, W$  be two vector spaces over  $K$ . Then  $\text{Hom}(V, W)$  has the structure of a vector space over  $K$ , if we endow it with the following operations:

- $\forall T_1, T_2 \in \text{Hom}(V, W)$ ,  $(T_1 + T_2)(v) := T_1(v) + T_2(v) \quad \forall v \in V$ .
- $\forall T \in \text{Hom}(V, W)$ ,  $\alpha \in K$ ,  $(\alpha T)(v) := \alpha T(v) \quad \forall v \in V$ .

The neutral element of  $\text{Hom}(V, W)$  is the map  $0 : V \rightarrow W$ ,  $v \mapsto 0$ .

**Theorem 4.2: The map  $\Psi_C^{\mathcal{B}}$** 

Let  $V, W$  be finite dimensional vector spaces over  $K$ ,  $\mathcal{B} = (v_1, \dots, v_n)$  a basis for  $V$ , and  $\mathcal{C} = (w_1, \dots, w_m)$  a basis for  $W$ . Then the map

$$\Psi_C^{\mathcal{B}} : \text{Hom}(V, W) \rightarrow M_{m \times n}(K)$$

defined by

$$T \mapsto [T]_{\mathcal{C}}^{\mathcal{B}} := \begin{pmatrix} | & & | \\ [Tv_1]_{\mathcal{C}} & \dots & [Tv_n]_{\mathcal{C}} \\ | & & | \end{pmatrix}$$

is a linear map and moreover an isomorphism.

**Corollary 4.3: Dimension of  $\text{Hom}(V, W)$** 

If  $V, W$  are finite dimensional vector spaces then so is  $\text{Hom}(V, W)$  and  $\dim \text{Hom}(V, W) = \dim V \cdot \dim W$ .

**4.0.1 Rings**

Recall that Rings have the same axioms as Fields except that we do **not** require that every non-zero element has an inverse.

**Definition 4.4: Homomorphism of Rings**

Let  $R, S$  be rings. A **homomorphism of rings** is a map  $\varphi : R \rightarrow S$  s.t.:

- $\forall r_1, r_2 \in R : \varphi(r_1 + r_2) = \varphi(r_1) + \varphi(r_2)$ .
- $\forall r_1, r_2 \in R : \varphi(r_1 \cdot r_2) = \varphi(r_1) \cdot \varphi(r_2)$
- $\varphi(1) = 1$ .
- $\varphi(0) = 0$ .

**Corollary 4.5:  $\Psi$  for an Endomorphism is an isomorphism of Rings**

Let  $V$  be a vector space over  $K$ . Then  $\text{End}(V) := \text{Hom}(V, V)$  is a ring with a unity. This ring is, in general, not commutative. The multiplication (product) in  $\text{End}(V)$  is given by composition, i.e.,

$$T_1 \cdot T_2 := T_1 \circ T_2 \quad \forall T_1, T_2 \in \text{Hom}(V, V).$$

The unity is  $\text{id}_V$ . Moreover, if  $V$  is finite dimensional and  $\dim V = n$ , and  $\mathcal{B} = (v_1, \dots, v_n)$  is a basis for  $V$ , then  $\Psi_{\mathcal{B}}^{\mathcal{B}} : \text{Hom}(V, V) \rightarrow M_{n \times n}(K)$  is an isomorphism of rings

**4.1 Dual Space**

An important special case of  $\text{Hom}(V, W)$  is when  $W = K$ .

**Definition 4.6: Dual Space**

Let  $V$  be a vector space over  $K$ . The **dual space** of  $V$ , is defined to be the vector space  $V^* := \text{Hom}(V, K)$ . The elements of  $V^*$  are linear maps  $\ell : V \rightarrow K$ , and are sometimes called **functionals** (on  $V$ ).

**Corollary 4.7: Dimension of the Dual Space**

Let  $V$  be a finite dimensional vector space over  $K$ . Then  $\dim V^* = \dim V$ .

Let  $V = K_{\text{col}}^n$ . Then we can identify  $V^*$  with  $K_{\text{col}}^n$ , as follows: If  $\ell \in K_{\text{row}}^n$ , then  $\ell$  defines a functional  $f_{\ell} : K_{\text{col}}^n \rightarrow K$  by  $f_{\ell}(v) := \ell \cdot v \in K$ . The map  $D : K_{\text{row}}^n \rightarrow (K_{\text{col}}^n)^*$  defined by  $D(\ell) := f_{\ell}$  is a linear map and is an isomorphism.

Let  $V$  be a finite dimensional vector space over  $K$  and  $\mathcal{B} = (v_1, \dots, v_n)$  a basis for  $V$ .  $\forall 1 \leq i \leq n$ ,

define a functional  $v_i^* \in V^*$ , i.e.,  $v_i^* : V \rightarrow K$  linear map, as follows:

$$v_i^*(v_j) := \delta_{ij} = \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}.$$

Since  $v_1, \dots, v_n$  form a basis for  $V$ , our definition determines uniquely a well-defined functional  $v_i^* : V \rightarrow K$ .

**Lemma 4.8: The Dual Basis**

*The elements  $v_1^*, \dots, v_n^* \in V^*$  form a basis for  $V^*$ . This basis is called the **dual basis** to  $\mathcal{B}$ . We denote this basis by  $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$ .*

*Moreover,*

$$\forall \ell \in V^* : \ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*.$$

*Proof.* Let  $\ell \in V^*$ . We claim that

$$\ell = \sum_{i=1}^n \ell(v_i) \cdot v_i^*. \quad (4.1)$$

Or in other words, we claim that

$$\forall v \in V : \ell(v) = \sum_{i=1}^n \ell(v_i) \cdot v_i^*(v). \quad (4.2)$$

Indeed, both sides of Equation (4.1) are linear maps  $V \rightarrow K$ , so it's enough to check that Equation (4.2) holds for  $v = v_1, \dots, v = v_n$ , b.c.  $v_1, \dots, v_n$  form a basis for  $V$ . And indeed, for  $v = v_k$  we have

$$\sum_{i=1}^n \ell(v_i) \cdot v_i^*(v_k) = \sum_{i=1}^n \ell(v_i) \cdot \delta_{ik} = \ell(v_k).$$

The claim just proven shows that  $v_1^*, \dots, v_n^*$  span  $V^*$ . Since  $\dim V^* = \dim V = n$ , it follows that  $v_1^*, \dots, v_n^*$  form a basis for  $V^*$ .  $\square$

**Question:** Let  $V$  be a finite dimensional vector space over  $K$ . Let  $\mathcal{B}$  and  $\mathcal{C}$  be two bases for  $V$ . Therefore,  $\mathcal{B}^*$  and  $\mathcal{C}^*$  are two bases for  $V^*$ . What is the relation between  $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$  and  $[id_V]_{\mathcal{B}}^{\mathcal{C}}$ .

**Proposition 4.9: Relation between  $[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*}$  and  $[id_V]_{\mathcal{B}}^{\mathcal{C}}$**

$$[id_{V^*}]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([id_V]_{\mathcal{C}}^{\mathcal{B}})^T = \left( ([id_V]_{\mathcal{B}}^{\mathcal{C}})^T \right)^{-1}.$$

## 4.2 The Dual Map

**Definition 4.10: The Dual Map**

Let  $V, W$  be vector spaces over  $K$ . Let  $T : V \rightarrow W$  be a linear map. Define a new map

$$T^* : W^* \rightarrow V^*, \quad T^*(\ell) = \ell \circ T \quad \forall \ell \in W^*.$$

Then  $T^*$  is a linear map. It is called **the dual map** to  $T$ .

*Proof.* Let  $\ell \in W^*$ , i.e.,  $\ell : W \rightarrow K$  a linear map. Clearly

$$\ell \circ T : V \rightarrow K$$

is a linear map, b.c. both  $\ell$  and  $T$  are linear. So,  $T^*(\ell) := \ell \circ T \in V^*$ . This shows that  $T^* : W^* \rightarrow V^*$  indeed has as it's target  $V^*$  as claimed, see also the Figure 4.2. We'll show now that  $T^*$  is a linear

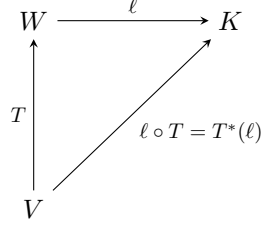


Figure 1: Map diagram of the dual map

map. Let  $\ell \in W^*$ ,  $\alpha \in K$ . We have that  $T^*(\alpha\ell) \in V^*$  is the map

$$\begin{aligned} T^*(\alpha\ell)(v) &= ((\alpha\ell) \circ T)(v) = (\alpha\ell)(Tv) = \alpha\ell(Tv) = \alpha(\ell \circ T)(v) = \alpha(T^*(\ell))(v) \\ \implies T^*(\alpha\ell) &= \alpha T^*(\ell). \end{aligned}$$

Now, let  $\ell_1, \ell_2 \in W^*$ . We have that  $T^*(\ell_1 + \ell_2) \in V^*$  is the map

$$\begin{aligned} T^*(\ell_1 + \ell_2)(v) &= ((\ell_1 + \ell_2) \circ T)(v) = (\ell_1 \circ T + \ell_2 \circ T)(v) = (\ell_1 \circ T)(v) + (\ell_2 \circ T)(v) \\ &= (T^*(\ell_1))(v) + (T^*(\ell_2))(v) \\ \implies T^*(\ell_1 + \ell_2) &= T^*(\ell_1) + T^*(\ell_2). \end{aligned}$$

This shows that  $T^*$  is linear as claimed. □

#### Proposition 4.11: Composition of the Dual Map

Let  $U, V, W$  be vector spaces over  $K$ . Let  $T : V \rightarrow W$ ,  $S : W \rightarrow U$  be linear maps. Consider  $T^* : W^* \rightarrow V^*$ ,  $S^* : U^* \rightarrow W^*$ , the dual maps. Then

$$(S \circ T)^* = T^* \circ S^*.$$

#### Lemma 4.12: Interpretation of the Matrix Transpose

Let  $S : V \rightarrow W$  be a linear map between two finite dimensional vector spaces  $V$  and  $W$  over  $K$ . Let  $\mathcal{B}$  be a basis for  $V$  and  $\mathcal{C}$  be a basis for  $W$ . Consider  $S^* : W^* \rightarrow V^*$  and the bases  $\mathcal{C}^*, \mathcal{B}^*$  for  $W^*, V^*$  respectively. Then

$$[S^*]_{\mathcal{B}^*}^{\mathcal{C}^*} = ([S]_{\mathcal{C}}^{\mathcal{B}})^T.$$

*Proof.* Write  $\mathcal{B} = (v_1, \dots, v_n), \mathcal{C} = (w_1, \dots, w_m)$ . Define  $A := [S]_{\mathcal{C}}^{\mathcal{B}}, A = (a_{ij})$ .  $A$  satisfies

$$Sv_j = \sum_{i=1}^m a_{ij}w_i.$$

Now, for  $w_j^* \in \mathcal{C}^*$  we have by Lemma 4.8

$$\begin{aligned} S^* w_j^* &= w_j^* \circ S = \sum_{i=1}^n (w_j^* \circ S)(v_i) \cdot v_i^* = \sum_{i=1}^n w_j^*(S(v_i)) \cdot v_i^* \\ &= \sum_{i=1}^n w_j^* \left( \sum_{k=1}^m a_{ki} w_k \right) \cdot v_i^* = \sum_{i=1}^n \sum_{k=1}^m a_{ki} w_j^*(w_k) \cdot v_i^* = \sum_{i=1}^n a_{ji} v_i^*. \end{aligned}$$

Therefore, the coordinates of  $S^* w_j^*$  in the basis  $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$  are

$$\begin{pmatrix} a_{j1} \\ \vdots \\ a_{jn} \end{pmatrix},$$

which is the row  $\#j$  in  $A$ , represented as a column, or equivalently the column  $\#j$  in the matrix  $A^T$ .  $\square$

### 4.3 Reflexivity

Let  $V$  be a vector space over  $K$ . We have  $V^*$ , but also  $V^{**} := (V^*)^*$  by applying 'dual' one more time, to  $V^*$ .  $(V^*)^*$  is sometimes called the bidual space of  $V$ . Is there any relation between  $(V^*)^*$  and  $V$ .

#### Theorem 4.13: The Canonical Isomorphism between $(V^*)^*$ and $V$

Let  $v$  be a *finite* dimensional vector space over  $K$ . Then there exists a canonical isomorphism  $\tau : V \rightarrow (V^*)^*$ , given by

$$\forall v \in V, \tau(v) : V^* \rightarrow K$$

is the map

$$\tau(v)(\ell) := \ell(v) \quad \forall \ell \in V^*.$$

In other words,

$$V \ni v \xrightarrow{\tau} (V^* \ni \ell \mapsto \ell(v) \in K).$$

**Remark 4.14.** The map  $\tau : V \rightarrow (V^*)^*$  exists and is defined in the same way as above regardless of  $V$  being finite or infinite dimensional. Moreover,  $\tau$  is always injective. However, when  $V$  is infinite dimensional,  $\tau$  fails to be surjective.

### 4.4 Naturality

The map  $\tau : V \rightarrow (V^*)^*$  exists for every space  $V$ , and is defined in the same way. To distinguish  $\tau$ 's for different spaces, denote it now by  $\tau_V : V \rightarrow (V^*)^*$ . Let  $V, W$  be two vector spaces over  $K$ . Then for every linear map  $T : V \rightarrow W$  we have

$$\begin{array}{ccc}
V & \xrightarrow{\tau_V} & (V^*)^* \\
\downarrow T & \text{\textcircled{C}} & \downarrow (T^*)^* \\
W & \xrightarrow{\tau_W} & (W^*)^*
\end{array}$$

which we call naturality.

#### Proposition 4.15: Image of $\tau$

Let  $V$  be a finite dimensional vector space over  $K$ , let  $\mathcal{B} = (v_1, \dots, v_n)$  be a basis for  $V$ . Let  $\mathcal{B}^* = (v_1^*, \dots, v_n^*)$  be the dual basis and  $(\mathcal{B}^*)^* = ((v_1^*)^*, \dots, (v_n^*)^*)$  the basis for  $(V^*)^*$ , which is dual to  $\mathcal{B}^*$ . Then  $\tau(\mathcal{B}) = (\mathcal{B}^*)^*$ , i.e.,

$$\tau(v_i) = (v_i^*)^* \quad \forall 1 \leq i \leq n.$$

## 4.5 Direct Sums

#### Proposition 4.16: The Vector Space of the Cartesian Product

Let  $V, W$  be a vector space over  $K$ . Then the cartesian product  $V \times W$  becomes a vector space over  $K$  if we endow it with the following operations:

- $\forall (v_1, w_1), (v_2, w_2) \in V \times W : (v_1, w_1) + (v_2, w_2) := (v_1 + v_2, w_1 + w_2).$
- $\forall \alpha \in K, (v, w) \in V \times W : \alpha \cdot (v, w) := (\alpha v, \alpha w) \in V \times W.$
- $0_{V \times W} := (0_V, 0_W).$

This new vector space  $V \times W$  is usually denoted  $V \oplus W$  and is called the direct sum of  $V$  and  $W$ . We write the elements of  $V \oplus W$  as  $(v, w)$  or  $v \oplus w$ .

**Remark 4.17.**  $V \oplus W$  comes together with the following **canonical maps**:

- 1)  $i_V : V \rightarrow V \oplus W, \quad i_V(v) := (v, 0).$
- 2)  $i_W : W \rightarrow V \oplus W, \quad i_W(w) := (0, w).$
- 3)  $p_V : V \oplus W \rightarrow V, \quad p_V(v, w) := v.$
- 4)  $p_W : V \oplus W \rightarrow W, \quad p_W(v, w) := w.$

The blue marked maps are canonical embeddings, while the orange marked maps are canonical projections. These maps are all linear. The maps  $p_V, p_W$  are surjective, and  $i_V, i_W$  are injective. Because of the maps  $i_V, i_W$  we can sometimes view  $V$  and  $W$  as subspaces of  $V \oplus W$ .



**Proposition 4.18:** The map  $\varphi$ 

Let  $V$  be a vector space over  $K$ ,  $U \subseteq V$  a subspace and  $U' \subseteq V$  another subspace which is a complement of  $U$  in  $V$  (recall Definition 1.42). Then the map

$$\varphi : U \oplus U' \rightarrow V, \quad \varphi(u, u') := u + u'$$

is a linear isomorphism.

**Remark 4.19.** When  $U'$  is a complement of  $U$  in  $V$ , then  $U + U' = V$  and we also have a canonical isomorphism  $U \oplus U' \xrightarrow{\cong} V$ . But formally speaking  $U \oplus U'$  is a different vector space than  $V$ .

**4.5.1 Generalization to more Summands**

Let  $U_1, \dots, U_m$  be **finitely** many vector spaces over  $K$ . We can define  $U_1 \oplus \dots \oplus U_m := U_1 \times \dots \times U_m$ , with the coordinatewise/componentwise operations.

For **infinite** families  $U_i, i \in \mathcal{I}$  one can define

$$\bigoplus_{i \in \mathcal{I}} U_i := \left\{ t : \mathcal{I} \rightarrow \bigcup_{i \in \mathcal{I}} U_i \mid t(i) \in U_i \quad \forall i \in \mathcal{I}, t(i) \neq 0 \text{ for at most finitely many } i\text{'s} \right\}.$$

**4.6 Quotient Spaces**

Let  $V$  be a vector space over  $K$ , and  $U \subseteq V$  a subspace. We will define a new vector space over  $K$ ,  $V/U$ , together with a linear map  $\pi : V \rightarrow V/U$  s.t.  $\pi$  is surjective and  $\ker(\pi) = U$ .  $V/U$  will have some other good properties.  $V/U$  is called the **quotient of  $V$  by  $U$**  or  $V$  **modulo  $U$** .  $\pi$  is called the **quotient map** or the projection onto the quotient space.

We begin with an equivalence relation on  $v$ . Define for  $v_1, v_2 \in V$

$$v_1 \sim v_2 \quad :\Longleftrightarrow \quad v_1 - v_2 \in U.$$

We claim that  $\sim$  defines an equivalence relation on  $V$ . Indeed

- **Reflexivity:**  $v \sim v$ , b.c.  $v - v = 0 \in U$ .
- **Symmetry:** If  $v_1 \sim v_2 \implies v_2 \sim v_1$ , b.c.  $v_2 - v_1 = -(v_1 - v_2) \in U$ .
- **Transitivity:** If  $v_1 \sim v_2$  and  $v_2 \sim v_3$ , then  $v_1 \sim v_3$ , b.c.  $v_1 - v_3 = \underbrace{(v_1 - v_2)}_{\in U} + \underbrace{(v_2 - v_3)}_{\in U} \in U$ .

We denote the equivalence class of  $v \in V$  by  $[v]$ . So

$$[v] = v + U = \{v + u \mid u \in U\},$$

or equivalently

$$[v] = \{w \in V \mid v \sim w\} = \{w \in V \mid v - w \in U\}.$$

**Definition 4.20:** Quotient Space

Let  $V$  be a vector space over  $K$  and  $U \subseteq V$  a subspace. Define  $V/U := V / \sim$  = the set of equivalence classes of the equivalence relation  $\sim$ . So, the elements of  $V/U$  are  $[v]$ , where  $v \in V$ . Alternatively, each element in  $V/U$  is a set of the form  $v + U$ , where  $v \in V$ .  $V/U$  is called the **quotient of  $V$  by  $U$**  (or modulo  $U$ ).

We want to turn  $V/U$  into a vector space over  $K$ . To achieve this, we define:

- **Addition:**  $\forall [v_1], [v_2] \in V/U, \quad [v_1] + [v_2] := [v_1 + v_2] \in V/U.$
- **Mult. by a scalar:**  $\forall \alpha \in K, [v] \in V/U, \quad \alpha \cdot [v] := [\alpha v] \in V/U.$  (Alternatively written:  $\alpha(v + U) := \alpha v + U.$ )
- **The zero:**  $0_{V/U} := [0_V].$  (Alternatively written:  $0_{V/U} := 0_V + U.$ )

#### Proposition 4.21: Well Definedness of The Operations

*The operations above are well defined. Moreover, they give  $V/U$  the structure of a vector space over  $K$ .*

We now define the map

$$\pi : V \rightarrow V/U, \quad v \mapsto [v] \in V/U.$$

#### Proposition 4.22: Properties of $\pi$

*$\pi$  is a linear map. Moreover,  $\text{Im}(\pi) = V/U$  (i.e.,  $\pi$  is surjective), and  $\ker(\pi) = U$ .*

#### Proposition 4.23: Dimension of $V/U$

*Suppose that  $V$  is finite dimensional. Then  $V/U$  is finite dimensional too, and  $\dim V/U = \dim V - \dim U$ .*

#### Theorem 4.24: The Isomorphism Theorem

*Let  $V, W$  be vector spaces over  $K$  and  $T : V \rightarrow W$  a linear map. Define a new map  $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$ , as follows. Let  $x \in V/\ker(T)$ . Choose a representative  $v \in V$  for  $x$ , i.e.,  $x = [v]$ . Define  $\bar{T}(x) = \bar{T}([v]) := T(v)$ . Then:*

1.  $\bar{T}$  is well defined, and is a linear map.
2.  $\bar{T} : V/\ker(T) \rightarrow \text{Im}(T)$  is an isomorphism.
3. The Diagram in Figure 4.6 commutes, where  $i : \text{Im}(T) \rightarrow W$  is the inclusion.

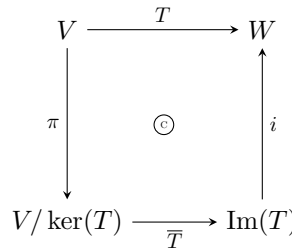


Figure 2: Map diagram of the map  $\bar{T}$ .

*Proof. 1)  $\bar{T}$  is well defined:* Suppose  $x = [v] = [v'] \in V/\ker(T)$ , we need to show that  $T(v) = T(v')$ . Indeed,  $[v] = [v']$  implies that  $v - v' \in \ker(T)$ . Therefore,  $T(v - v') = 0$ . Since  $T$  is linear we conclude that  $T(v) = T(v')$ .

**$\bar{T}$  is linear:** Let  $\alpha \in K, [v] \in V/\ker(T)$ . By the linearity of  $T$ , the definition of the operations on the quotient space  $V/\ker(T)$  and the definition of  $\bar{T}$ , we have that

$$\bar{T}(\alpha[v]) = \bar{T}([\alpha v]) = T(\alpha v) = \alpha T(v) = \alpha \bar{T}([v]).$$

Now, let  $[v_1], [v_2] \in V/\ker(T)$ . Then

$$\bar{T}([v_1] + [v_2]) = \bar{T}([v_1 + v_2]) = T(v_1 + v_2) = T(v_1) + T(v_2) = \bar{T}([v_1]) + \bar{T}([v_2]).$$

**2)  $\bar{T}$  is injective:** Let  $x \in \ker(\bar{T})$ . Write  $x$  as  $x = [v]$  for some  $v \in V$ . Then

$$\underbrace{\bar{T}(x)}_{=0} = T(v).$$

Therefore,  $v \in \ker(T)$ . Thus,  $x = [v] = [0_V] = 0_{V/\ker(T)}$ .

**$\bar{T}$  is surjective:** Let  $w \in \text{Im}(T)$ . Then there exists a  $v \in V$  s.t.  $T(v) = w$ . Therefore,  $\bar{T}([v]) = T(v) = w$ . This shows that  $\text{Im}(T) \subseteq \text{Im}(\bar{T})$ .

But by the definition of  $\bar{T}$  we also have that  $\text{Im}(\bar{T}) \subseteq \text{Im}(T)$ .

**3) The diagram commutes:** Let  $v \in V$  and  $w \in \text{Im}(T) \subseteq W$  s.t.  $T(v) = w$ . Then by the definition of  $\pi, \bar{T}$  and  $i$ , we have that

$$(i \circ \bar{T} \circ \pi)(v) = (i \circ \bar{T})([v]) = i(T(v)) = i(w) = w.$$

□

#### 4.6.1 Universal Property of Quotient Spaces

##### Theorem 4.25: Universal Property of Quotient Spaces

Let  $V$  be a vector space over  $K$  and  $U \subseteq V$  a subspace. The quotient space  $V/U$  has the following universal property:

For every space  $W$  and every linear map  $T : V \rightarrow W$ , for which  $T(U) = 0$  (i.e., for which  $U \subseteq \ker(T)$ ), there exists a **unique** linear map  $T' : V/U \rightarrow W$  s.t. the diagram in Figure 4.6.1 commutes. Moreover,  $\ker(T') = \ker(T)/U$ .

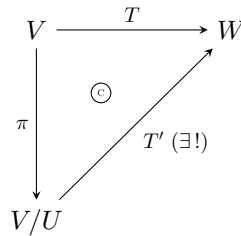


Figure 3: Map diagram of the map  $T'$ .

**Jargon:** Every linear map  $T : V \rightarrow W$  that maps  $U$  to 0 factors through  $V/U$  (i.e.,  $T = T' \circ \pi$ ), and moreover in a unique way. We say that  $T$  **induces** a map  $T' : V/U \rightarrow W$ . We also say that  $T$  **descends** to a map  $T' : V/U \rightarrow W$ .

### 4.6.2 Quotient Spaces & Complements

#### Proposition 4.26: A Canonical Isomorphism to a Complement

Let  $V$  be a vector space over  $K$  and  $U \subseteq V$  a subspace. Let  $W \subseteq V$  be a complement of  $U$ . Then there exists a canonical isomorphism  $Q : W \xrightarrow{\cong} V/U$ , defined by

$$Q(w) := [w] \in V/U \quad \forall w \in W.$$

**Remark 4.27.** The isomorphism  $Q$  is canonical, only once  $W$  is specified. In general there is no canonical choice of a complement  $W$  to  $U$ . (In general there exist many choices of complements  $W$  to  $U$ , and there is no preferred choice.)

## 4.7 Relations to Dual Spaces

#### Proposition 4.28: Perpendicular Subspace

Let  $V$  be a vector space over  $K$  and  $U \subseteq V$  a subspace. Define

$$U^\perp := \{\ell \in V^* \mid \ell|_U = 0\},$$

i.e., all the functionals on  $V$  that vanish on  $U$ . Then:

1.  $U^\perp \subseteq V^*$  is a linear subspace.
2. There exists a **canonical** isomorphism  $(V/U)^* \cong U^\perp$ . If we assume that  $V$  is finite dimensional then there exists a **canonical** isomorphism  $U^* \cong V^*/U^\perp$ .

#### Proposition 4.29

Let  $T : V \rightarrow W$  be a linear map. Then

$$(\text{Im}(T))^\perp = \ker(T^*).$$

## 5 Determinants

Let

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in M_{2 \times 2}(K).$$

There is a very simple criterion to check whether  $A$  is invertible or not.

#### Proposition 5.1: Criterion for Invertibility of a $2 \times 2$ Matrix

$A$  is invertible if and only if  $ad - bc \neq 0$ . Moreover, if  $A$  is invertible, then

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}.$$

The number  $ad - bc$  is  $\det(A)$ .

**Definition 5.2: N-Linear Function**

Let  $D : M_{n \times n}(K) \rightarrow K$  be a function. We say that  $D$  is  **$n$ -linear** if for every  $1 \leq i \leq n$ ,  $D$  is a linear function of the  $i$ -th row when the other  $(n - 1)$  rows are held fixed.

We can think of  $M_{n \times n}(K)$  as  $K_{\text{row}}^n \times \dots \times K_{\text{row}}^n$  and if

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}$$

we write  $D(A) = D(\alpha_1, \dots, \alpha_n)$ . Saying that  $D$  is  $n$ -linear means that for all  $1 \leq i \leq n$ , we have that

1.

$$\begin{aligned} D(\alpha_1, \dots, \alpha_i + \alpha'_i, \dots, \alpha_n) &= D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) + D(\alpha_1, \dots, \alpha'_i, \dots, \alpha_n) \\ \forall \alpha_1, \dots, \alpha_i, \alpha'_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

2.

$$\begin{aligned} D(\alpha_1, \dots, c \cdot \alpha_i, \dots, \alpha_n) &= c \cdot D(\alpha_1, \dots, \alpha_i, \dots, \alpha_n) \\ \forall c \in K, \alpha_1, \dots, \alpha_i, \dots, \alpha_n &\in K_{\text{row}}^n \end{aligned}$$

**Lemma 5.3: Linear Combinations of  $n$ -linear Functions**

*Any linear combination of  $n$ -linear functions is an  $n$ -linear function again.*

**Definition 5.4: Alternating Functions**

Let  $D : M_{n \times n}(K) \rightarrow K$  be an  $n$ -linear function. We say that  $D$  is **alternating** if for every matrix  $A \in M_{n \times n}(K)$  in which some two rows are equal we have that  $D(A) = 0$ .

**Proposition 5.5**

*Let  $D$  be an  $n$ -linear function. Assume that  $D$  has the property that whenever  $A$  has two equal adjacent (aufeinanderfolgende) rows then  $D(A) = 0$ . Then*

1.  *$D$  is alternating.*
2. *For every matrix  $B$ , if  $B'$  is obtained from  $B$  by interchanging any two of its rows, then  $D(B') = -D(B)$ .*

**Definition 5.6: Determinant Function**

A function  $D : M_{n \times n}(K) \rightarrow K$  is called **determinant function** if  $D$  is  $n$ -linear, is alternating and  $D(I) = 1$ .

**Definition 5.7: Row&Column-Deletion Matrix**

Let  $A \in M_{n \times n}(K)$ ,  $n \geq 2$ ,  $1 \leq i \leq n$ ,  $1 \leq j \leq n$ . We denote by

$$A(i|j) \in M_{(n-1) \times (n-1)}(K)$$

the matrix obtained from  $A$  by deleting row  $\#i$  and column  $\#j$ .

If  $D$  is an  $(n-1)$ -linear function and  $A \in M_{n \times n}(K)$  we denote

$$D_{ij}(A) := D(A(i|j)).$$

**Notation:** Let  $A \in M_{m \times n}(K)$  and  $1 \leq i \leq m$ ,  $1 \leq j \leq n$ . Denote the element at position  $(i, j)$  in the matrix  $A$  by  $A(i, j)$  or  $A_{ij}$ .

**Theorem 5.8: Entwicklungssatz**

Let  $n \geq 2$  and  $D$  an  $(n-1)$ -linear function, which is alternating. Let  $1 \leq j \leq n$ . Define a function  $E_j : M_{n \times n}(K) \rightarrow K$  by

$$E_j(A) := \sum_{i=1}^n (-1)^{i+j} A_{ij} D_{ij}(A).$$

Then  $E_j$  is an  $n$ -linear function and is alternating. If  $D$  is a determinant function then so is  $E_j$ .

**Corollary 5.9: Existence of a Determinant Function**

$\forall n \geq 1$ , there exists at least one determinant function  $M_{n \times n}(K) \rightarrow K$ .

**5.1 Uniqueness of the Determinant**

Let  $D : M_{n \times n}(K) \rightarrow K$  be an  $n$ -linear alternating function. Let

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix} \in M_{n \times n}(K).$$

Denote

$$I = \begin{pmatrix} 1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & 1 \end{pmatrix} = \begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}.$$

(So,  $\varepsilon_1, \dots, \varepsilon_n$  is the standard basis of  $K_{\text{row}}^n$ .) Clearly,

$$\alpha_i = \sum_{j=1}^n A(i, j) \varepsilon_j \quad \forall 1 \leq i \leq n.$$

Therefore,

$$\begin{aligned} D(A) &= D(\alpha_1, \dots, \alpha_n) = D\left(\sum_{j=1}^n A(1, j) \varepsilon_j, \alpha_2, \dots, \alpha_n\right) = \sum_{j=1}^n A(1, j) \cdot D(\varepsilon_j, \alpha_2, \dots, \alpha_n) \\ &= \sum_{j=1}^n A(1, j) \cdot D\left(\varepsilon_j, \sum_{k=1}^n A(2, k) \varepsilon_k, \dots, \alpha_n\right) = \sum_{j,k=1}^n A(1, j) \cdot A(2, k) \cdot D(\varepsilon_j, \varepsilon_k, \dots, \alpha_n). \end{aligned}$$

Applying these steps to every row we get that

$$D(A) = \sum_{k_1, \dots, k_n=1}^n A(1, k_1) \cdot A(2, k_2) \cdot \dots \cdot A(n, k_n) \cdot D(\varepsilon_{k_1}, \varepsilon_{k_2}, \dots, \varepsilon_{k_n}), \quad (5.1)$$

where each of the indices  $k_1, \dots, k_n$  runs between 1 and  $n$ . So far we haven't used the fact that  $D$  is alternating. Now, since  $D$  is alternating  $D(\varepsilon_{k_1}, \dots, \varepsilon_{k_n}) = 0$ , whenever two of the indices  $k_1, \dots, k_n$  are equal. So, we are interested only in ordered sequences of indices  $(k_1, \dots, k_n)$  in which  $k_i \in \{1, \dots, n\} \quad \forall i$  and  $k_i \neq k_j \quad \forall i \neq j$ . Such a sequence is called a permutation of  $\{1, \dots, n\}$ , or a permutation of degree  $n$ .

#### Definition 5.10: Permutation

A **permutation** is a bijective map  $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ .

So Alternatively, we can think of a permutation of degree  $n$  as a bijective function  $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ . Then the sequence  $(k_1, \dots, k_n) = (\sigma 1, \dots, \sigma n)$  is a permutation as defined earlier. (Notation:  $\sigma(i) := \sigma i$ .) In other words, a permutation  $\sigma$  of degree  $n$  is a re-ordering of a list with  $n$  elements  $(1, \dots, n) \xrightarrow{\sigma} (\sigma 1, \dots, \sigma n)$ .

We can rewrite Equation (5.1) as

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}),$$

where  $S_n$  is the set of all permutations of degree  $n$ .

#### 5.1.1 Important Thing About Permutations

Let  $\sigma$  be a permutation of degree  $n$ . Then one can pass from  $(1, \dots, n)$  to  $(\sigma 1, \dots, \sigma n)$  by performing a finite sequence of interchanges of pairs, called transpositions.

#### Definition 5.11: Transposition

A **transposition** is a permutation  $\theta : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , where  $\exists i, j \in \{1, \dots, n\}, i \neq j$  s.t.  $\theta(i) = j, \theta(j) = i$  and  $\theta(k) = k \quad \forall k \neq i, j$ .

Now, every permutation  $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$  can be written as a composition of transpositions

$$\sigma = \theta_1 \circ \dots \circ \theta_m, \quad (5.2)$$

where  $\theta_1, \dots, \theta_m$  are transpositions. But these  $\theta_i$ 's and  $m$  are not unique. In other words, the same  $\sigma$  can be written as in Equation (5.2) in many different ways

$$\sigma = \theta_1 \circ \dots \circ \theta_m = \theta'_1 \circ \dots \circ \theta'_{m'}.$$

Why Can  $\sigma$  be written as in (5.2). Start with  $(1, \dots, n)$  and if  $\sigma 1 \neq 1$  then interchange 1 and

$\sigma 1$ . We now have  $(\sigma 1, \dots, 1, \dots, n)$ . If  $\sigma 2 \neq 2$ , then interchange 1 and  $\sigma 2$ ; otherwise proceed to 3, etc. But of course there are other ways to do it, e.g.,  $(3, 2, 1)$  can be obtained in 3 steps:  $(1, 2, 3) \rightarrow (2, 1, 3) \rightarrow (2, 3, 1) \rightarrow (3, 2, 1)$ . But also in one step  $(1, 2, 3) \rightarrow (3, 2, 1)$ .

#### Definition 5.12: Sign of a Permutation

Let  $\sigma$  be a permutation and let  $\theta_1, \dots, \theta_m$  be transpositions s.t.  $\sigma = \theta_1 \circ \dots \circ \theta_m$ . The permutation  $\sigma$  is called **even** if  $m$  is even, and is called **odd** if  $m$  is odd. The number  $(-1)^m \in \{-1, 1\}$  is called the **sign** of  $\sigma$ , i.e.,  $\text{sgn}(\sigma) := (-1)^m$ .

#### Lemma 5.13: Well Definedness of the Sign of a Permutation

*If  $(\sigma 1, \dots, \sigma n)$  is obtained from  $(1, \dots, n)$  in two different ways, one time by a sequence of  $m$  transpositions, and another time by a sequence of  $m'$  transpositions, then  $m'$  and  $m$  have the same parity, i.e.,  $(-1)^{m'} = (-1)^m$ .*

*(So the parity of  $m$  depends only on  $\sigma$  and not on the particular way we performed a sequence of transpositions  $(1, \dots, n) \rightarrow \dots \rightarrow (\sigma 1, \dots, \sigma n)$ .)*

*Proof.* We know that determinant functions  $M_{n \times n}(K) \rightarrow K$  exists for every  $n \geq 1$ . Fix a determinant function  $E : M_{n \times n}(K) \rightarrow K$ . Consider  $E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n})$ . If  $(\sigma 1, \dots, \sigma n)$  can be obtained from  $(1, \dots, n)$  by a sequence of  $m$  transpositions, then the matrix

$$\begin{pmatrix} - & \varepsilon_{\sigma 1} & - \\ & \vdots & \\ - & \varepsilon_{\sigma n} & - \end{pmatrix}$$

can be obtained from

$$I = \begin{pmatrix} - & \varepsilon_1 & - \\ & \vdots & \\ - & \varepsilon_n & - \end{pmatrix}$$

by a sequence of  $m$  interchanges of pairs of rows. Therefore, since  $E$  is a determinant function, we have that

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^m E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^m E(I) = (-1)^m.$$

But this applies also to  $m'$ . So,

$$E(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}) = (-1)^{m'} E(\varepsilon_1, \dots, \varepsilon_n) = (-1)^{m'} E(I) = (-1)^{m'}.$$

So we have that

$$(-1)^m = (-1)^{m'}.$$

□

#### 5.1.2 Back to our $D(A)$

We have seen that if  $D$  is  $n$ -linear and alternating then

$$D(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdot \dots \cdot A(n, \sigma n) \cdot D(\varepsilon_{\sigma 1}, \dots, \varepsilon_{\sigma n}).$$



But

$$D(\varepsilon_{\sigma 1}, \varepsilon_{\sigma n}) = (\operatorname{sgn} \sigma) \cdot D(\varepsilon_1, \dots, \varepsilon_n) = (\operatorname{sgn} \sigma) \cdot D(I) \quad (5.3)$$

#### Theorem 5.14: Uniqueness of the Determinant Function

$\forall n \geq 1$ , there exists a unique determinant function  $M_{n \times n}(K) \rightarrow K$ . We will denote it by  $\det$ . It is given by the formula

$$\det(A) = \sum_{\sigma \in S_n} A(1, \sigma 1) \cdots A(n, \sigma n).$$

#### Corollary 5.15

For every  $n$ -linear and alternating function  $D : M_{n \times n}(K) \rightarrow K$  we have that

$$D(A) = \det(A) \cdot D(I) \quad \forall A \in M_{n \times n}(K).$$

#### 5.1.3 A Bit more on Permutations

Denote by  $S_n$  the set of permutations  $\sigma : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ .  $S_n$  has the structure of a group:

- We can compose permutations, i.e., if  $\sigma, \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ , then  $\sigma \cdot \tau := \sigma \circ \tau$ , where  $\sigma \circ \tau : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$ .
- The identity  $id : \{1, \dots, n\} \rightarrow \{1, \dots, n\}$  is the neutral element  $1 \in S_n$ .
- The inverse  $\sigma^{-1}$  of  $\sigma$ , is given by the inverse of the function  $\sigma$  (recall that  $\sigma$  is bijective, so  $\sigma^{-1}$  does exist).

Also, note that  $|S_n| = n!$

#### Lemma 5.16: Sign of the Composition of Permutations

$\forall \sigma, \tau \in S_n$  we have

$$\operatorname{sgn}(\sigma \cdot \tau) = \operatorname{sgn}(\sigma) \cdot \operatorname{sgn}(\tau).$$

#### Theorem 5.17: The Determinant of the Product of Two Matrices

$\forall A, B \in M_{n \times n}(K)$  we have that

$$\det(A \cdot B) = \det(A) \cdot \det(B).$$

*Proof.* Fix a matrix  $B \in M_{n \times n}(K)$ , and consider a function  $D : M_{n \times n}(K) \rightarrow K$  defined by

$$D(A) := \det(A \cdot B).$$

**Claim:**  $D$  is  $n$ -linear and alternating.

**Proof of Claim:** Write

$$A = \begin{pmatrix} - & \alpha_1 & - \\ & \vdots & \\ - & \alpha_n & - \end{pmatrix}.$$

If we view  $\alpha_i$  as a  $1 \times n$  matrix, then  $\alpha_i B$  is also  $1 \times n$ , and in fact

$$A \cdot B = \begin{pmatrix} - & \alpha_1 B & - \\ & \vdots & \\ - & \alpha_n B & - \end{pmatrix}.$$

So,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B).$$

Clearly, by the rules of matrix multiplication, for every two row vectors  $\alpha_i, \alpha'_i$  we have that  $(\alpha_i + \alpha'_i)B = \alpha_i B + \alpha'_i B$  and

$$\forall c \in K : (c \cdot \alpha_i)B = c \cdot (\alpha_i B).$$

Since  $\det$  is  $n$ -linear it follows that  $D$  is also  $n$ -linear.

If  $\alpha_i = \alpha_j$ , then  $\alpha_i B = \alpha_j B$ . Therefore,

$$D(A) = \det(\alpha_1 B, \dots, \alpha_n B) = 0.$$

This proves the claim.

We continue with the proof of the Theorem. Since we found that  $D$  is  $n$ -linear and alternating, by Corollary 5.15, we have that

$$D(A) = \det(A) \cdot D(I).$$

But  $D(I) = \det(I \cdot B) = \det(B)$ . Therefore,

$$D(A) = \det(A \cdot B) = \det(A) \cdot \det(B). \quad \square$$

### Corollary 5.18: Equivalent Condition for The Invertibility of a Matrix

*If  $A$  is invertible then  $\det(A) \neq 0$ . Moreover,*

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

## 5.2 More on Determinants

### Theorem 5.19: Determinant of a Transposed Matrix

$$\forall A \in M_{n \times n}(K) : \det(A^T) = \det(A).$$

### Corollary 5.20

*If  $D : M_{n \times n}(K) \rightarrow K$  is an alternating  $n$ -linear function (of the rows of a matrix) then  $D$  is also an alternating and  $n$ -linear function of the columns of a matrix. In fact,  $D(A^T) = D(A) \quad \forall A \in M_{n \times n}(K)$ . In particular, this holds also for  $D = \det$ .*

### Theorem 5.21: Determinant after Row/Column Operations

1. If  $B$  is obtained from  $A$  by adding a multiple of one row of  $A$  to another row (or by doing the analogous operation on columns), then  $\det(B) = \det(A)$ .
2. If  $A \xrightarrow{R_i \leftrightarrow R_j} B$ ,  $i \neq j$ , (or the analogous operation on columns) then  $\det(B) = -\det(A)$ .
3. If  $A \xrightarrow{c \cdot R_i \rightarrow R_i} B$ , then  $\det(B) = c \cdot \det(A)$ .

#### 5.2.1 Determinants of Block Matrices

Consider a block matrix  $M$  of the type

$$M = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where  $A \in M_{r \times r}(K)$ ,  $C \in M_{s \times s}(K)$ ,  $B \in M_{r \times s}(K)$ .

### Proposition 5.22: Determinant of a Block Matrix

Let  $M$  be as above. Then

$$\det(M) = \det(A) \cdot \det(C).$$

### Corollary 5.23: Entwicklungssatz with Cofactors

$\forall A \in M_{n \times n}(K)$  we have the following  $n$  different formulas for  $\det(A)$ , i.e.,

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} A_{ij} \det(A(i|j)), \quad j = 1, \dots, n.$$

The scalars  $c_{ij} := (-1)^{i+j} \det(A(i|j))$  are called the  $i, j$  **cofactor** of  $A$ . We have

$$\det(A) = \sum_{i=1}^n A_{ij} \cdot c_{ij}. \tag{5.4}$$

**Claim:** If we replace  $A_{ij}$  in Equation (5.4) by  $A_{ik}$ , where  $k$  is fixed, we always get 0. In other words,  $\forall A \in M_{n \times n}(K)$ ,  $\forall j$  we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \begin{cases} \det(A) & j = k \\ 0 & j \neq k \end{cases}.$$

**Proof of Claim:** Let  $B \in M_{n \times n}(K)$  be the matrix obtained from  $A$  by replacing column  $j$  of  $A$  by column  $k$  of  $A$ . So, in  $B$  column  $j$  equals column  $k$ . Clearly,  $\det(B) = 0$  since  $B$  has two equal columns ( $\iff B^T$  has two equal rows).

$$\implies 0 = \det(B) = \sum_{i=1}^n (-1)^{i+j} B_{ij} \det(B(i|j)) = \sum_{i=1}^n = \sum_{i=1}^n A_{ik} \det(A(i|j)) = \sum_{i=1}^n A_{ik} \cdot c_{ij}.$$

This proves the claim.

So,  $\forall A \in M_{n \times n}(K)$ ,  $\forall j, k$  we have

$$\sum_{i=1}^n A_{ik} \cdot c_{ij} = \delta_{jk} \cdot \det(A). \quad (5.5)$$

#### Definition 5.24: Classical Adjoint of $A$

The matrix  $(c_{ij})^T$  is called the **classical adjoint of  $A$** , i.e.,

$$\text{adj}(A) \in M_{n \times n}(K), \quad (\text{adj}(A))_{ij} = c_{ji} = (-1)^{i+j} \cdot \det(A(j|i)).$$

The Equation (5.5) says that

$$(\text{adj } A) \cdot A = \det(A) \cdot I.$$

Indeed,

$$\begin{aligned} \det(A) \cdot \delta_{jk} &= \sum_{i=1}^n c_{ij} \cdot A_{ik} = \sum_{i=1}^n (\text{adj}(A))_{ji} \cdot A_{ik} \\ \implies \det(A) \cdot I &= \text{adj}(A) \cdot A. \end{aligned}$$

#### Theorem 5.25: Formula for the inverse of $A$

Let  $A \in M_{n \times n}(K)$ . Then  $A$  is invertible if and only if  $\det(A) \neq 0$ . When  $A$  is invertible we have

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

### 5.3 Cramer's Rule

Let  $A \in M_{n \times n}(K)$ . We would like a formula to solve the system of equations  $Ax = b$ , where

$$x = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

are the unknowns and

$$b = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} \in K^n$$

is given. Assume  $Ax = b$ . Then

$$\begin{aligned} \text{adj } A \cdot Ax &= \text{adj } A \cdot b \\ \implies \det(A) \cdot x &= \text{adj } A \cdot b \\ \implies \forall 1 \leq j \leq n : \det(A) \cdot x_j &= \sum_{i=1}^n (\text{adj } A)_{ji} \cdot b_i = \sum_{i=1}^n (-1)^{i+j} b_i \det(A(i|j)), \end{aligned}$$

where the last expression is the determinant of the matrix obtained from  $A$  by replacing column  $j$  by  $b$ .

**Theorem 5.26: Cramer's Rule**

Let  $A \in M_{n \times n}(K)$  with  $\det(A) \neq 0$ . Let  $b \in K^n$ . Then the system  $Ax = b$  has a unique solution, and it is given by the formula

$$x_j = \frac{\det(A(j, b))}{\det(A)} \quad \forall 1 \leq j \leq n,$$

where  $A(j, b)$  is the matrix obtained from  $A$  by replacing column  $j$  by  $b$ .

**5.4 Determinants of Endomorphisms**

Let  $A \in M_{n \times n}(K)$  and  $P \in GL_n(K)$ , then

$$\det(P^{-1}AP) = \det(P^{-1}) \cdot \det(A) \cdot \det(P) = \frac{1}{\det(P)} \cdot \det(A) \cdot \det(P) = \det(A).$$

In other words, similar matrices have the same determinant.

Let  $V$  be a finite dimensional vector space over  $K$ . Put  $n := \dim V$ . Let  $T : V \rightarrow V$  be a linear map. Pick a basis  $\mathcal{B}$  for  $V$ . We have a matrix  $[T]_{\mathcal{B}}^{\mathcal{B}} \in M_{n \times n}(K)$ . Consider  $\det[T]_{\mathcal{B}}^{\mathcal{B}} \in K$ .

**Question:** Does  $\det[T]_{\mathcal{B}}^{\mathcal{B}}$  depend on the basis  $\mathcal{B}$ ?

**Answer:** Let  $\mathcal{B}'$  be another basis for  $V$ . Denote by  $P := [id_V]_{\mathcal{B}}^{\mathcal{B}'}$  the transition matrix between the basis  $\mathcal{B}'$  and  $\mathcal{B}$ . Recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'},$$

and we also have  $[id_V]_{\mathcal{B}'}^{\mathcal{B}} = ([id_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1}$ . So,

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = P^{-1} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot P. \quad \implies \quad \det[T]_{\mathcal{B}'}^{\mathcal{B}'} = \det[T]_{\mathcal{B}}^{\mathcal{B}}.$$

Recall also that if  $T, S : V \rightarrow V$  are two linear maps then  $[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}}$ . Recall also that if  $T : V \rightarrow V$  is an isomorphism then  $[T^{-1}]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^{-1}$ . Finally,  $[id_V]_{\mathcal{B}}^{\mathcal{B}} = I$ .

**Corollary 5.27: Well Definedness of the Determinant of an Endomorphism**

Let  $V$  be a finite dimensional vector space over  $K$ . Let  $T : V \rightarrow V$  be a linear map. Then  $\det[T]_{\mathcal{B}}^{\mathcal{B}}$  does **not** depend on the choice of basis  $\mathcal{B}$  of  $V$ . We denote this scalar by  $\det(T) \in K$ . The function

$$\det : \text{End}(V) \rightarrow K, \quad T \mapsto \det(T)$$

has the following properties:

1.  $\det(S \circ T) = \det(S) \cdot \det(T) \quad \forall S, T \in \text{End}(V)$ .
2.  $T$  is an isomorphism if and only if  $\det(T) \neq 0$ . If  $T$  is an isomorphism then

$$\det(T^{-1}) = \frac{1}{\det(T)}.$$

3.  $\det(id_V) = 1$ .
4.  $\det(0) = 0 \in K$ .

**Remark 5.28.** In contrast to  $T : V \rightarrow V$ , if we take  $T : V \rightarrow W$ , and  $\mathcal{B}, \mathcal{C}$  bases of  $V, W$ , then  $\det[T]_{\mathcal{C}}^{\mathcal{B}}$  **depends** on  $\mathcal{B}, \mathcal{C}$  (unless  $W = V$  and  $\mathcal{C} = \mathcal{B}$ ).

## 6 Eigenvalues & Eigenvectors

### Definition 6.1: Eigenvalues & Eigenvectors

Let  $V$  be a vector space over  $K$  and  $T : V \rightarrow V$  a linear map (an endomorphism of  $V$ ). We say that  $\lambda \in K$  is an **eigenvalue** of  $T$  if  $\exists v \in V \setminus \{0\}$  s.t.

$$Tv = \lambda v.$$

Any vector  $v \in V \setminus \{0\}$  with the property that  $Tv = \lambda v$  is called an **eigenvector** of  $T$  for the eigenvalue  $\lambda$ .

### Definition 6.2: Diagonalizability

Let  $V$  be a finite dimensional vector space over  $K$  and  $T : V \rightarrow V$  a linear map. We say that  $T$  is diagonalizable if there exists a basis  $\mathcal{B}$  of  $V$  s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}. \quad (6.1)$$

### Lemma 6.3: Equivalent Condition for Diagonalizability

*$T$  is diagonalizable if and only if there exists a basis  $\mathcal{B} = (v_1, \dots, v_n)$  consisting of eigenvectors of  $T$  (i.e.,  $\forall i, Tv_i = \lambda_i v_i$  for some  $\lambda_i \in K$ ). Moreover, if for some basis  $\mathcal{B} = (v_1, \dots, v_n)$  the matrix  $[T]_{\mathcal{B}}^{\mathcal{B}}$  has the shape as in Equation (6.1), then  $\forall i, \lambda_i$  is an eigenvalue of  $T$  and  $v_i$  is an eigenvector of  $\lambda_i$ .*

**Remark 6.4.** We saw that  $\forall A \in M_{n \times n}(K)$  we have a linear map  $T_A : K^n \rightarrow K^n$ ,  $T_A(v) := Av$ . We say that  $A \in M_{n \times n}(K)$  is diagonalizable if the linear map  $T_A$  is diagonalizable.

### Proposition 6.5: Linear Independence of Eigenvectors of Distinct Eigenvalues

*Let  $V$  be a vector space over  $K$  and  $T : V \rightarrow V$  a linear map. Suppose that  $\lambda_1, \dots, \lambda_n \in K$  are eigenvalues of  $T$  which are pairwise distinct (i.e.,  $\lambda_i \neq \lambda_j \ \forall i \neq j$ ). Let  $v_1, \dots, v_n$  be eigenvectors for  $\lambda_1, \dots, \lambda_n$  respectively. Then  $v_1, \dots, v_n$  are linearly independent.*

*Proof.* Induction on  $n$ .

$n = 1$ :  $v_1$  is an eigenvector for  $\lambda_1$ . By definition of  $v_1 \neq 0 \implies v_1$  is linearly independent.

**Induction step:** Let  $n \geq 1$  and suppose the proposition holds for any list of  $n$  eigenvalues and eigenvectors. We'll prove now that the proposition holds also for a list of  $n + 1$  eigenvalues and eigenvectors. Let  $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$  be pairwise distinct eigenvalues of  $T$ , and  $v_1, \dots, v_n, v_{n+1}$  a given list of eigenvectors of  $\lambda_1, \dots, \lambda_n, \lambda_{n+1}$ , respectively. We need to prove that  $v_1, \dots, v_n, v_{n+1}$  are linearly independent.

Assume that

$$a_1 v_1 + \dots + a_n v_n + a_{n+1} v_{n+1} = 0. \quad (6.2)$$

We need to show that  $a_1 = \dots = a_n = a_{n+1} = 0$ . Apply  $T$  to Equation (6.2). Since  $T$  is linear, we have that

$$a_1 T v_1 + \dots + a_n T v_n + a_{n+1} T v_{n+1} = 0.$$

But  $\forall i: T v_i = \lambda_i v_i$ . Therefore,

$$a_1 \lambda_1 v_1 + \dots + a_n \lambda_n v_n + a_{n+1} \lambda_{n+1} v_{n+1} = 0. \quad (6.3)$$

Consider now Equation (6.3) -  $\lambda_{n+1} \cdot (6.2)$ , i.e.,

$$a_1(\lambda_1 - \lambda_{n+1})v_1 + \dots + a_n(\lambda_n - \lambda_{n+1})v_n + a_{n+1} \underbrace{(\lambda_{n+1} - \lambda_{n+1})}_{=0} v_{n+1} = 0.$$

By the induction hypothesis  $v_1, \dots, v_n$  are linearly independent. Therefore,

$$a_1(\lambda_1 - \lambda_{n+1}) = \dots = a_n(\lambda_n - \lambda_{n+1}) = 0.$$

By assumption  $\lambda_i \neq \lambda_{n+1} \quad \forall 1 \leq i \leq n$ . So we have that  $a_1 = \dots = a_n = 0$ . Going back to Equation (6.2) we get that  $a_{n+1} \cdot v_{n+1} = 0$ . But by assumption we also have that  $v_{n+1} \neq 0$ . Therefore,  $a_{n+1} = 0$ .  $\square$

## 6.1 The Characteristic Polynomial

### Proposition 6.6: Equivalent Condition for Eigenvalue

*Let  $V$  be a vector space over  $K$  and  $T : V \rightarrow V$  a linear map. Then  $\lambda \in K$  is an eigenvalue of  $T$  if and only if the linear map  $T - \lambda \cdot \text{id}$  is **not** injective.*

### Definition 6.7: The Characteristic Polynomial

Let  $V$  be a finite dimensional vector space over  $K$  and  $T : V \rightarrow V$  a linear map. We define the **characteristic polynomial of  $T$**  as

$$p_T(x) := \det(T - x \cdot \text{id}_V).$$

We define the characteristic polynomial of a matrix  $A \in M_{n \times n}(K)$  in the same way, i.e.,

$$p_A(x) := \det(A - x \cdot I_n).$$

### Definition 6.8: Eigenspace

Let  $V$  be a vector space over  $K$  and  $T \in \text{End}(V)$ . Let  $\lambda \in K$  be an eigenvalue of  $T$ . The **eigenspace** corresponding to  $\lambda$  is defined as

$$\text{Eig}_T(\lambda) := \{v \in V \mid T v = \lambda v\} = \ker(T - \lambda \text{id}_V) = \{v \in V \mid v \text{ is an eigenvector of } \lambda\} \cup \{0\}.$$

Clearly, since  $\text{Eig}_T(\lambda)$  is the kernel of a linear map,  $\text{Eig}_T(\lambda) \subseteq V$  is a linear subspace of  $V$ .

### Lemma 6.9: Properties of the Characteristic Polynomial

Let  $A \in M_{n \times n}(K)$ . Then:

1.  $p_A(x)$  is a polynomial of degree  $n$ , with leading coefficient  $(-1)^n$ , i.e.,  $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$ , with  $c_i \in K \quad \forall i$ , and  $c_n = (-1)^n$ .
2.  $p_{A^T}(x) = p_A(x)$ .
3. If  $A$  has upper triangular form as in Definition 3.12, then

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x).$$

4. If

$$A = \begin{pmatrix} B_1 & * \\ 0 & B_2 \end{pmatrix}$$

is a block matrix, then

$$p_A(x) = p_{B_1}(x) \cdot p_{B_2}(x).$$

5. If  $A$  and  $B$  are similar matrices, i.e.,  $B = P^{-1}AP$  for some  $P \in GL_n(K)$ , then

$$p_A(x) = p_B(x).$$

**Remark 6.10.** If  $V$  is an  $n$ -dimensional vector space over  $K$  and  $T \in \text{End}(V)$ , then  $p_T(x)$  is a polynomial of degree  $n$  with leading coefficient  $(-1)^n$ .

### Definition 6.11: The Trace

Let  $A \in M_{n \times n}(K)$ . Define the **trace** (Spur) of  $A$  as

$$\text{Tr}(A) := \sum_{i=1}^n A_{ii} \in K,$$

i.e., the sum of the entries along the diagonal of  $A$ . If  $V$  is a finite dimensional vector space over  $K$  and  $T \in \text{End}(V)$ , we define the trace of  $T$  as

$$\text{Tr}(T) := \text{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}),$$

where  $\mathcal{B}$  is some basis for  $V$ . We'll see soon that  $\text{Tr}(T)$  is well defined, i.e., it does not depend on the choice of  $\mathcal{B}$ .



### Lemma 6.12: Properties of the Trace

1.

$$\forall A, B \in M_{n \times n}(K) : \operatorname{Tr}(A \cdot B) = \operatorname{Tr}(B \cdot A).$$

2. If  $A$  and  $B$  are similar matrices, then

$$\operatorname{Tr}(A) = \operatorname{Tr}(B).$$

In particular, if  $\mathcal{B}$  and  $\mathcal{B}'$  are two bases for  $V$ , then

$$\operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}) = \operatorname{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}).$$

Thus,  $\operatorname{Tr}(T)$  is well defined.

*Proof. 1):*

$$\begin{aligned} \operatorname{Tr}(A \cdot B) &= \sum_{i=1}^n (A \cdot B)_{ii} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} \cdot B_{ji} \\ &= \sum_{j=1}^n \sum_{i=1}^n A_{ij} \cdot B_{ji} = \sum_{j=1}^n \sum_{i=1}^n B_{ji} A_{ij} = \sum_{j=1}^n (B \cdot A)_{jj} \\ &= \operatorname{Tr}(B \cdot A). \end{aligned}$$

**2):** If  $B = P^{-1}AP$  for some  $P \in GL_n(K)$ , then by 1) we have that

$$\operatorname{Tr}(B) = \operatorname{Tr}(P^{-1}AP) = \operatorname{Tr}(P^{-1} \cdot (AP)) = \operatorname{Tr}((AP) \cdot P^{-1}) = \operatorname{Tr}(A).$$

Regarding the claim about  $\operatorname{Tr}(T)$ , recall that

$$[T]_{\mathcal{B}'}^{\mathcal{B}'} = [id_V]_{\mathcal{B}'}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [id_V]_{\mathcal{B}}^{\mathcal{B}'} \quad (6.4)$$

and  $[id_V]_{\mathcal{B}}^{\mathcal{B}'} = ([id_V]_{\mathcal{B}'}^{\mathcal{B}})^{-1}$ . So,  $[T]_{\mathcal{B}'}^{\mathcal{B}'}$  and  $[T]_{\mathcal{B}}^{\mathcal{B}}$  are similar matrices. Therefore,

$$\operatorname{Tr}([T]_{\mathcal{B}'}^{\mathcal{B}'}) = \operatorname{Tr}([T]_{\mathcal{B}}^{\mathcal{B}}).$$

□

### Lemma 6.13: Coefficients of the Characteristic Polynomial

Let  $A \in M_{n \times n}(K)$  and  $p_A(x) = c_n x^n + c_{n-1} x^{n-1} + \dots + c_1 x + c_0$  its characteristic polynomial. We've already seen that  $c_n = (-1)^n$ . We also have that

$$c_{n-1} = (-1)^{n-1} \cdot \operatorname{Tr}(A), \quad c_0 = \det(A).$$

In other words,

$$p_A(x) = (-1)^n x^n + (-1)^{n-1} \cdot \operatorname{Tr}(A) x^{n-1} + c_{n-2} x^{n-2} + \dots + c_1 x + \det(A).$$

**Definition 6.14: Direct Sum**

Let  $V$  be a vector space over  $K$  and  $U_1, \dots, U_r \subseteq V$  subspaces. Denote  $W := U_1 + \dots + U_r \subseteq V$ . We say that  $W$  is a **direct sum** of  $U_1, \dots, U_r$  (or that  $U_1, \dots, U_r$  is a direct sum) if every  $w \in W$  has a **unique** representation as  $w = u_1 + \dots + u_r$  with  $u_i \in U_i \quad \forall i$ .

**Proposition 6.15: Eigenspaces are a Direct Sum**

Let  $V$  be a vector space over  $K$  and  $T \in \text{End}(V)$ . Let  $\lambda_1, \dots, \lambda_r \in K$  be pairwise distinct eigenvalues of  $T$ . Then

$$\text{Eig}_T(\lambda_1) + \dots + \text{Eig}_T(\lambda_r)$$

is a direct sum.

**Corollary 6.16: Equivalent Condition for Diagonalizability of an Endomorphism**

Assume  $V$  is finite dimensional,  $T \in \text{End}(V)$  and let  $\lambda_1, \dots, \lambda_r \in K$  be all the eigenvalues of  $T$ . Assume additionally that  $\lambda_1, \dots, \lambda_r$  are pairwise distinct. Then  $T$  is diagonalizable if and only if

$$\sum_{i=1}^r \dim \text{Eig}_T(\lambda_i) = \dim V.$$

**Lemma 6.17: Decomposition of the Characteristic Polynomial**

Let  $A \in M_{n \times n}(K)$  (or  $T \in \text{End}(V)$ ) be diagonalizable. Then  $p_A(x)$  decomposes as a product of linear factors, i.e., degree 1

$$p_A(x) = (\lambda_1 - x) \cdot \dots \cdot (\lambda_n - x),$$

where  $\lambda_1, \dots, \lambda_n$  are the eigenvalues of  $A$  (but there might be repetitions among the  $\lambda_i$ 's).

**6.1.1 A Quick Digression about Polynomials**

Let  $0 \neq p(x) \in K[x]$ ,  $\lambda \in K$ . By dividing  $p(x)$  by  $(x - \lambda)$  we can write

$$p(x) = (x - \lambda) \cdot q(x) + R,$$

where  $\deg q(x) = \deg p(x) - 1$  and  $R \in K$ . But if we substitute  $x = \lambda$ , we obtain

$$p(\lambda) = 0 \cdot q(\lambda) + R = R \implies p(x) = (x - \lambda) \cdot q(x) + p(\lambda).$$

**Conclusion:**  $p(\lambda) = 0 \iff p(x) = (x - \lambda) \cdot q(x)$  for some  $q(x) \in K[x]$ . We say that  $\lambda$  is a zero of  $p(x)$  (Nullstelle) or a root of  $p(x)$ .

Sometimes  $\lambda$  is also a zero of  $q(x)$ , so  $q(x) = (x - \lambda) \cdot h(x)$ . Therefore,  $p(x) = (x - \lambda)^2 \cdot h(x)$ , etc.

**Definition 6.18: Multiplicity of Zeroes**

Let  $0 \neq p(x) \in K[x]$ ,  $\lambda \in K$ . We say that  $\lambda$  is a **zero of multiplicity**  $m \in \mathbb{N}_{>0}$  if

$$p(x) = (x - \lambda)^m \cdot g(x),$$

where  $g(x) \in K[x]$  and  $g(\lambda) \neq 0$ .

If  $m \geq 2$  we say  $\lambda$  is a **multiple zero** of  $p(x)$ .

If  $m = 1$  we say  $\lambda$  is a **simple zero** of  $p(x)$ .

**6.2 Geometric & Algebraic Multiplicities of Eigenvalues****Definition 6.19: Geometric & Algebraic Multiplicities**

Let  $V$  be a vector space over  $K$  and  $T \in \text{End}(V)$ . Let  $\lambda \in K$  be an eigenvalue of  $T$ . We define the **geometric multiplicity** of  $\lambda$  to be

$$m_g(T; \lambda) := \dim \text{Eig}_T(\lambda).$$

Assume now, in addition, that  $V$  is finite dimensional. We define the **algebraic multiplicity** of  $\lambda$  to be the multiplicity of the zero  $\lambda$  in the characteristic polynomial  $p_T(x)$  of  $T$ . We denote it by  $m_A(T; \lambda)$ .

**Proposition 6.20: Inequality Regarding Geometric & Algebraic Multiplicity**

$$m_g(T; \lambda) \leq m_A(T; \lambda).$$

*Proof.* Let  $v_1, \dots, v_k$  be a basis for  $\text{Eig}_T(\lambda)$ . We extend this basis to a basis  $\mathcal{B} = (v_1, \dots, v_k, w_{k+1}, \dots, w_n)$  of  $V$ . Then we have

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} | & & | & & | & & | \\ [Tv_1]_{\mathcal{B}} & \dots & [Tv_n]_{\mathcal{B}} & [Tw_{k+1}]_{\mathcal{B}} & \dots & [Tw_n]_{\mathcal{B}} \\ | & & | & & | & & | \end{pmatrix} = \begin{pmatrix} \lambda & \dots & 0 & * \\ \vdots & \ddots & \vdots & \vdots \\ 0 & \dots & \lambda & * \\ 0 & \dots & 0 & C \end{pmatrix},$$

where  $C \in M_{(n-k) \times (n-k)}(K)$ . Therefore,

$$p_T(x) = \det([T]_{\mathcal{B}}^{\mathcal{B}} - xI_n) = (\lambda - x)^k \cdot \det(C - xI_{n-k}) = (\lambda - x)^k \cdot p_C(x) = (-1)^k \cdot (x - \lambda)^k \cdot p_C(x).$$

This shows that the multiplicity of  $\lambda$  as a zero of  $p_T(x)$  is at least  $k$ . In other words,

$$m_A(T; \lambda) \geq k = m_g(T; \lambda).$$

□

### 6.3 Diagonalizability

#### Theorem 6.21: Equivalent Statements for Diagonalizability

Let  $V$  be an  $n$ -dimensional vector space over  $K$  and  $T \in \text{End}(V)$ . The following statements are equivalent:

1.  $T$  is diagonalizable.
2. There exists a basis for  $V$  whose elements are eigenvectors of  $T$ .
3. The characteristic polynomial  $p_T(x)$  splits into a product of linear factors, and for every eigenvalue of  $T$  the geometric and algebraic multiplicities are equal.

4.

$$\sum_{i=1}^k \dim \text{Eig}_T(\lambda_i) = \dim V,$$

where  $\lambda_1, \dots, \lambda_k \in K$  are the eigenvalues of  $T$ .

5.

$$V \cong \bigoplus_{i=1}^k \text{Eig}_T(\lambda_i).$$

#### Corollary 6.22

If  $p_T(x)$  splits as a product of linear factors and each eigenvalue  $\lambda$  of  $T$  has algebraic multiplicity of 1, then  $T$  is diagonalizable.

### 6.4 Trigonalization

#### Definition 6.23: Trigonalizability

We say that  $T \in \text{End}(V)$  is **trigonalizable** if there exists a basis  $\mathcal{B}$  for  $V$  s.t.  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is upper triangular, i.e.,

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \lambda_1 & \dots & * \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_n \end{pmatrix}.$$

#### Theorem 6.24: Equivalent Condition for Trigonalizability

Let  $V$  be a finite dimensional vector space over  $K$  and  $T \in \text{End}(V)$ . Then  $T$  is trigonalizable if and only if  $p_T(x)$  splits as a product of linear factors, i.e.,

$$p_T(x) = (\lambda_1 - x)^{l_1} \cdots (\lambda_k - x)^{l_k}.$$

**Theorem 6.25: Fundamental Theorem of Algebra**

Every polynomial  $p(x) \in \mathbb{C}[x]$  with complex coefficients, splits as a product of linear factors and a constant, i.e.,

$$p(x) = c \cdot (x - a_1)^{n_1} \cdot \dots \cdot (x - a_r)^{n_r},$$

with  $c \in \mathbb{C}$ ,  $r \geq 0$ ,  $a_i \in \mathbb{C}$ ,  $n_j \in \mathbb{N}_{>0}$ .

**Corollary 6.26: Trigonalizability over  $\mathbb{C}$** 

Let  $V$  be a finite dimensional vector space over  $\mathbb{C}$ . Then every  $T \in \text{End}(V)$  is trigonalizable.

**6.5 Minimal Polynomial & Cayley-Hamilton Theorem****Definition 6.27: Matrix Polynomial**

Let  $A \in M_{n \times n}(K)$ . Let  $g(x) = a_d x^d + \dots + a_1 x + a_0 \in K[x]$  be a polynomial. We can define  $g(A) \in M_{n \times n}(K)$ , as

$$g(A) := a_d A^d + \dots + a_1 A + a_0 I_n.$$

**Remark 6.28.**  $\forall A \in M_{n \times n}(K) \exists g \in K[x]$  s.t.  $g(A) = 0$ . Indeed, consider

$$I_n, A, A^2, \dots, A^r \in M_{n \times n}(K). \quad (6.5)$$

Since  $M_{n \times n}(K)$  is a vector space of dimension  $n^2$ , then if we take  $r > n^2$  then the elements in (6.5) cannot be linearly independent. Therefore,  $\exists a_0, a_1, \dots, a_r \in K$  s.t.

$$a_0 I_n + a_1 A + \dots + a_r A^r = 0.$$

A similar thing applies to  $T \in \text{End}(V)$ , where  $V$  is a finite dimensional vector space over  $K$ .  
 $(T^k = \underbrace{T \circ \dots \circ T}_{\times k}, g(T) := a_d T^d + \dots + a_1 T + a_0 \text{id}_V.)$

**Definition 6.29: Minimal Polynomial**

Let  $V$  be a vector space over  $K$  and  $T \in \text{End}(V)$ . A **minimal polynomial** for  $T$  is a polynomial  $g(x) \in K[x]$  of the minimal possible degree s.t.  $g(T) = 0$ .

**Theorem 6.30: Cayley-Hamilton**

1. Let  $A \in M_{n \times n}(K)$ . Then

$$p_A(A) = 0.$$

2. Let  $V$  be a finite dimensional vector space over  $K$ , and  $T \in \text{End}(V)$ . Then

$$p_T(T) = 0.$$

The proof of this Theorem may be important for the exam but will be skipped in this Document. The proof can be found in the lecture notes.

## 7 Euclidean & Hermitian (unitary) Spaces

At last we endow our spaces with a geometric structure. We work here with vector spaces  $V$  over  $K = \mathbb{R}$  or  $K = \mathbb{C}$ . Over  $\mathbb{R}$  we'll have Euclidean spaces. Over  $\mathbb{C}$  we'll have Hermitian (a.k.a unitary) spaces.

### Definition 7.1: Scalar Product

Let  $V$  be a vector space over  $\mathbb{R}$ . A **scalar product** (a.k.a inner product) is a function  $V \times V \rightarrow \mathbb{R}$ ,  $(v, w) \mapsto \langle v, w \rangle$  that has the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

2. Linearity in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{R}, v, w \in V.\end{aligned}$$

3. Symmetry:

$$\langle v, w \rangle = \langle w, v \rangle \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V \setminus \{0\}.$$

We call  $(V, \langle \cdot, \cdot \rangle)$  an **Euclidean space**, or a space endowed with a scalar product.

**Definition 7.2: Hermitian Product**

Let  $V$  be a vector space over  $\mathbb{C}$ . A **Hermitian product** (or inner product) on  $V$  is a function  $V \times V \rightarrow \mathbb{C}$ ,  $(v, w) \mapsto \langle v, w \rangle$  with the following properties:

1. Linearity in the first variable:

$$\begin{aligned}\langle v_1 + v_2, w \rangle &= \langle v_1, w \rangle + \langle v_2, w \rangle & \forall v_1, v_2, w \in V \\ \langle \alpha v, w \rangle &= \alpha \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

2. Sesquilinearity (complex anti-linearity) in the second variable:

$$\begin{aligned}\langle v, w_1 + w_2 \rangle &= \langle v, w_1 \rangle + \langle v, w_2 \rangle & \forall v, w_1, w_2 \in V \\ \langle v, \alpha w \rangle &= \bar{\alpha} \langle v, w \rangle & \forall \alpha \in \mathbb{C}, v, w \in V.\end{aligned}$$

3. Hermitian property:

$$\langle v, w \rangle = \overline{\langle w, v \rangle} \quad \forall v, w \in V.$$

4. Positivity:

$$\langle v, v \rangle > 0 \quad \forall v \in V.$$

We call  $(V, \langle \cdot, \cdot \rangle)$  a **Hermitian (or unitary) space**.

**Lemma 7.3: Properties of Euclidean/Hermitian Spaces**

Let  $V$  be an Euclidean/Hermitian vector space. Then:

1.  $\forall v \in V : \langle 0, v \rangle = \langle v, 0 \rangle = 0$ .
2. Let  $w \in V$ . If  $\langle v, w \rangle = 0 \quad \forall v \in V$ , then  $w = 0$ .
3. Let  $w_1, w_2 \in V$ . If  $\langle v, w_1 \rangle = \langle v, w_2 \rangle \quad \forall v \in V$ , then  $w_1 = w_2$ .

### Definition 7.4: Some Definitions to Euclidean/Hermitian Spaces

1. Let  $V$  be an Euclidean/Hermitian vector space.  $\forall v \in V$ , define  $\|v\| := \sqrt{\langle v, v \rangle} \in \mathbb{R}_{\geq 0}$ .  $\|\cdot\|$  is called the **norm** on  $V$  induced by  $\langle \cdot, \cdot \rangle$ .

2. A vector  $u \in V$  is called a **unit vector** if  $\|u\| = 1$ . Note that if  $0 \neq v \in V$  is any vector then

$$\frac{v}{\|v\|}$$

is a unit vector (which is positively proportional to  $v$ , so it points in the same direction as  $v$ ). We call  $\frac{v}{\|v\|}$  the normalization of  $v$ .

3. For  $v, w \in V$  we define the **distance** between  $v$  and  $w$  by

$$d(v, w) := \|v - w\|.$$

4. Two vectors  $u, v \in V$  are called **orthogonal** (or perpendicular one to the other) if  $\langle u, v \rangle = 0$ . In this case we sometimes write  $u \perp v$ .

5. A subset  $S \subseteq V$  is called an **orthogonal system** if  $0 \notin S$  and for every two distinct elements  $u, v \in S$  we have  $u \perp v$ .

6. An orthogonal system  $S \subseteq V$  is called **orthonormal** if  $\forall v \in S$  we have  $\|v\| = 1$ .

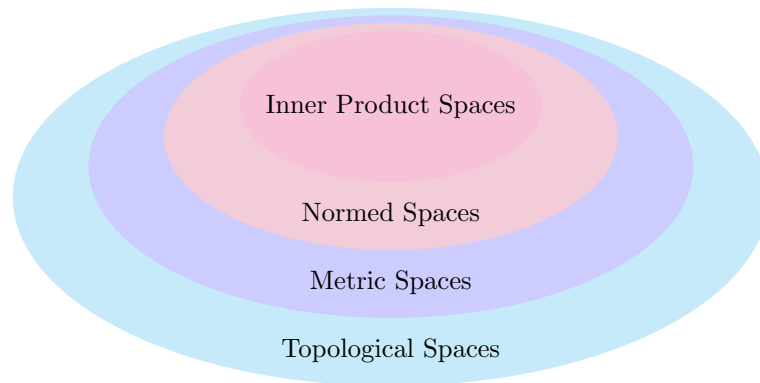


Figure 4: Overview of the generality of different spaces

### Theorem 7.5: Pythagoras Theorem

Let  $V$  be an Euclidean/Hermitian vector space and  $u, v \in V$  with  $u \perp v$ . Then

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$



**Lemma 7.6: Induced Norm**

Let  $V$  be an Euclidean space. Then

$$\forall u, v \in V : \langle u, v \rangle = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

This means that  $\|\cdot\|$  determines  $\langle \cdot, \cdot \rangle$ .

If  $V$  is an Hermitian space, we have that

$$\forall u, v \in V : \frac{1}{2}(\langle u, v \rangle + \overline{\langle u, v \rangle}) = \frac{1}{2}(\|u + v\|^2 - \|u\|^2 - \|v\|^2).$$

**Warning:** There exist norms that do **not** come from scalar products.

**Definition 7.7: Complex Conjugated Matrix/Adjoint Matrix/Hermitian Matrix**

1. Let  $A \in M_{n \times m}(\mathbb{C})$ . We write  $\overline{A}$  for the matrix whose entry at  $i, j$  is  $\overline{(A_{ij})}$ , i.e.,

$$(\overline{A})_{ij} := \overline{(A_{ij})}.$$

2. Let  $A \in M_{n \times n}(\mathbb{C})$ . Define

$$A^* := \overline{A}^T.$$

$A^*$  is called the **adjoint matrix** of  $A$ . (Has nothing to do with the classical adjoint  $\text{adj } A$ ).

3.  $A \in M_{n \times n}(\mathbb{C})$  is called **Hermitian** if  $A^* = A$  ( $\iff A^T = \overline{A}$ ).

**Important Example:** Let  $A \in M_{n \times n}(\mathbb{R})$ . Define

$$\langle \cdot, \cdot \rangle_A : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}, \quad \langle v, w \rangle_A = v^T A w.$$

We can quickly see that  $\langle \cdot, \cdot \rangle_A$  is bilinear. Assume now that  $A$  is symmetric. Then  $\langle v, w \rangle_A = \langle w, v \rangle_A \quad \forall v, w \in V$ . Indeed,

$$\langle w, v \rangle_A = w^T A v = w^T A^T v = (v^T A w)^T = (\langle v, w \rangle_A)^T = \langle v, w \rangle_A,$$

where we've used that a scalar can be viewed as a  $1 \times 1$  matrix.

Is  $\langle \cdot, \cdot \rangle_A$  in general a scalar product?

In general it is not, e.g., for

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

we have that

$$\left\langle \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\rangle_A = -1.$$

But for a scalar product we need that  $\langle v, v \rangle > 0 \quad \forall v \neq 0$ .

**Definition 7.8: Positive Definite**

A symmetric matrix  $A \in M_{n \times n}(\mathbb{R})$  is called **positive definite** if

$$v^T A v > 0 \quad \forall v \in \mathbb{R}^n \setminus \{0\}.$$

**Proposition 7.9: Scalar Product with Matrices**

Let  $A \in M_{n \times n}(\mathbb{R})$ . Then  $\langle \cdot, \cdot \rangle_A$  is a scalar product if and only if  $A$  is positive definite.

The situation for Hermitian products is similar.

**Definition 7.10: Positive Definite for Hermitian Matrices**

Let  $A \in M_{n \times n}(\mathbb{C})$  be a Hermitian matrix. We say that  $A$  is **positive definite** if

$$v^T A \bar{v} > 0 \quad \forall v \in \mathbb{C}^n \setminus \{0\}.$$

Note that  $v^T A \bar{v} \in \mathbb{R}$ , b.c.

$$\overline{(v^T A \bar{v})} = \bar{v}^T \bar{A} v = (\bar{v}^T \bar{A} v)^T = v^T \bar{A}^T \bar{v} = v^T A \bar{v}.$$

Given a positive definite complex matrix  $A$  we can define a Hermitian product on  $\mathbb{C}^n$ , i.e.,

$$\langle v, w \rangle_A = v^T A \bar{w}.$$

**Proposition 7.11: Hermitian Product with Matrices**

Let  $A \in M_{n \times n}(\mathbb{C})$ . Then  $\langle \cdot, \cdot \rangle_A$  is a Hermitian product if and only if  $A$  is positive definite.

**7.1 Norms & Angles****Definition 7.12: Norm**

Let  $V$  be a vector space over  $K$ , where  $K = \mathbb{R}$  or  $\mathbb{C}$ . A **norm** on  $V$  is a function  $\|\cdot\| : V \rightarrow \mathbb{R}$  s.t.:

1. Triangle inequality:

$$\|u + v\| \leq \|u\| + \|v\| \quad \forall u, v \in V.$$

2. Homogeneity:

$$\|\alpha v\| = |\alpha| \cdot \|v\| \quad \forall \alpha \in K, v \in V.$$

3. Non-degeneracy:

$$\|v\| = 0 \implies v = 0.$$

**Theorem 7.13: Cauchy-Schwarz Inequality**

Let  $(V, \langle \cdot, \cdot \rangle)$  be a vector space endowed with an inner product  $\langle \cdot, \cdot \rangle$ . Then

$$|\langle u, v \rangle| \leq \|u\| \cdot \|v\| \quad \forall u, v \in V,$$

where equality holds if and only if  $u$  and  $v$  are linearly dependent.

**Corollary 7.14: Induced Norm**

Let  $V$  be a vector space and  $\langle \cdot, \cdot \rangle$  an inner product on  $V$ , then

$$V \ni v \mapsto \|v\| := \sqrt{\langle v, v \rangle}$$

is a norm on  $V$ .

**Definition 7.15: Angle**

Let  $(V, \langle \cdot, \cdot \rangle)$  be an Euclidean space. Let  $u, v \in V$  be two non-zero vectors. By the Cauchy-Schwarz inequality 7.13, we have that

$$-1 \leq \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \leq 1.$$

We define the **angle** between  $u$  and  $v$  to be the unique  $\alpha \in [0, \pi]$  s.t.

$$\cos \alpha = \frac{\langle u, v \rangle}{\|u\| \cdot \|v\|} \left( = \left\langle \frac{u}{\|u\|}, \frac{v}{\|v\|} \right\rangle \right)$$

**7.2 The Gram-Schmidt Orthogonalization Process****Theorem 7.16: Properties of Orthogonal Systems**

Let  $(V, \langle \cdot, \cdot \rangle)$  be an inner product space.

1. If  $S \subseteq V$  is an orthogonal system then  $S$  is linearly independent.
2. If  $v_1, \dots, v_n$  is an orthogonal system and  $v \in \text{Sp}(v_1, \dots, v_n)$  and write  $v = a_1 v_1 + \dots + a_n v_n$ , for  $a_j \in K$ . Then the coefficients  $a_j$  are given by

$$a_j = \frac{\langle v, v_j \rangle}{\langle v_j, v_j \rangle} \quad \forall 1 \leq j \leq n.$$

**Corollary 7.17: Matrix for an Orthonormal Basis**

Let  $V, W$  be a vector space over  $K = \mathbb{R}$  or  $\mathbb{C}$  and let  $\langle \cdot, \cdot \rangle$  be an inner product on  $W$ . Let  $\mathcal{B} = (v_1, \dots, v_n)$  be a basis for  $V$  and  $\mathcal{C} = (w_1, \dots, w_m)$  be an orthonormal basis for  $W$ . Let  $T : V \rightarrow W$  be a linear map. Write  $[T]_{\mathcal{C}}^{\mathcal{B}} = (a_{ij})$ . Then

$$a_{ij} = \langle T v_j, w_i \rangle \quad \forall i, j.$$

(If  $\mathcal{C}$  is only a orthogonal basis then  $a_{ij} = \langle T v_j, w_i \rangle / \langle w_i, w_i \rangle$ ).

So having an ortho-gonal/normal basis for our spaces is useful!

**Theorem 7.18: Gram-Schmidt Orthogonalization Process**

Let  $(V, \langle \cdot, \cdot \rangle)$  be an inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$  and let  $n = \dim V$ . Let  $(v_1, \dots, v_n)$  be a basis for  $V$ . Define a new list of vectors  $w_1, \dots, w_n$  by induction, as follows

$$w_1 := v_1,$$

$$w_j := v_j - \sum_{i=1}^{j-1} \frac{\langle v_j, w_i \rangle}{\langle w_i, w_i \rangle} \cdot w_i \quad \text{for } j = 2, \dots, n.$$

Then:

1.  $(w_1, \dots, w_n)$  is an orthogonal basis for  $V$ .
2.  $(\frac{w_1}{\|w_1\|}, \dots, \frac{w_n}{\|w_n\|})$  is an orthonormal basis for  $V$ .
3.  $\forall 1 \leq j \leq n$  we have  $\text{Sp}(v_1, \dots, v_j) = \text{Sp}(w_1, \dots, w_j) = \text{Sp}(\frac{w_1}{\|w_1\|}, \dots, \frac{w_j}{\|w_j\|})$ .

**Corollary 7.19: Existence of an Orthonormal Basis for Finite Dimensional Spaces**

Every finite dimensional inner product space has an orthonormal basis.

**7.3 The Orthogonal Complement**

Let  $(V, \langle \cdot, \cdot \rangle)$  be an inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ .

**Definition 7.20: Orthogonal Complement**

Let  $\emptyset \neq S \subseteq V$  be a subset. We define the **orthogonal complement of  $S$**  as

$$S^\perp := \{v \in V \mid v \perp s \quad \forall s \in S\} = \{v \in V \mid \langle v, s \rangle = 0 \quad \forall s \in S\}.$$

When  $S = \{v\}$  we write  $v^\perp := \{v\}^\perp$ .

**Lemma 7.21: The Orthogonal Complement is a Subspace**

Let  $\emptyset \neq S \subseteq V$  be a subset. Then  $S^\perp$  is a linear subspace.

*Proof.*  $0 \in S^\perp$  b.c.  $\langle 0, s \rangle = 0$  for every  $s \in S$ . Furthermore, if  $\alpha, \beta \in K$  and  $v, w \in S^\perp$  then for every  $s \in S$  we have that

$$\langle \alpha v + \beta w, s \rangle = \underbrace{\alpha \langle v, s \rangle}_{=0} + \underbrace{\beta \langle w, s \rangle}_{=0} = 0.$$

Therefore,  $\alpha v + \beta w \in S^\perp$ . □

### Lemma 7.22: Further Properties of the Orthogonal Complement

Let  $\emptyset \neq S, T \subseteq V$ . Then:

1.  $0^\perp = V$ ,  $V^\perp = \{0\}$ .
2.  $S \cap S^\perp$  is either  $\emptyset$  or equal to  $\{0\}$ .
3. If  $S \subseteq T$  then  $T^\perp \subseteq S^\perp$ .
4.  $(\text{Sp}(S))^\perp = S^\perp$ .
5.  $S \subseteq (S^\perp)^\perp$ .

### Theorem 7.23: The Orthogonal Complement is a Complement

Let  $V$  be an inner product space (possibly of infinite dimension), and  $U \subseteq V$  a **finite dimensional** subspace. Then  $U^\perp$  is a complement of  $U$  in  $V$  (, i.e.,  $U + U^\perp = V$  and  $U \cap U^\perp = \{0\}$ ).

*Proof.* By Lemma 7.22 2., we have that  $U \cap U^\perp = \{0\}$ , since  $U \cap U^\perp \neq \emptyset$ . So it's enough to show that  $U + U^\perp = V$ .

Let  $r := \dim U$  and pick an orthonormal basis  $(e_1, \dots, e_r)$  for  $U$ , which we can do by the Gram-Schmidt Orthogonalization Process 7.18. Let  $v \in V$  be arbitrary. Define

$$P_U(v) := \langle v, e_1 \rangle e_1 + \dots + \langle v, e_r \rangle e_r \in U.$$

We have that

$$v = \underbrace{P_U(v)}_{\in U} + (v - P_U(v)). \quad (7.1)$$

We claim that  $v - P_U(v) \in U^\perp$ . Indeed,

$$\langle v - P_U(v), e_i \rangle = \langle v, e_i \rangle - \sum_{k=1}^r \langle v, e_k \rangle \cdot \langle e_k, e_i \rangle = \langle v, e_i \rangle - \langle v, e_i \rangle = 0.$$

This holds for every  $1 \leq i \leq r$ , hence

$$v - P_U(v) \perp \text{Sp}(e_1, \dots, e_r) = U,$$

i.e.,  $v - P_U(v) \in U^\perp$ . Going back to Equation (7.1) we see that

$$v = \underbrace{P_U(v)}_{\in U} + \underbrace{(v - P_U(v))}_{\in U^\perp}.$$

Since  $v \in V$  was arbitrary, we have that  $U + U^\perp = V$ . □

**Remark 7.24.** If  $U \subseteq V$  is a finite dimensional subspace, then since  $U^\perp$  is a complement of  $U$  in  $V$ , we have a canonical isomorphism  $V \cong U \oplus U^\perp$ .

**Definition 7.25: The Orthogonal Projection onto a Subspace**

Let  $V$  be an inner product space and  $U \subseteq V$  a finite dimensional subspace. We have seen that  $U^\perp$  is a complement of  $U$  in  $V$  ( $\implies \forall v \in V, \exists! u \in U, w \in U^\perp$  s.t.  $v = u + w$ ).

Define a map  $P_U : V \rightarrow U$  as follows:

$$\forall v \in V, \exists! u \in U, w \in U^\perp \text{ s.t. } v = u + w.$$

We define

$$P_U(v) := u.$$

$P_U$  is called **the orthogonal projection onto  $U$** .

**Proposition 7.26: Properties of  $P_U$** 

$P_U$  has the following properties:

1.  $P_U$  is a linear map.
2.  $\text{Im } P_U = U, \text{ ker } P_U = U^\perp$ .
3.  $\forall v \in V : v - P_U(v) \in U^\perp$ , i.e.,  $P_U + P_{U^\perp} = \text{id}_V$ .
4.  $\forall u \in U : P_U(u) = u$ .
5. If  $e_1, \dots, e_r$  is any orthonormal basis for  $U$ , then

$$P_U(v) = \sum_{i=1}^r \langle v, e_i \rangle e_i.$$

**Corollary 7.27: The Complement of the Complement of a Subspace**

Let  $V$  be an inner product space and  $U \subseteq V$  a finite dimensional subspace. Then

$$(U^\perp)^\perp = U.$$

**Corollary 7.28: Dimension of a Subspace and its Complement**

If  $V$  is a finite dimensional inner product space and  $U \subseteq V$  is a subspace, then

$$\dim U + \dim U^\perp = \dim V.$$

## 7.4 Orthogonal & Unitary Matrices

### Definition 7.29: Orthogonal & Unitary Matrices

1. A matrix  $A \in M_{n \times n}(\mathbb{R})$  is called **orthogonal** if its columns form an orthonormal basis in  $\mathbb{R}^n$ , w.r.t. the standard scalar product of  $\mathbb{R}^n$ . We denote the set of orthogonal  $n \times n$  matrices by  $O(n) \subseteq M_{n \times n}(\mathbb{R})$ .
2. A matrix  $A \in M_{n \times n}(\mathbb{C})$  is called **unitary** if its columns form an orthonormal basis in  $\mathbb{C}^n$ , w.r.t. the standard hermitian product on  $\mathbb{C}^n$ . We denote the set of all unitary  $n \times n$  matrices by  $U(n) \subseteq M_{n \times n}(\mathbb{C})$ .

### Lemma 7.30: Equivalent Conditions for Orthogonal/Unitary Matrices

- $A \in M_{n \times n}(\mathbb{R})$  is orthogonal  $\iff A^T A = I_n \iff A A^T = I_n \iff A^{-1} = A^T$ .
- $A \in M_{n \times n}(\mathbb{C})$  is unitary  $\iff A^* A = I_n \iff A A^* = I_n \iff A^{-1} = A^*$ .

### Corollary 7.31

1. Let  $A \in M_{n \times n}(\mathbb{R})$ . Then  $A \in O(n) \iff$  the rows of  $A$  form an orthonormal basis for  $\mathbb{R}^n$ .
2. Let  $A \in M_{n \times n}(\mathbb{C})$ . Then  $A \in U(n) \iff$  the rows of  $A$  form an orthonormal basis for  $\mathbb{C}^n$ .

## 7.5 QR-Decomposition

### Theorem 7.32: QR-Decomposition

1. Let  $A \in GL_n(\mathbb{R})$ . Then  $\exists Q \in O(n)$  and an upper triangular matrix  $R \in M_{n \times n}(\mathbb{R})$  s.t.

$$A = QR.$$

2. Let  $A \in GL_n(\mathbb{C})$ . Then  $\exists Q \in U(n)$  and an upper triangular matrix  $R \in M_{n \times n}(\mathbb{C})$  s.t.

$$A = QR.$$

*Proof.* We'll prove 1 & 2 together.

Write

$$A = \begin{pmatrix} | & & | \\ v_1 & \dots & v_n \\ | & & | \end{pmatrix}$$

with  $v_k \in K_{\text{col}}^n$ , where  $K = \mathbb{R}$  or  $\mathbb{C}$ . Since  $A \in GL_n(K)$ ,  $v_1, \dots, v_n$  form a basis for  $K_{\text{col}}^n$ . Apply the Gram-Schmidt orthogonalization process 7.18 to  $v_1, \dots, v_n$  and obtain an orthonormal basis  $e_1, \dots, e_n$  for  $K_{\text{col}}^n$ . Recall that  $\forall 1 \leq j \leq n : \text{Sp}(v_1, \dots, v_j) = \text{Sp}(e_1, \dots, e_j)$ . By Theorem 7.16 we also have that

$$v_j = \sum_{i=1}^j \langle v_j, e_i \rangle e_i \quad \forall 1 \leq j \leq n.$$

Therefore, we can write

$$\underbrace{\begin{pmatrix} | & & | & & | \\ v_1 & \dots & v_j & \dots & v_n \\ | & & | & & | \end{pmatrix}}_{=A} = \underbrace{\begin{pmatrix} | & & | & & | \\ e_1 & \dots & e_j & \dots & e_n \\ | & & | & & | \end{pmatrix}}_{=Q} \cdot \underbrace{\begin{pmatrix} \langle v_1, e_1 \rangle & \langle v_2, e_1 \rangle & \dots & \langle v_j, e_1 \rangle & \dots & \langle v_n, e_1 \rangle \\ 0 & \langle v_2, e_2 \rangle & \dots & \langle v_j, e_2 \rangle & \dots & \langle v_n, e_2 \rangle \\ \vdots & \ddots & \ddots & \vdots & & \vdots \\ \vdots & & \ddots & \langle v_j, e_j \rangle & \dots & \langle v_n, e_j \rangle \\ \vdots & & & \ddots & \ddots & \vdots \\ 0 & \dots & \dots & \dots & 0 & \langle v_n, e_n \rangle \end{pmatrix}}_{=R},$$

as desired.  $\square$

Extension of QR-Decomposition for all matrices  $A \in M_{n \times n}(K)$ .

### Theorem 7.33: Extension of QR-Decomposition

Let  $A \in M_{n \times n}(K)$ , where  $K = \mathbb{R}$  or  $\mathbb{C}$  and let  $r := \text{rank}(A)$ . Then  $\exists Q \in O(n)/U(n)$  and

$$R = \begin{pmatrix} C & * \\ 0 & 0 \end{pmatrix} \in M_{n \times n}(K),$$

where  $C \in M_{r \times r}(K)$  is upper triangular, s.t.

$$A = QR.$$

## 8 Dual Spaces & Inner Products

Let  $V$  be a vector space over  $K$  (any field). Recall that we have the dual space  $V^* := \text{Hom}(V, K) =$  the space of linear maps  $\varphi : V \rightarrow K$  (functionals).  $V^*$  is also a vector space over  $K$ .

Consider now an inner product space  $(V, \langle \cdot, \cdot \rangle)$  over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Let  $u \in V$ . Define

$$\varphi_u : V \rightarrow K, \quad V \ni v \mapsto \langle v, u \rangle \in K.$$

$\varphi_u$  is linear since,  $v$  is the first argument in the scalar/hermitian product.

### Theorem 8.1: Canonical 'Almost' Isomorphism between $V$ and $V^*$

Let  $(V, \langle \cdot, \cdot \rangle)$  be a finite dimensional inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Let  $\varphi \in V^*$ . Then there exists a **unique**  $u \in V$  s.t.  $\varphi = \varphi_u$ , i.e.,  $\varphi(v) = \langle v, u \rangle \quad \forall v \in V$ .

In fact, for  $K = \mathbb{R}$  the map  $\Phi : V \rightarrow V^*$ , given by  $\Phi(u) := \varphi_u$  is an isomorphism.

For  $K = \mathbb{C}$ , this map is injective and surjective but only  $\mathbb{C}$ -anti linear, i.e.,

$$\begin{aligned} \Phi(u_1 + u_2) &= \Phi(u_1) + \Phi(u_2) & \forall u_1, u_2 \in V \\ \Phi(cu) &= \bar{c} \cdot \Phi(u) & \forall c \in \mathbb{C}, u \in V. \end{aligned}$$

*Proof.* Let  $n := \dim V$  and fix an orthonormal basis  $e_1, \dots, e_n$  for  $V$ , which exists by the Gram-



Schmidt orthogonalization process 7.18. Let  $\varphi \in V^*$ . Let  $v \in V$ . By Theorem 7.16 we have that

$$v = \sum_{i=1}^n \langle v, e_i \rangle e_i.$$

Therefore,

$$\varphi(v) = \varphi\left(\sum_{i=1}^n \langle v, e_i \rangle e_i\right) = \sum_{i=1}^n \langle v, e_i \rangle \varphi(e_i) = \langle v, \sum_{i=1}^n \overline{\varphi(e_i)} e_i \rangle.$$

Define

$$u := \sum_{i=1}^n \overline{\varphi(e_i)} e_i.$$

Then we have that  $\varphi(v) = \varphi_u(v) \quad \forall v \in V$ .

We'll show now the uniqueness of  $u$ . Suppose that  $\varphi_{u_1} = \varphi_{u_2}$  for some  $u_1, u_2 \in V$ . Then we have that

$$\langle v, u_1 \rangle = \langle v, u_2 \rangle \quad \forall v \in V.$$

By a previous result, this implies that  $u_1 = u_2$ . This shows that  $\Phi$  is surjective and injective.

For  $K = \mathbb{R}$ , we have that the scalar product  $\langle \cdot, \cdot \rangle$  is bilinear. So it follows that  $\Phi$  is also linear, hence  $\Phi$  is an isomorphism.

For  $K = \mathbb{C}$ , we have that the hermitian product  $\langle \cdot, \cdot \rangle$  is  $\mathbb{C}$ -anti linear in the second variable. So, it follows that  $\Phi$  is also  $\mathbb{C}$ -anti linear.  $\square$

So, at least for  $K = \mathbb{R}$ , a scalar product on  $V$  gives us a canonical isomorphism  $V \cong V^*$ . (But this isomorphism depends on the scalar product!)

## 8.1 The Adjoint Map

(Everything in this Subsection should be learned by heart.) Let  $V, W$  be vector spaces over  $K$ . Let  $T : V \rightarrow W$  be a linear map. We have seen in Subsection 4.2 that  $T$  induces a linear map (the dual map)

$$T^* : W^* \rightarrow V^*, \quad T^*(\varphi) := \varphi \circ T.$$

Assume now that  $(V, \langle \cdot, \cdot \rangle_V)$  and  $(W, \langle \cdot, \cdot \rangle_W)$  are inner product spaces over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Assume also that  $\dim V < \infty$ . Given  $T$  as above, we can define  $T' : W \rightarrow V$  using our 'canonical' identifications  $W \cong W^*, V \cong V^*$ . (No problem for  $K = \mathbb{R}$ , but we have to be careful with  $K = \mathbb{C}$ .)

$$\begin{array}{ccc} W^* & \xrightarrow{T^*} & V^* \\ \uparrow \Phi_W & & \downarrow \Phi_V^{-1} \\ W & \xrightarrow{\quad T' \quad} & V \end{array}$$

i.e.  $T' := \Phi_V^{-1} \circ T^* \circ \Phi_W$ . We've also used that  $\dim V < \infty$ , b.c. then  $\Phi_V$  is bijective.

What properties does  $T'$  have? Note that  $T'$  is uniquely determined by

$$\begin{aligned}
 \Phi_V \circ T' &= T^* \circ \Phi_W && \iff \forall w \in W : \Phi_V \circ T'(w) = T^* \circ \Phi_W(w) \\
 &&& \iff \varphi_{T'(w)} = T^* \circ \varphi_w \quad (\in V^*) \\
 &&& \iff \forall v \in V : \varphi_{T'(w)}(v) = (T^* \circ \varphi_w)(v) \\
 &&& \iff \langle v, T'(w) \rangle_V = \varphi_w(T(v)) = \langle T(v), w \rangle_W.
 \end{aligned}$$

In other words,

$$\langle Tv, w \rangle_W = \langle v, T'w \rangle_V \quad \forall v \in V, w \in W.$$

**Claim:**  $T'$  is a linear map.

*Proof.* For  $K = \mathbb{R}$ ,  $\Phi_W, \Phi_V$  and  $T^*$  are linear. Therefore,  $T'$  is linear too.

For  $K = \mathbb{C}$ ,  $\Phi_W, \Phi_V$  are additive but only  $\mathbb{C}$ -anti linear. Therefore,  $\Phi_V^{-1}$  is also additive and  $\mathbb{C}$ -anti linear. Indeed, let  $y_1, y_2 \in V^*$ . Since  $\Phi_V$  is bijective, we can find unique  $x_1, x_2 \in V$  s.t.  $\Phi_V(x_1) = y_1$  and  $\Phi_V(x_2) = y_2$ . Since  $\Phi_V$  is additive we have that

$$\Phi_V^{-1}(y_1 + y_2) = \Phi_V^{-1}(\Phi_V(x_1) + \Phi_V(x_2)) = \Phi_V^{-1}(\Phi_V(x_1 + x_2)) = x_1 + x_2 = \Phi_V^{-1}(y_1) + \Phi_V^{-1}(y_2).$$

Similarly, let  $c \in \mathbb{C}, y \in V^*$ . We again can find a unique  $x \in V$  s.t.  $\Phi_V(x) = y$ . Since  $\Phi_V$  is  $\mathbb{C}$ -anti linear we have that

$$\Phi_V^{-1}(cy) = \Phi_V^{-1}(c \cdot \Phi_V(x)) = \Phi_V^{-1}(\Phi_V(\bar{c}x)) = \bar{c} \cdot x = \bar{c} \cdot \Phi_V^{-1}(y).$$

So, we have shown that  $\Phi_V^{-1}$  is additive and  $\mathbb{C}$ -anti linear as well.

Now, we return to the proof of the claim. We recall that  $T^*$  is linear. Let  $w_1, w_2 \in W$  be arbitrary. Since  $T^*$  is in particular additive and  $T' = \Phi_V^{-1} \circ T^* \circ \Phi_W$ , we can conclude that  $T'$  is also additive. Now, let  $c \in \mathbb{C}, w \in W$ . Since  $T^*$  is linear and  $\Phi_V^{-1}, \Phi_W$  are  $\mathbb{C}$ -anti linear we have that

$$\begin{aligned}
 T'(cw) &= \Phi_V^{-1} \circ T^* \circ \Phi_W(cw) = \Phi_V^{-1}(T^*(\bar{c} \cdot \Phi_W(w))) = \Phi_V^{-1}(\bar{c} \cdot T^*(\Phi_W(w))) \\
 &= c \cdot \Phi_V^{-1}(T^*(\Phi_W(w))) = c \cdot T'(w).
 \end{aligned}$$

This proves the claim. □

**Notation:** The linear map  $T' : W \rightarrow V$  is denoted many times by  $T^* : W \rightarrow V$ . This makes sense b.c. of the 'canonical' identifications. But we have to assume additionally that  $\dim W < \infty$ . The map  $T^* : W \rightarrow V$  is called the **adjoint** of  $T$ .

## 8.2 Basic Properties of Adjoint Maps

### Lemma 8.2: Properties of Adjoint Maps

Let  $V, W, U$  be finite dimensional inner product spaces over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Let  $S, T : V \rightarrow W$  and  $R : W \rightarrow U$  be linear maps. Then:

1.  $T^*$  is a linear map.
2.  $(S + T)^* = S^* + T^*$ .
3.  $\forall \lambda \in K : (\lambda T)^* = \bar{\lambda} T^*$ .
4.  $(T^*)^* = T$ .
5.  $(id_V)^* = id_V$ .
6.  $(R \circ T)^* = T^* \circ R^*$ .

*Proof.* Let  $y_1, y_2 \in V$ . We'll repeatedly use that if  $\forall x \in V : \langle x, y_1 \rangle = \langle x, y_2 \rangle$ , then  $y_1 = y_2$ . This is because

$$\begin{aligned} \forall x \in V : \langle x, y_1 \rangle &= \langle x, y_2 \rangle &\implies & \forall x \in V : \langle x, y_1 - y_2 \rangle = 0 \\ &\implies y_1 - y_2 = 0 &\implies & y_1 = y_2, \end{aligned}$$

where we've used 2. of Lemma 7.3

- 1) Has already been proven in Subsection 8.1.
- 2) Let  $w \in W, v \in V$ . By the defining properties of  $T^*, S^*$  we have that

$$\langle v, (S + T)^* w \rangle = \langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^* w \rangle + \langle v, T^* w \rangle = \langle v, (S^* + T^*) w \rangle.$$

This holds for every  $v \in V$ . Therefore, we have that

$$(S + T)^* w = (S^* + T^*) w.$$

Since, this holds for every  $w \in W$ , we conclude that  $(S + T)^* = (S^* + T^*)$ .

- 3) Let  $\lambda \in K, v \in V$  and  $w \in W$ . Then we have that

$$\begin{aligned} \langle v, (\lambda T)^* w \rangle &= \langle (\lambda T)v, w \rangle = \lambda \langle Tv, w \rangle = \lambda \langle v, T^* w \rangle \\ &= \langle v, \bar{\lambda} T^* w \rangle. \end{aligned}$$

Since this holds for every  $v \in V$ , we get that

$$(\lambda T)^* w = \bar{\lambda} T^* w.$$

Since this holds for every  $w \in W$ , we conclude that  $(\lambda T)^* = \bar{\lambda} T^*$ .

- 4) Let  $v \in V, w \in W$ . Then we have that

$$\langle w, (T^*)^* v \rangle = \langle T^* w, v \rangle = \overline{\langle v, T^* w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle.$$

This holds for every  $w \in W$ . Therefore, we have that  $(T^*)^* v = Tv$ . And since this holds for every  $v \in V$ , we get that  $(T^*)^* = T$ .

5) Let  $v \in V$ . Then we have that

$$\langle v, (id_V)^* v \rangle = \langle id_V v, v \rangle = \langle v, v \rangle = \langle v, id_V v \rangle.$$

Since this holds for every  $v \in V$  we conclude that

$$(id_V)^* v = id_V v \quad \implies \quad (id_V)^* = id_V.$$

6) Let  $v \in V, u \in U$ . Then we have that

$$\begin{aligned} \langle v, (R \circ T)^* u \rangle &= \langle (R \circ T)v, u \rangle = \langle R(T(v)), u \rangle = \langle Tv, R^* u \rangle \\ &= \langle v, T^*(R^*(u)) \rangle = \langle v, T^* \circ R^* u \rangle. \end{aligned}$$

This holds for every  $v \in V$ . Therefore,

$$(R \circ T)^* u = T^* \circ R^* u.$$

Since this holds for every  $u \in U$ , we conclude that  $(R \circ T)^* = T^* \circ R^*$ .  $\square$

#### Lemma 8.3: Kernel & Image of Adjoint Maps

Let  $V, W$  be finite dimensional inner product spaces. Let  $T : V \rightarrow W$  and  $T^* : W \rightarrow V$  be the adjoint of  $T$ . Then

1.  $\ker T^* = (\operatorname{Im} T)^\perp$ .
2.  $\operatorname{Im} T^* = (\ker T)^\perp$ .
3.  $\ker T = (\operatorname{Im} T^*)^\perp$ .
4.  $\operatorname{Im} T = (\ker T^*)^\perp$ .

### 8.3 Matrix Representation of Adjoint Maps

#### Proposition 8.4: Matrix Representation of Adjoint Maps

Let  $V, W$  be finite dimensional inner product spaces and  $T : V \rightarrow W$  a linear map. Let  $\mathcal{B} = (e_1, \dots, e_n)$  be an orthonormal basis for  $V$  and  $\mathcal{C} = (f_1, \dots, f_m)$  an orthonormal basis for  $W$ . Then

$$[T^*]_{\mathcal{B}}^{\mathcal{C}} = \left( \overline{[T]_{\mathcal{C}}^{\mathcal{B}}} \right)^T = \left( [T]_{\mathcal{C}}^{\mathcal{B}} \right)^*,$$

where the last asterisk denotes the adjoint matrix.

#### Corollary 8.5

Let  $A \in M_{m \times n}(K)$ , where  $K = \mathbb{R}$  or  $\mathbb{C}$  and let  $T_A : K^n \rightarrow K^m$  be the linear map  $T_A(v) = Av$ . Endow  $K^n$  and  $K^m$  with their standard inner product. Then

$$(T_A)^* = T_{A^*}.$$

### Proposition 8.6: Properties of the Adjoint Matrix

Let  $K = \mathbb{R}$  or  $\mathbb{C}$ ,  $A, B \in M_{n \times m}(K)$ ,  $C \in M_{m \times p}(K)$ . Then:

1.  $(A + B)^* = A^* + B^*$ .
2.  $(\lambda A)^* = \bar{\lambda} A^* \quad \forall \lambda \in K$ .
3.  $(A^*)^* = A$ .
4.  $I_n^* = I_n$ .
5.  $(AC)^* = C^* A^*$ .

## 9 Spectral Theorems

### Definition 9.1: Orthogonal Diagonalizability

Let  $V$  be an inner product space and  $T : V \rightarrow V$  an endomorphism. We say that  $T$  is **orthogonally diagonalizable** if there exists an orthonormal basis for  $V$  consisting of eigenvectors of  $T$ .

### Lemma 9.2: Consequence of Orthogonal Diagonalizability

If  $T$  is orthogonally diagonalizable, then

$$T \circ T^* = T^* \circ T.$$

### Definition 9.3: Normal Endomorphism/Matrix

Let  $V$  be an inner product space. An endomorphism  $T : V \rightarrow V$  is called **normal** if

$$T \circ T^* = T^* \circ T.$$

A matrix  $A \in M_{n \times n}(K)$  ( $K = \mathbb{R}, \mathbb{C}$ ) is called **normal** if

$$AA^* = A^*A.$$

This is equivalent to saying that  $T_A : K^n \rightarrow K^n$  is normal, if we endow  $K^n$  with its standard inner product.

We've seen in Lemma 9.2 that if  $T : V \rightarrow V$  is orthogonally diagonalizable, then  $T$  is normal. Is the converse statement also true?

Over  $K = \mathbb{R}$  the answer is **no!**, e.g.,

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

$T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is a clockwise rotation by  $90^\circ$ , so there exist no eigenvalues in  $\mathbb{R}$ . But  $AA^* = A^*A = I$ .

### Lemma 9.4: Properties of a Normal Endomorphism

Let  $V$  be a finite dimensional inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Let  $T : V \rightarrow V$  be a normal endomorphism. Then:

1.  $\forall v \in V : \|Tv\| = \|T^*v\|$ .
2.  $\forall \lambda \in K$  the endomorphism  $T - \lambda \cdot id_V$  is also normal.
3. If  $v$  is an eigenvector of  $T$  with eigenvalue  $\lambda$ , then  $v$  is also an eigenvector of  $T^*$  with eigenvalue  $\bar{\lambda}$ .
4. Let  $v_1, v_2$  be eigenvectors of  $T$  with eigenvalues  $\lambda_1, \lambda_2$  respectively. If  $\lambda_1 \neq \lambda_2$ , then  $\langle v_1, v_2 \rangle = 0$ .

*Proof.* **1)** Let  $v \in V$ . Then by the properties of the adjoint map  $T^*$  and since  $T$  is normal, we have that

$$\begin{aligned} \|Tv\|^2 &= \langle Tv, Tv \rangle = \langle v, T^* \circ Tv \rangle = \langle v, T \circ T^*v \rangle \\ &= \overline{\langle T \circ T^*v, v \rangle} = \overline{\langle T^*v, T^*v \rangle} = \langle T^*v, T^*v \rangle = \|T^*v\|^2. \end{aligned}$$

Since the norm is non-negative, we conclude that

$$\|Tv\| = \|T^*v\|.$$

**2)** Let  $\lambda \in K$ . By properties of adjoint maps we have that

$$(T - \lambda \cdot id_V) \circ (T - \lambda \cdot id_V)^* = (T - \lambda \cdot id_V) \circ (T^* - \bar{\lambda} \cdot id_V) = T \circ T^* - \bar{\lambda} \cdot T - \lambda \cdot T^* + \lambda \bar{\lambda} \cdot id_V, \quad (9.1)$$

and

$$(T - \lambda \cdot id_V)^* \circ (T - \lambda \cdot id_V) = (T^* - \bar{\lambda} \cdot id_V) \circ (T - \lambda \cdot id_V) = T^* \circ T - \lambda \cdot T^* - \bar{\lambda} \cdot T + \bar{\lambda} \lambda \cdot id_V. \quad (9.2)$$

By the normality of  $T$  we get that Equation (9.1) = (9.2), so  $(T - \lambda \cdot id_V)$  is normal.

**3)** Let  $v$  be an eigenvector of  $T$  with eigenvalue  $\lambda$ . Then  $(T - \lambda \cdot id_V)v = 0$ . By **2)**,  $(T - \lambda \cdot id_V)$  is normal. Hence by **1)** we have

$$\begin{aligned} \|(T^* - \bar{\lambda} \cdot id_V)v\| &= \|(T - \lambda \cdot id_V)^*v\| = \|(T - \lambda \cdot id_V)v\| = 0 \\ \implies (T^* - \bar{\lambda} \cdot id_V)v &= 0. \end{aligned}$$

This proves that  $v$  is an eigenvector of  $T^*$  with eigenvalue  $\bar{\lambda}$ .

**4)** Let  $v_1, v_2$  be eigenvectors of  $T$  with eigenvalues  $\lambda_1, \lambda_2$  respectively, where  $\lambda_1 \neq \lambda_2$ . Then, by properties of adjoint maps and **3)**, we have that

$$\begin{aligned} \lambda_1 \langle v_1, v_2 \rangle &= \langle \lambda_1 v_1, v_2 \rangle = \langle Tv_1, v_2 \rangle = \langle v_1, T^*v_2 \rangle = \langle v_1, \bar{\lambda}_2 v_2 \rangle = \lambda_2 \langle v_1, v_2 \rangle \\ \implies (\lambda_1 - \lambda_2) \langle v_1, v_2 \rangle &= 0. \end{aligned}$$

Since  $\lambda_1 \neq \lambda_2$ , we conclude that  $\langle v_1, v_2 \rangle = 0$ . □

### Theorem 9.5: Spectral Theorem over $\mathbb{C}$

*Let  $V$  be a finite dimensional inner product space over  $\mathbb{C}$  and  $T : V \rightarrow V$  a linear map. Then  $T$  is orthogonally diagonalizable if and only if  $T$  is normal.*

*Proof.  $\Rightarrow$ :* Let  $\mathcal{B}$  be an orthonormal basis for  $V$  s.t. all the elements in  $\mathcal{B}$  are eigenvectors of  $T$ . Therefore,  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is a diagonal matrix. By Proposition 8.4 we have that

$$[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*.$$

So,  $[T^*]_{\mathcal{B}}^{\mathcal{B}}$  is also diagonal. Since diagonal matrices commute and by a property of matrix multiplication, we have that

$$\begin{aligned} [T]_{\mathcal{B}}^{\mathcal{B}} \cdot [T^*]_{\mathcal{B}}^{\mathcal{B}} &= [T^*]_{\mathcal{B}}^{\mathcal{B}} \cdot [T]_{\mathcal{B}}^{\mathcal{B}} \\ \Rightarrow [T \circ T^*]_{\mathcal{B}}^{\mathcal{B}} &= [T^* \circ T]_{\mathcal{B}}^{\mathcal{B}}. \end{aligned}$$

By a result from the lecture, we conclude that  $T \circ T^* = T^* \circ T$ .

*$\Leftarrow$ :* By the fundamental Theorem of Algebra 6.26,  $T$  is trigonalizable, i.e., there exists a basis  $\mathcal{C} = (u_1, \dots, u_n)$  for  $V$  s.t.  $[T]_{\mathcal{C}}^{\mathcal{C}}$  is an upper triangular matrix. By applying the Gram-Schmidt orthogonalization process 7.18 to the basis  $\mathcal{C}$  we obtain a new basis  $\mathcal{B} = (v_1, \dots, v_n)$  which is orthonormal and s.t.  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is still upper triangular, b.c.  $\text{Sp}(u_1, \dots, u_j) = \text{Sp}(v_1, \dots, v_j)$  for every  $1 \leq j \leq n$ .

We claim that  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is actually diagonal. To see this write  $[T]_{\mathcal{B}}^{\mathcal{B}} = (a_{ij})$ . Since  $\mathcal{B}$  is orthonormal, we have  $\|Tv_1\|^2 = |a_{11}|^2$ . Indeed,

$$\langle Tv_1, Tv_1 \rangle = \langle a_{11}v_1, a_{11}v_1 \rangle = a_{11} \overline{a_{11}} \underbrace{\langle v_1, v_1 \rangle}_{=1} = |a_{11}|^2.$$

Since  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*$  we also have

$$\begin{aligned} [T^*]_{\mathcal{B}}^{\mathcal{B}} &= \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ \bar{a}_{12} & \bar{a}_{22} & \ddots & \vdots \\ \vdots & \vdots & \ddots & 0 \\ \bar{a}_{1n} & \bar{a}_{2n} & \dots & \bar{a}_{nn} \end{pmatrix} \\ \Rightarrow \|T^*v_1\|^2 &= \sum_{k=1}^n |\bar{a}_{1k}|^2 = |a_{11}|^2 + |a_{12}|^2 + \dots + |a_{1n}|^2. \end{aligned}$$

As  $T$  is normal, we have that  $\|Tv_1\|^2 = \|T^*v_1\|^2$ , hence  $a_{12} = \dots = a_{1n} = 0$ . This shows that

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} \bar{a}_{11} & 0 & \dots & 0 \\ 0 & * & \dots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \dots & 0 & * \end{pmatrix}.$$

By the same argumentation, we have that  $\|Tv_2\|^2 = |a_{22}|^2$  and  $\|T^*v_2\|^2 = |a_{22}|^2 + |a_{23}|^2 + \dots + |a_{2n}|^2$ . As  $\|Tv_2\| = \|T^*v_2\|$  we obtain  $a_{23} = \dots = a_{2n} = 0$ .

We can continue the proof by induction. Suppose the claim holds for  $k$ . So, we have that

$\|Tv_{k+1}\|^2 = |a_{k+1,k+1}|^2$ , as  $a_{l,k+1} = 0 \quad \forall l < k+1$ . Since  $[T^*]_{\mathcal{B}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{B}})^*$ , we have that

$$\|T^*v_{k+1}\|^2 = |a_{k+1,k+1}|^2 + \dots + |a_{k+1,n}|^2.$$

Since  $T$  is normal, we get that  $\|T^*v_{k+1}\| = \|Tv_{k+1}\|$ . Therefore, we conclude that

$$|a_{k+1,k+2}| = \dots = |a_{k+1,n}| = 0.$$

This proves the claim.  $\square$

**Remark 9.6.** In the proof above we used  $K = \mathbb{C}$  only, when we said that  $T$  is trigonalizable. Therefore, we arrive at the following corollary.

#### Corollary 9.7

Let  $V$  be a finite dimensional inner product space over  $\mathbb{R}$  and  $T : V \rightarrow V$  an endomorphism, which is normal. If  $T$  is trigonalizable, then  $T$  is orthogonally diagonalizable.

We've seen that over  $\mathbb{R}$   $T \in \text{End}(V)$  is normal  $\not\Rightarrow T$  is orthogonally diagonalizable.

#### Definition 9.8: Self Adjoint

Let  $V$  be an inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ , and  $T \in \text{End}(V)$ .  $T$  is called **self adjoint** if  $T^* = T$ .

A matrix  $A \in M_{n \times n}(K)$  is called **self adjoint** if  $A^* = A$ . (Over  $\mathbb{R}$  a self adjoint matrix is a symmetric matrix.)

#### Lemma 9.9: A Condition for Self Adjoint

Let  $V$  be a finite dimensional inner product space over  $\mathbb{R}$  and  $T \in \text{End}(V)$  which is orthogonally diagonalizable. Then  $T$  is self adjoint.

#### Lemma 9.10

Let  $V$  be an inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ . Then  $T \in \text{End}(V)$  is self adjoint if and only if for every orthonormal basis  $\mathcal{B}$  of  $V$ , the matrix  $[T]_{\mathcal{B}}^{\mathcal{B}}$  is self adjoint.

#### Theorem 9.11: Spectral Theorem over $\mathbb{R}$

Let  $V$  be an Euclidean space(, i.e., inner product space over  $\mathbb{R}$ ) of finite dimension. Then  $T$  is orthogonally diagonalizable if and only if  $T$  is self adjoint.

For the proof of the above Theorem the following Lemma is needed.

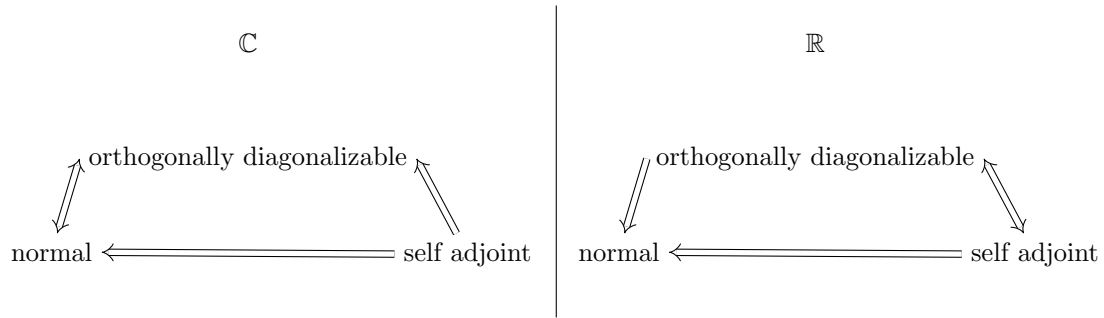
#### Lemma 9.12: Properties of Self Adjoint Maps

Let  $V$  be an inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$  and  $T \in \text{End}(V)$  be self adjoint. Then

1. All eigenvalues of  $T$  are real.
2.  $p_T$  factors into a product of linear factors over  $K$ .

In the Figure below we illustrated the logical relations regarding the Spectral theorem.





### Theorem 9.13: Spectral Theorem for Matrices

1. If  $A \in M_{n \times n}(\mathbb{C})$  is normal, then  $\exists U \in U(n)$  s.t.

$$\underbrace{U^{-1} A U}_{= U^*}$$

is diagonal.

2. If  $A \in M_{n \times n}(\mathbb{R})$  is a symmetric matrix, then  $\exists O \in O(n)$  s.t.

$$\underbrace{O^{-1} A O}_{= O^T}$$

is diagonal.

## 10 Orthogonal & Unitary maps / Isometries

### Definition 10.1: Linear Isometry

Let  $V, W$  be inner product spaces over  $K = \mathbb{R}$  or  $\mathbb{C}$ . A map  $T : V \rightarrow W$  is called a **linear isometry** if  $T$  is linear and

$$\forall v_1, v_2 \in V : \langle T v_1, T v_2 \rangle_W = \langle v_1, v_2 \rangle_V,$$

i.e.,  $T$  preserves the inner product.

**Remark 10.2.** If  $T : V \rightarrow W$  is a linear isometry, then:

1.  $\forall v \in V : \|T v\| = \|v\|$ .
2.  $\forall p, q \in V : d(T p, T q) = d(p, q)$ , where  $d(p, q) := \|p - q\|$ .
3.  $T$  is injective. Indeed, if  $v \in \ker T$ , then

$$0 = \|T v\| = \|v\| \implies v = 0.$$

So,  $\ker T = \{0\}$ .

From now on we'll concentrate on the case  $W = V$ , i.e.,  $T \in \text{End}(V)$ . If such a map  $T$  is an isometry we also call it **orthogonal endomorphism** if  $K = \mathbb{R}$ , or **unitary endomorphism** if

$K = \mathbb{C}$ .

### Theorem 10.3: Equivalent Statements to Isometry

Let  $V$  be a finite dimensional inner product space over  $K = \mathbb{R}$  or  $\mathbb{C}$ , and  $T \in \text{End}(V)$ . The following are equivalent:

1.  $T$  is an isometry.
2.  $\forall v \in V : \|Tv\| = \|v\|$ .
3. For every orthonormal system  $e_1, \dots, e_k$  of  $V$ , the list of vectors  $Te_1, \dots, Te_k$  is also an orthonormal system.
4. There exists an orthonormal basis  $e_1, \dots, e_n$  of  $V$  s.t.  $Te_1, \dots, Te_n$  is also an orthonormal basis.
5.  $TT^* = id_V$ .
6.  $T^*T = id_V$ .
7.  $T^* = T^{-1}$ .
8.  $T^*$  is an isometry.

Back to matrices.

### Proposition 10.4: Equivalent Condition for a Orthogonal/Unitary Map

Let  $V$  be an  $n$ -dimensional inner product space over  $K = \mathbb{R}$  (respectively  $\mathbb{C}$ ). Let  $\mathcal{B}$  be an orthonormal basis for  $V$ . Let  $T \in \text{End}(V)$ . Then  $T$  is orthogonal (respectively unitary) if and only if  $[T]_{\mathcal{B}}^{\mathcal{B}} \in O(n)$  (respectively  $[T]_{\mathcal{B}}^{\mathcal{B}} \in U(n)$ ).

Note that  $\forall A \in O(n)$  we have  $\det A = \pm 1$ . Indeed,

$$1 = \det I = \det(AA^T) = \det A \cdot \det A.$$

Similarly,  $\forall A \in U(n)$ ,  $|\underbrace{\det A}_{\in \mathbb{C}}| = 1$ . Indeed,

$$1 = \det(AA^*) = \det A \cdot \det \overline{A}^T = \det A \cdot \det \overline{A} = \det A \cdot \overline{\det A} = |\det A|^2.$$

### Definition 10.5: Special Orthogonal/Unitary Group

The set

$$SO(n) := \{A \in O(n) \mid \det A = 1\} \subseteq O(n) \subseteq GL_n(\mathbb{R})$$

is called the **special orthogonal group**.

The set

$$SU(n) := \{A \in U(n) \mid \det A = 1\} \subseteq U(n) \subseteq GL_n(\mathbb{C})$$

is called the **special unitary group**.

**Proposition 10.6: Group Structure of  $O(n)/SO(n)/U(n)/SU(n)$**

Let  $G = O(n), SO(n), U(n), SU(n)$ . Then:

1.  $I_n \in G$ .
2.  $\forall A, B \in G : A \cdot B \in G$ .
3.  $\forall A \in G : A^{-1} \in G$ .

### 10.1 Classification of the Elements of $O(2), O(3)$ etc.

**Lemma 10.7: Eigenvalue of an Orthogonal Matrix**

Let  $A \in O(n)$ . If  $\lambda$  is an eigenvalue of  $A$ , then  $\lambda = \pm 1$ .

**Proposition 10.8: Classification of the Elements of  $O(2)$**

Let  $A \in O(2)$ .

1. If  $\det A = 1$ , then  $\exists \theta \in [0, 2\pi)$  s.t.

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} := R_\theta.$$

The map  $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  is an anti-clockwise rotation by the angle  $\theta$ .

2. If  $\det A = -1$ , then there exists a basis  $\mathcal{B} = (v_1, v_2)$  for  $\mathbb{R}^2$  s.t.  $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^2$  has the following representation in the basis  $\mathcal{B}$

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (10.1)$$

In other words  $T_A$  performs a reflection w.r.t. the line  $l = \text{Sp}(v_1)$ . Moreover,  $\exists \theta \in [0, 2\pi)$  s.t.

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} = R_\theta \cdot \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The basis  $\mathcal{B}$  for which Equation (10.1) holds, is given by

$$v_1 = \begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} \end{pmatrix}, v_2 = \begin{pmatrix} \cos(\frac{\theta}{2} + \frac{\pi}{2}) \\ \sin(\frac{\theta}{2} + \frac{\pi}{2}) \end{pmatrix}.$$

**Summary:**

$$\begin{aligned} SO(2) &= \text{rotations} \\ O(2) \setminus SO(2) &= \text{reflections.} \end{aligned}$$

## Corollary 10.9

Let  $V$  be a 2-dimensional Euclidean space and  $T : V \rightarrow V$  a linear isometry. Then  $\det T = \pm 1$ . Moreover, there exists an orthonormal basis  $\mathcal{B}$  for  $V$  s.t.

1. If  $\det T = 1$ , then  $[T]_{\mathcal{B}}^{\mathcal{B}} = R_{\theta}$ .

2. If  $\det T = -1$ , then

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Proposition 10.10: **Classification of the Elements of  $SO(3)$** 

Let  $A \in SO(3)$ . Then 1 is an eigenvalue of  $A$ . Moreover, there exists an orthonormal basis  $\mathcal{B} = (v_1, v_2, v_3)$  and  $\theta \in [0, 2\pi)$  s.t.

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left( \begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & & \end{array} \begin{array}{cc} R_{\theta} \end{array} \right).$$

So,  $T_A$  is a rotation with angle  $\theta$  around the axis  $\text{Sp}(v_1)$ . Moreover,

(a) If  $-1$  is an eigenvalue of  $A$ , then  $\theta = \pi$ , hence

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

(b) If  $-1$  is **not** an eigenvalue of  $A$  ( $\iff 1$  is the only eigenvalue), then  $\theta \neq \pi$ .

*Proof.* The characteristic polynomial  $p_A(x)$  of  $A$  is a polynomial of degree 3 and with real coefficients. By the intermediate value theorem (Analysis I) there exists a real zero  $\lambda \in \mathbb{R}$  (i.e.,  $p_A(\lambda) = 0$ ). Therefore,  $A$  has at least one real eigenvalue. By Lemma 10.7,  $\lambda = \pm 1$ . Let  $v_1 \in \mathbb{R}^3 \setminus \{0\}$  be an eigenvector of  $\lambda$ . We can normalize  $v_1$  and still have an eigenvector for  $\lambda$ . So, let's assume  $\|v_1\| = 1$ . Consider the subspace

$$L := v_1^\perp = \{v \in \mathbb{R}^3 \mid \langle v, v_1 \rangle = 0\}.$$

Since  $A$  is orthogonal, we have that

$$\forall v \in L : Av \in L,$$

i.e.,  $T_A(L) \subseteq L$ , i.e.,  $L$  is invariant under  $T_A$ . Indeed, if  $v \in L$ , then since  $A \in O(3)$

$$\begin{aligned} 0 &= \langle v, v_1 \rangle = \langle Av, Av_1 \rangle = \langle Av, \pm v_1 \rangle = \pm \langle Av, v_1 \rangle \\ \implies Av &\in L. \end{aligned}$$

We also have  $\mathbb{R}^3 = \text{Sp}(v_1) \oplus L$ , hence  $\dim L = 2$ .

Now  $T_A|_L : L \rightarrow L$  is a linear isometry. Let  $\mathcal{L} = (w_1, w_2)$  be an orthonormal basis for  $L$ . By Proposition 10.4, we have that

$$Q := [T_A|_L]_{\mathcal{L}}^{\mathcal{L}} \in O(2).$$

Case (a):  $\lambda = 1$ . Take  $\mathcal{B} = (v_1, w_1, w_2)$ . Then

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left( \begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & Q & \end{array} \right).$$

Now, since  $A \in SO(3)$ , we have

$$\begin{aligned} 1 &= \det A = \det(T_A) = 1 \cdot \det Q \\ \implies \det Q &= 1. \end{aligned}$$

So,  $Q \in SO(2)$ . By Proposition 10.10,  $\exists \theta \in [0, 2\pi)$  s.t.  $Q = R_\theta$ . Therefore,

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left( \begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\theta & \end{array} \right).$$

Case (b):  $\lambda = -1$ . By the same argument as for case (a), we get that

$$\begin{aligned} 1 &= \det A = (-1) \cdot \det(T_A|_L) \\ \implies \det(T_A|_L) &= -1. \end{aligned}$$

By Corollary 10.9, there exists an orthonormal basis  $\mathcal{L} = (w_1, w_2)$  for  $L$  s.t.

$$[T_A|_L]_{\mathcal{L}}^{\mathcal{L}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Take  $\mathcal{B}' = (v_1, w_1, w_2)$ . Then

$$[T_A]_{\mathcal{B}'}^{\mathcal{B}'} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

And for  $\mathcal{B} = (w_1, v_1, w_2)$  we have

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \left( \begin{array}{c|cc} 1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\pi & \end{array} \right).$$

Note that in case (b), 1 is an eigenvalue of  $A$  as well. □

### Proposition 10.11: Classification of $O(3) \setminus SO(3)$

Let  $A \in O(3) \setminus SO(3)$ . Then there exists an orthonormal basis  $\mathcal{B} = (v_1, v_2, v_3)$  for  $\mathbb{R}^3$  s.t.

$$[T_A]_{\mathcal{B}}^{\mathcal{B}} = \left( \begin{array}{c|cc} -1 & 0 & 0 \\ \hline 0 & & \\ 0 & R_\theta & \end{array} \right)$$

for some  $\theta \in [0, 2\pi)$ . (So  $T_A$  does a rotations by  $\theta$  around  $\text{Sp}(v_1)$ , followed by a reflection w.r.t.  $\text{Sp}(v_2, v_3)$ .)

## 11 Bilinear & Quadratic Forms

### Definition 11.1: Bilinear Form

Let  $V$  be a vector space over  $K$ . A function  $B : V \times V \rightarrow K$  is called a **bilinear form** if the following two conditions are satisfied:

1.  $\forall w \in V$  the map  $B(\cdot, w) : V \rightarrow K, \quad v \mapsto B(v, w)$  is linear.
2.  $\forall v \in V$  the map  $B(v, \cdot) : V \rightarrow K, \quad w \mapsto B(v, w)$  is linear.

### Lemma 11.2: Matrix Representation of a Bilinear Form

Let  $V$  be a vector space over  $K$  and  $\mathcal{C} = (e_1, \dots, e_n)$  a basis for  $V$ . Let  $B : V \times V \rightarrow K$  be a bilinear form. Then  $\exists [B]_{\mathcal{C}} \in M_{n \times n}(K)$  s.t.

$$B\left(\sum_{i=1}^n c_i e_i, \sum_{j=1}^n d_j e_j\right) = \begin{pmatrix} c_1 & \dots & c_n \end{pmatrix} \cdot [B]_{\mathcal{C}} \cdot \begin{pmatrix} d_1 \\ \vdots \\ d_n \end{pmatrix} \quad \forall c_i, d_j \in K.$$

The matrix  $[B]_{\mathcal{C}} = B(e_i, e_j)$  is called the matrix representation of  $B$  in the basis  $\mathcal{C}$ .

We saw also that for  $T \in \text{End}(V)$  and  $\mathcal{C}$  a basis for  $V$  we have a matrix representation  $[T]_{\mathcal{C}}^{\mathcal{C}} \in M_{n \times n}(K)$ , and that if  $\mathcal{D}$  is another basis for  $V$ , then

$$\begin{aligned} [T]_{\mathcal{D}}^{\mathcal{D}} &= \underbrace{[id_V]_{\mathcal{D}}^{\mathcal{C}}}_{= ([id_V]_{\mathcal{D}}^{\mathcal{D}})^{-1}} \cdot [T]_{\mathcal{C}}^{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}. \end{aligned} \tag{11.1}$$

(Corollary 3.17). What happens for bilinear forms?

### Lemma 11.3: Change of Bases for a Bilinear Form

Let  $V$  be a finite dimensional vector space over  $K$  and  $\mathcal{C}, \mathcal{D}$  be two bases for  $V$ . Let  $B : V \times V \rightarrow K$  be a bilinear form. Then

$$[B]_{\mathcal{D}} = ([id_V]_{\mathcal{C}}^{\mathcal{D}})^T \cdot [B]_{\mathcal{C}} \cdot [id_V]_{\mathcal{C}}^{\mathcal{D}}.$$

(This is quite different from Equation (11.1)).

### Definition 11.4: Symmetric/Positive Definite Bilinear Form & Quadratic Form

1. A bilinear form  $B : V \times V \rightarrow K$  is called **symmetric** if

$$\forall v, w \in V : B(v, w) = B(w, v).$$

2. Assume that  $K = \mathbb{R}$ . A bilinear form  $B : V \times V \rightarrow K$  is called **positive definite** if

$$\forall v \in V : B(v, v) > 0.$$

3. For a bilinear form  $B : V \times V \rightarrow K$ , we define

$$q_B : V \rightarrow K, \quad q_B(v) := B(v, v).$$

We call  $q_B$  the **quadratic form** associated to  $B$ . Sometimes we just denote it by  $q$ .

**Remark 11.5.** Note that  $B \in M_{n \times n}(K)$  and  $B^T \in M_{n \times n}(K)$  give the same quadratic form  $q : K^n \rightarrow K$ . Indeed,  $\forall v \in K^n$

$$q_B(v) = v^T B v = (v^T B v)^T = v^T B^T v = q_{B^T}(v),$$

where we used that  $q_B(v)$  is a scalar.

Assume now that  $2 \neq 0$  in  $K$ , then (i.e.,  $1 + 1 \neq 0$ ). Then also  $\frac{1}{2}(B + B^T)$  gives the same quadratic form as  $B$ .

**Conclusion:** if  $2 \neq 0$  in  $K$ , then for the discussion on quadratic forms  $q : K^n \rightarrow K$ , we can assume they all come from symmetric matrices / bilinear forms.

### Proposition 11.6: Polarization Identity

Assume  $2 \neq 0$  in  $K$ . (note that then also  $4 \neq 0$ .) If  $B$  is symmetric and defines  $q := q_B$ , then  $\forall v, w \in V$  we have

- 1.

$$B(v, w) = \frac{1}{4}q(v + w) - \frac{1}{4}q(v - w).$$

- 2.

$$B(v, w) = \frac{1}{2}(q(v + w) - q(v) - q(w)).$$

### Theorem 11.7: Sylvester's Inertia Theorem

Let  $B : V \times V \rightarrow K$  be a symmetric bilinear form on a vector space  $V$  over  $\mathbb{R}$ , of dimension  $n$ . Then there exists a basis  $\mathcal{C}$  for  $V$  s.t.

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_k & 0 & 0 \\ 0 & -I_l & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

for some  $k, l \in \mathbb{Z}_{\geq 0}$  with  $k + l \leq n$ . Moreover,  $k$  and  $l$  do **not** depend on the basis  $\mathcal{C}$  (they depend only on  $B$ ). In fact

$$\begin{aligned} k &= \max\{\dim W \mid W \subseteq V \text{ is a subspace and } B|_{W \times W} \text{ is positive definite}\}, \\ l &= \max\{\dim U \mid U \subseteq V \text{ is a subspace and } B|_{U \times U} \text{ is positive definite}\}, \\ n - (k + l) &= \max\{\dim Z \mid Z \subseteq V \text{ is a subspace and } B|_{Z \times V} \equiv 0\}. \end{aligned}$$

**Remark 11.8.** 1. In the basis  $\mathcal{C}$ , the quadratic form  $q$  associated to  $B$  looks like

$$q(x_1, \dots, x_n) = x_1^2 + \dots + x_k^2 - x_{k+1}^2 - \dots - x_{k+l}^2.$$

2.  $k$  is called the **positivity index** of  $B$ ,  $l$  the **negativity index**,  $n$  the **nullity** and  $\sigma(B) := k - l$  is called the **signature** of  $B$ . We say  $B$  is a bilinear form of type  $(k, l)$ .

### Theorem 11.9: Sylvester's Inertia Theorem for Euclidean Spaces

Let  $V$  be a finite dimensional Euclidean space of dimension  $n$ . Let  $B : V \times V \rightarrow K$  be a symmetric bilinear form. Then there exists an **orthonormal** basis  $\mathcal{C}$  for  $V$  in which

$$[B]_{\mathcal{C}} = \begin{pmatrix} \eta_1 & 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 \\ 0 & \ddots & \ddots & & & & & & \vdots \\ \vdots & \ddots & \eta_k & \ddots & & & & & \vdots \\ \vdots & & \ddots & \eta_{k+1} & \ddots & & & & \vdots \\ \vdots & & & \ddots & \eta_{k+l} & \ddots & & & \vdots \\ \vdots & & & & \ddots & \eta_{k+l} & \ddots & & \vdots \\ \vdots & & & & & \ddots & 0 & \ddots & \vdots \\ \vdots & & & & & & \ddots & \ddots & 0 \\ 0 & \dots & \dots & \dots & \dots & \dots & \dots & 0 & 0 \end{pmatrix},$$

where  $k, l \in \mathbb{Z}_{\geq 0}$  and  $k + l \leq n$ ,  $\eta_1, \dots, \eta_k > 0$ ,  $\eta_{k+1}, \dots, \eta_{k+l} < 0$ . As before,  $k, l$  are the positivity and negativity indices of  $B$ . Moreover, the scalars  $\eta_1, \dots, \eta_k, \eta_{k+1}, \dots, \eta_{k+l}, 0, \dots, 0$  are independent of  $\mathcal{C}$ . They depend only on  $B$  (and on the scalar product on  $V$ ). They are the eigenvalues of  $[B]_{\mathcal{D}}$  for every **orthonormal** basis  $\mathcal{D}$  of  $V$  (independent of  $\mathcal{D}$  and on whether or not  $[B]_{\mathcal{D}}$  is diagonal).

**Remark 11.10.** This theorem has a geometric meaning. If  $q : \mathbb{R}^n \rightarrow \mathbb{R}$  is a quadratic form

$$q(x_1, \dots, x_n) = \sum_{i=1}^n a_{ij} x_i x_j$$



for some  $a_{ij} \in \mathbb{R}$ , then the set

$$Q = \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid q(x_1, \dots, x_n) = C \in \mathbb{R}\}$$

is called a quadric. The theorem tells us that there exists a linear isometry  $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$  s.t.

$$T(Q) = \{(y_1, \dots, y_n) \in \mathbb{R}^n \mid \underbrace{\eta_1 y_1^2 + \dots + \eta_k y_k^2}_{\text{pos. coeffs.}} + \underbrace{\eta_{k+1} y_{k+1}^2 + \dots + \eta_{k+l} y_{k+l}^2}_{\text{neg. coeffs.}} = C \in \mathbb{R}\}.$$

### Theorem 11.11: Sylvester's Theorem for 'almost' general Fields

Let  $V$  be a finite dimensional vector space over a field  $K$ , in which  $2 \neq 0$ . Let  $B : V \times V \rightarrow K$  be a symmetric bilinear form. Then there exists a basis  $\mathcal{C}$  of  $V$  s.t.  $[B]_{\mathcal{C}}$  is diagonal.

What happens over  $\mathbb{C}$ ?

### Theorem 11.12: Sylvester's Theorem over $\mathbb{C}$

Let  $B : V \times V \rightarrow \mathbb{C}$  be a symmetric bilinear form on a vector space over  $\mathbb{C}$  of dimension  $n$ . Then there exists a basis  $\mathcal{C}$  of  $V$  s.t.

$$[B]_{\mathcal{C}} = \begin{pmatrix} I_r & 0 \\ 0 & 0 \end{pmatrix},$$

for some  $0 \leq r \leq n$ . The number  $r$  is independent of the basis  $\mathcal{C}$ . It depends only on  $B$ . In fact

$$n - r = \max\{\dim W \mid W \subseteq V \text{ is a subspace s.t. } B|_{W \times V} \equiv 0\}.$$

$r$  is sometimes called the rank of  $B$ .

## 11.1 More on positive definite Bilinear Forms

### Theorem 11.13: Equivalent Conditions for a Positive Definite Bilinear Form

Let  $V$  be a finite dimensional vector space over  $\mathbb{R}$  and  $B : V \times V \rightarrow \mathbb{R}$  a symmetric bilinear form. Then the following conditions are equivalent:

1.  $B$  is positive definite
2. There exists a basis  $\mathcal{C} = (e_1, \dots, e_n)$  for  $V$  s.t. the principal minors  $\det(B(e_j, e_k))_{j,k=1, \dots, l}$  for all  $1 \leq l \leq n$  are positive.
3. Same as 2. But instead of  $\exists$  a basis put for  $\forall$  bases.

## 12 Jordan Normal Form

### Definition 12.1: Jordan Matrix

Let  $\lambda \in K, n \in \mathbb{Z}_{\geq 1}$ . Define

$$J_{\lambda,n} = \begin{pmatrix} \lambda & 1 & 0 & \dots & 0 \\ 0 & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ \vdots & & \ddots & \ddots & 1 \\ 0 & \dots & \dots & 0 & \lambda \end{pmatrix} \in M_{n \times n}(K).$$

For  $n = 1$ ,  $J_{\lambda,1} = (\lambda)$ . For  $n = 2$ ,

$$J_{\lambda,2} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}.$$

For  $n = 3$ ,

$$J_{\lambda,3} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix}.$$

$J_{\lambda,n}$  is called the  $n$ -dimensional **Jordan matrix** with eigenvalue  $\lambda$ .

### Lemma 12.2: Properties of the Jordan Matrix

1.  $\lambda$  is the only eigenvalue of  $J_{\lambda,n}$
2.  $p_{J_{\lambda,n}}(x) = (\lambda - x)^n$ , and  $m_a(J_{\lambda,n}, \lambda) = n$ .
3.  $m_g(J_{\lambda,n}, \lambda) = 1$ . In fact  $\text{Eig}_{\lambda}(J_{\lambda,n}) = \text{Sp}(e_1)$ .

### Theorem 12.3: Jordan Normal Form (JNF)

Let  $V$  be an  $n$ -dimensional vector space over  $K$  and  $T \in \text{End}(V)$  s.t.  $p_T(x)$  splits as a product of linear factors in  $K[x]$  (e.g. for  $K = \mathbb{C}$  this always happens). Then there exists a basis  $\mathcal{B}$  for  $V$  s.t.

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1, n_1} & 0 & \dots & 0 \\ 0 & J_{\lambda_2, n_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & J_{\lambda_k, n_k} \end{pmatrix},$$

where  $k \geq 1, \lambda_1, \dots, \lambda_k \in K, n_1, \dots, n_k \geq 1, n_1 + \dots + n_k = n$ .  $\mathcal{B}$  is called a **Jordan basis** for  $T$ . Moreover, up to permutation, the blocks  $J_{\lambda_1, n_1}, \dots, J_{\lambda_k, n_k}$  are unique.

For an example see the lecture notes.

We can write  $J_{\lambda,n} = \lambda \cdot I_n + N$ , where

$$N = \begin{pmatrix} 0 & 1 & 0 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ \vdots & & & \ddots & 1 \\ 0 & \dots & \dots & \dots & 0 \end{pmatrix} = J_{0,n}$$

#### Lemma 12.4: Powers of $N$

1.

$$N^2 = \begin{pmatrix} 0 & 0 & 1 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 1 \\ \vdots & & & \ddots & 0 \\ 0 & \dots & \dots & \dots & 0 \end{pmatrix}, \quad N^3 = \dots, \quad \dots, N^n = 0,$$

$N^k = 0 \quad \forall k \geq n$ . In other words,  $N^k$  has 1's in the  $k$ -th diagonal above the central diagonal and 0's everywhere else. Note also that  $N$  is nilpotent.

2.  $\forall k \geq 1$  we have

$$J_{\lambda,n}^k = \sum_{j=0}^k \binom{k}{j} \lambda^{k-j} N^j.$$

#### Corollary 12.5: Geometric Multiplicity and Minimal Polynomial of JNF

Suppose  $T \in \text{End}(V)$ , admits a matrix representation

$$[T]_{\mathcal{B}}^{\mathcal{B}} = \begin{pmatrix} J_{\lambda_1, n_1} & 0 & \dots & 0 \\ 0 & J_{\lambda_2, n_2} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & J_{\lambda_k, n_k} \end{pmatrix}$$

in some basis  $\mathcal{B}$ . Then for every eigenvalue  $\lambda$  of  $T$  the geometric multiplicity  $m_g(\lambda)$  equals the number of blocks in  $[T]_{\mathcal{B}}^{\mathcal{B}}$  labeled with  $\lambda$ . In other words,

$$m_g(\lambda) = |\{j \mid \lambda_j = \lambda, \quad 1 \leq j \leq k\}|.$$

The minimal polynomial  $\mu_T(x)$  is

$$\mu_T(x) = \prod_{\lambda = \text{eigenvalue}} (x - \lambda)^{S(\lambda)},$$

where

$$\begin{aligned} S(\lambda) &= \text{size of the largest block labeled by } \lambda \\ &= \max\{n_j \mid \lambda_j = \lambda, \quad 1 \leq j \leq k\}. \end{aligned}$$